The CPU: A very tall pile of simple

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Part 1: Foundations

You can hear the phrase “computers think in 1s and 0s” a hundred times and still not understand how a computer actually works. By itself, this explains basically nothing. Sure, a wire can be high or low. Sure, a light can be on or off. But how does that become addition?

How does that become memory?

How does that become a program sitting in `RAM`, one instruction after another, telling a machine what to do?

Some resources stay extremely high-level, so you never really understand how a CPU actually works. The deeper resources are amazing, but they are long, dense, and intimidating. And frankly, for someone who doesn’t want that level of detail, a lot of it can often feel like too much.

My goal is to help you understand what is going on under the hood, without exploding your brain or eating weeks of time.

We start with a simple circuit turning a light bulb on and off, then work our way through logic gates, memory, and the basic circuits underneath them.

The key point is that nothing here is smart in isolation. A CPU is not one hard idea. It is a very tall pile of simple ones.

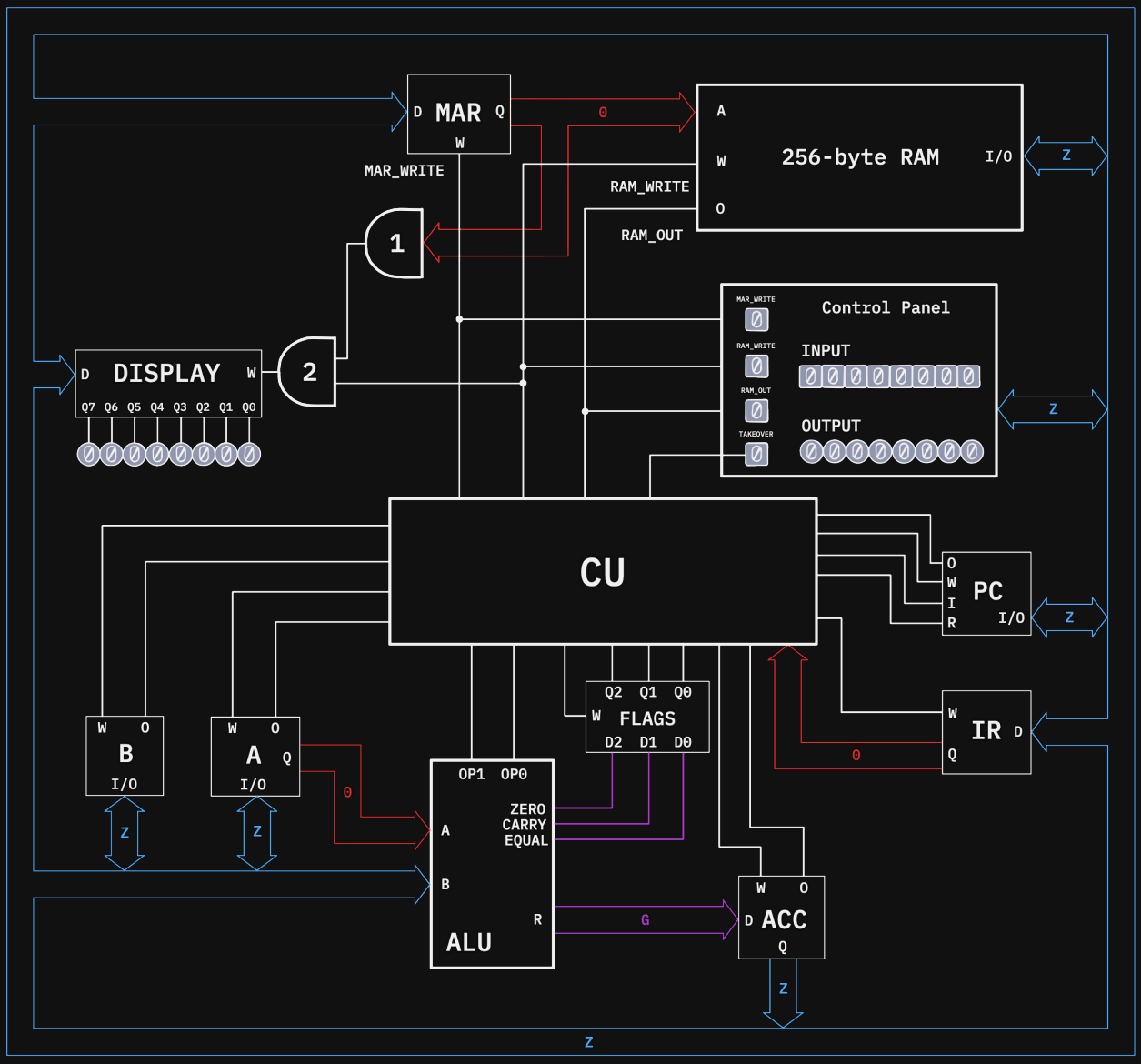


Diagram 1.1. The whole CPU.

This probably looks like an overwhelming mess, but don't worry, that's normal. Throughout this guide we are going to try to understand how this simple CPU works, from the ground up. It is not a modern CPU with decades of optimization, but it has the same core functionality. So if you have ever wondered how a computer truly works, buckle up, we are going on a journey.

1.1. How to Read This

- If a diagram is hard to understand, click it and step through each frame one by one using the arrow keys, with the provided descriptions. This only works if you are reading on my website, bytecolony.computer.

- Don't try to "memorize" every layout. Focus on the mental models, concepts, and what part each piece has to play.
- If a section feels dense or hard to understand, follow the diagrams first and try to get a feel for what is happening.

1.2. Circuits & Switches

First, we need the basics of how electricity and circuits work. Here is a simple circuit:

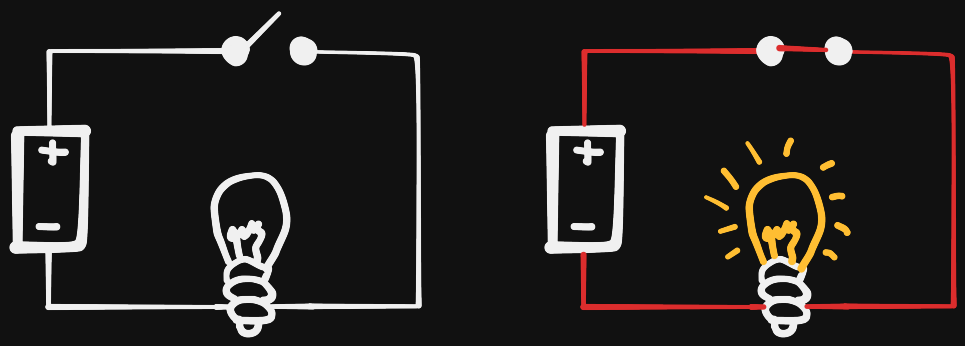


Diagram 3.1. A basic circuit with a battery, switch, and bulb.

The wire connected to the `-` of the battery is being driven low, and the wire connected to the `+` of the battery is being driven high. This difference in voltage causes current to flow through the filament of the light bulb, turning the light on. But current can only flow when the loop is complete, from `+` through the bulb to `-`.

If the loop is broken, nothing flows. A switch is simply a controlled break in the loop, allowing us to break and complete the loop whenever we want. And a light bulb is just a simple light bulb. It glows when current flows through the filament. Now we have a circuit that can do one yes/no thing.

Let's see if we can combine more switches so the circuit can "answer" slightly more interesting questions. Say we want to build a simple dog washer circuit: a circuit that, based on some inputs, can tell us whether to wash our dog or not. Our simple circuit is going to use a light bulb being on to mean yes, wash the dog. Light bulb off means no, don't wash the dog. So we start with an extremely simple version with two switches. In this first version, the switches are directly inside the bulb circuit. The person using the circuit can open or close each switch to answer a yes/no question.

Let's say switch 1 represents `STINKY`: whether the dog is stinky or not. Switch 2 represents `OLD_WASH`: has it been more than 5 days since the last wash. The rules for our first circuit are:

If `STINKY AND OLD_WASH`, the bulb is on. Or in other words, if the dog is stinky and its last wash was over 5 days ago, then wash the dog. Here is the circuit:

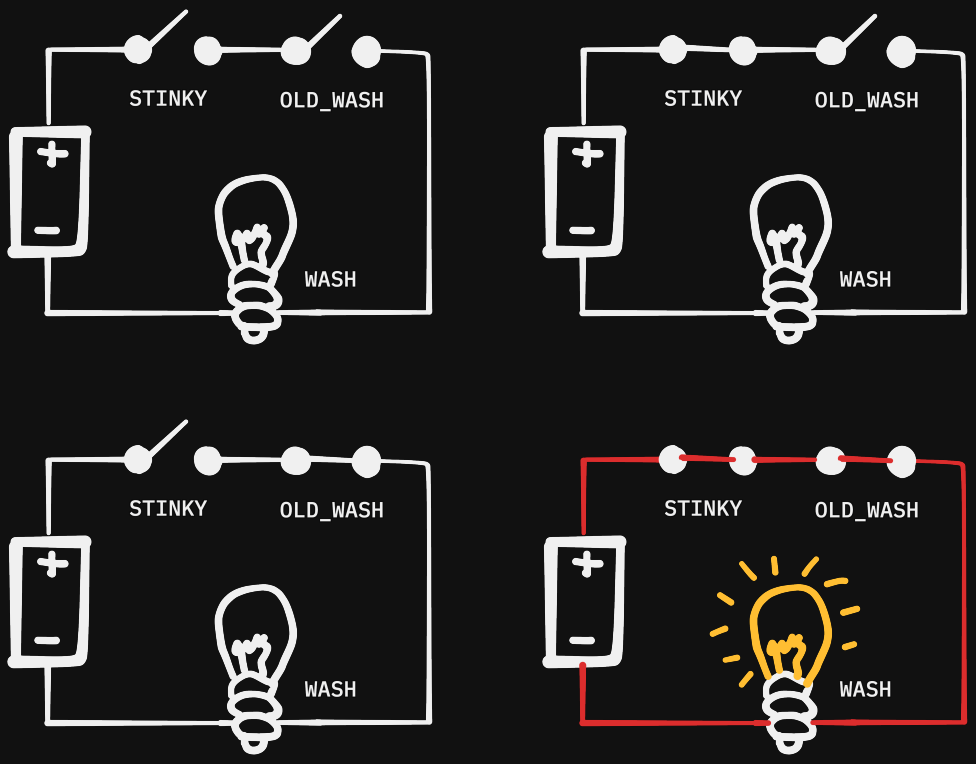


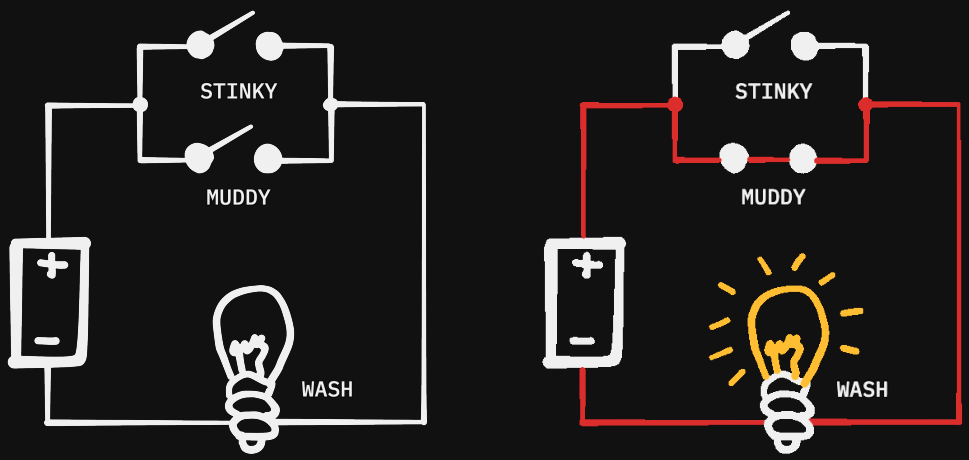
Diagram 3.2. The hand-switch version of AND.

This circuit shows a logical AND operation. A person is flipping the switches manually. The output turns on only when both inputs are true.

Let's add a new input: `MUDDY`, if the dog is muddy. The rules of the circuit change:

If `(MUDDY OR STINKY) AND OLD_WASH`, the bulb is on.

All this says is, if the dog is muddy or stinky and it's been more than 5 days since the dog's last wash, you should wash the dog. Let's focus on the `MUDDY OR STINKY` part of this circuit:



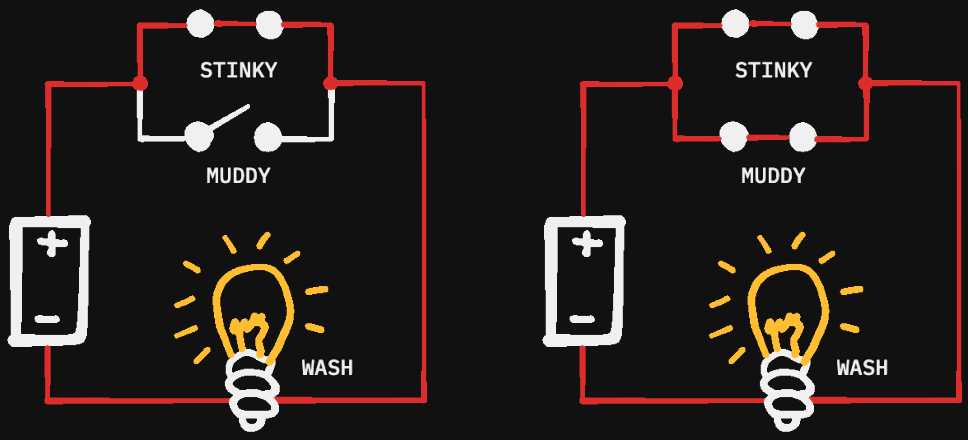


Diagram 3.3. The hand-switch version of OR.

This is a logical OR: at least one of `MUDDY` or `STINKY` needs to be on for the bulb to turn on. Both of them could also be on, and that is fine.

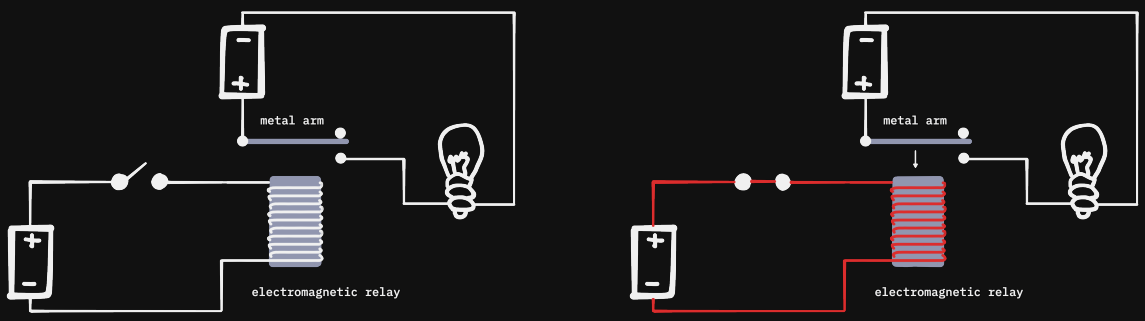
Before we continue, let's tackle a bigger problem. So far, every switch is flipped by a human. If we ever want to build a computer, it needs to run by itself: electricity has to be able to flip a switch by itself in order to chain circuits! But how?

1.3. Relays & Logic Gates

Electromagnetic relays, that's how. Or at least, that is one early solution to this problem. We will talk about other solutions a little more later on. This probably sounds quite complicated, but it is just a magnet powered by electricity.

One thing to mention before the next diagram: if you see many little batteries in a circuit, don't interpret that as several power sources. The battery symbols show if a wire is connected to power.

Anyway, here is how a relay works:



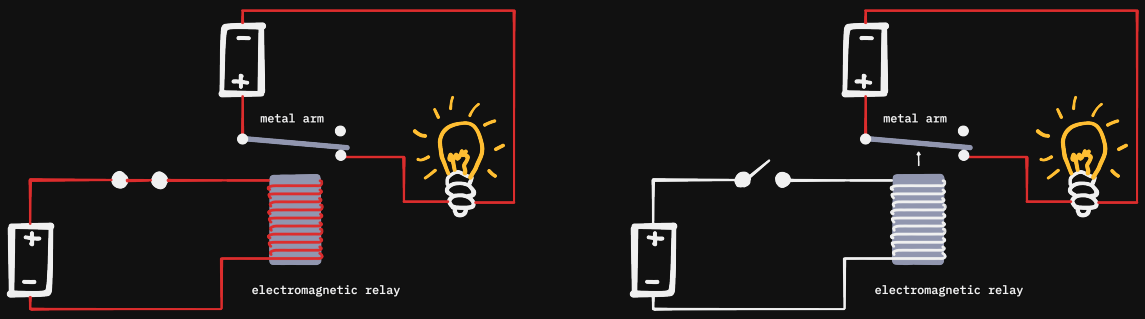


Diagram 4.1. An electromagnetic relay.

This relay is made from a coil of wire and a movable metal arm. When current flows through the coil, the coil becomes a magnet and pulls the arm down. When current stops, a spring pulls the arm back up.

A relay lets one circuit open or close a switch in another circuit.

In this example, we end up using a switch in the input circuit anyway, but any kind of electrical signal could be used, like the output of another circuit. The switch is just there to demonstrate how the relay works.

As you can also tell by the diagram, there is a slight delay between the coil turning on and the metal arm moving. Relays are mechanical, so they do not switch instantly.

Now let's see how we can build an actual electrical AND gate that takes two input wires and outputs an electrical signal.

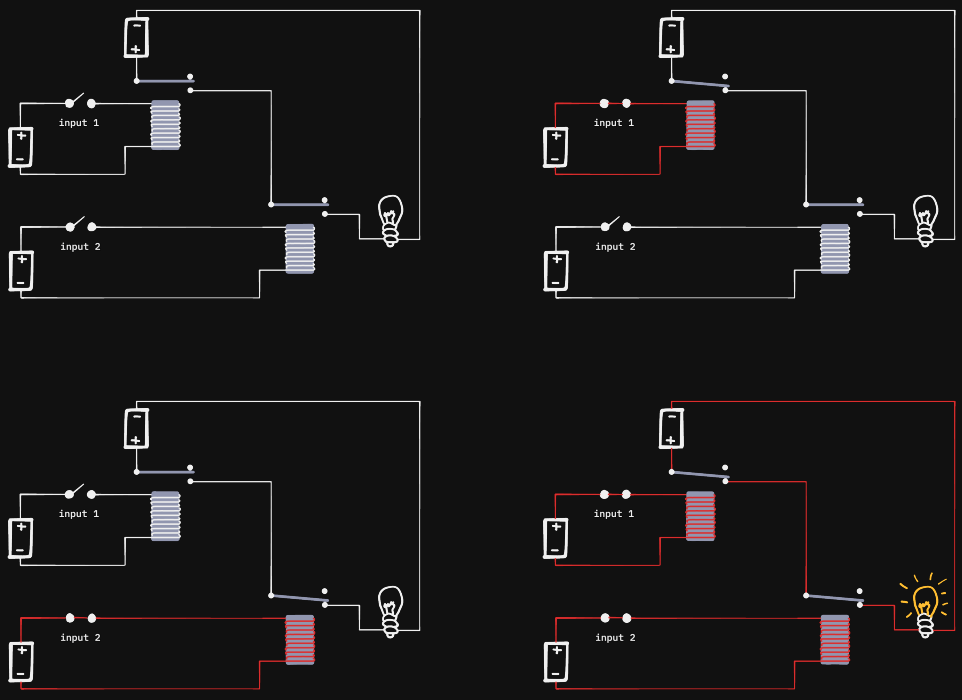


Diagram 4.2. An AND gate.

The output circuit has two breaks in it, one controlled by each input relay. Only when both inputs are on do both relays close, completing the output loop.

That is the big difference from the hand-switch version. The inputs are still drawn as switches here, but they don't have to be. The input could just be a wire: driven high, current flows through the coil and the relay pulls. Driven low, no current flows and the relay lets go. Using these relays chained in clever ways, you can create every fundamental logic gate, such as the OR gate.

But before the next diagram, I am going to use one more new symbol: ground. For our purposes here, the ground symbol will simply refer to the common return point of the circuit, usually connected to the negative side of the battery.

Every point marked with the ground symbol is connected together, as if there were hidden wires joining them underneath the drawing. It is not a new component. It is a less messy way to draw the return path of the circuit. The circuits are still loops. I am not explicitly drawing the return wire anymore.

In a real schematic, the ground symbol itself would usually stay white. In these diagrams, I color it red when that return point is part of the active path for that frame. I think it makes the current path easier to follow visually, and so the concept is easier to understand, especially for beginners.

This is how the ground symbol looks:

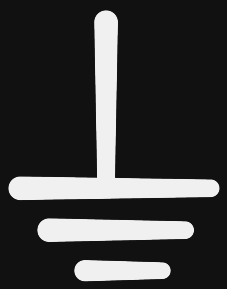


Diagram 4.3. The ground symbol.

Now here is the OR gate:

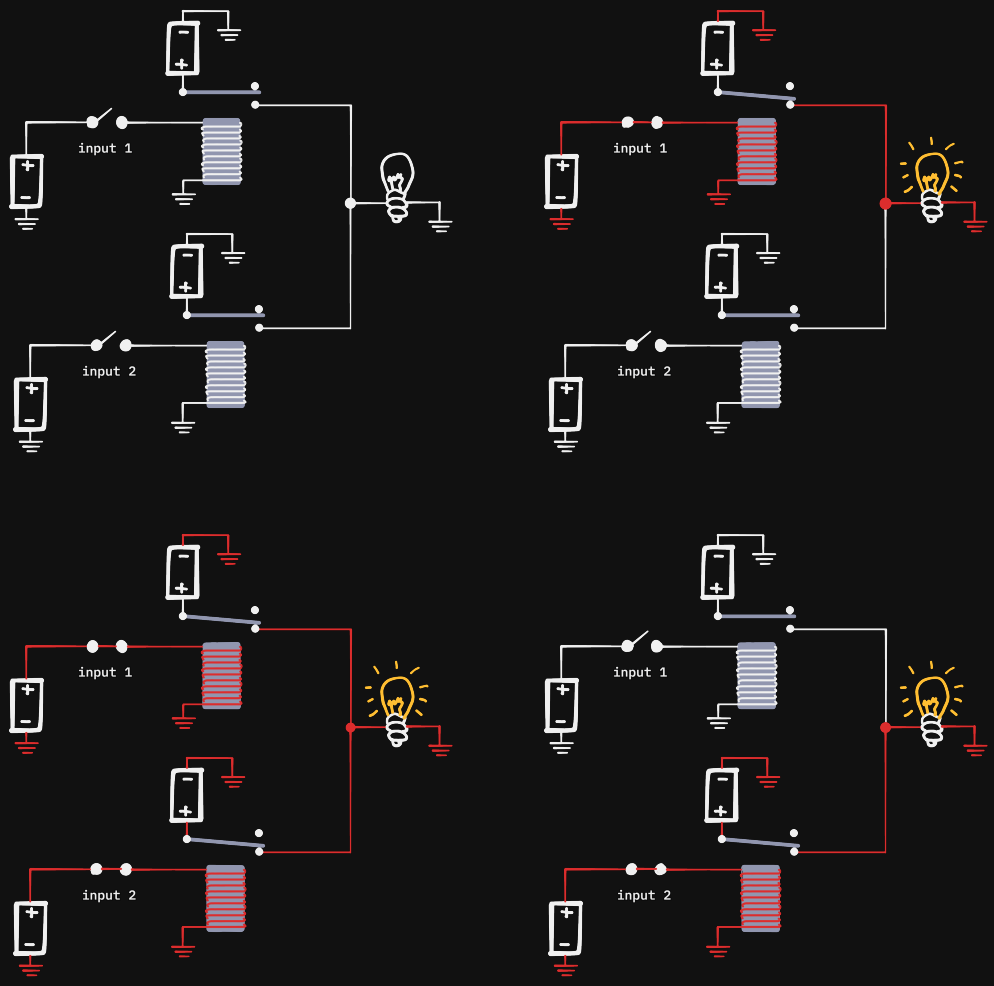
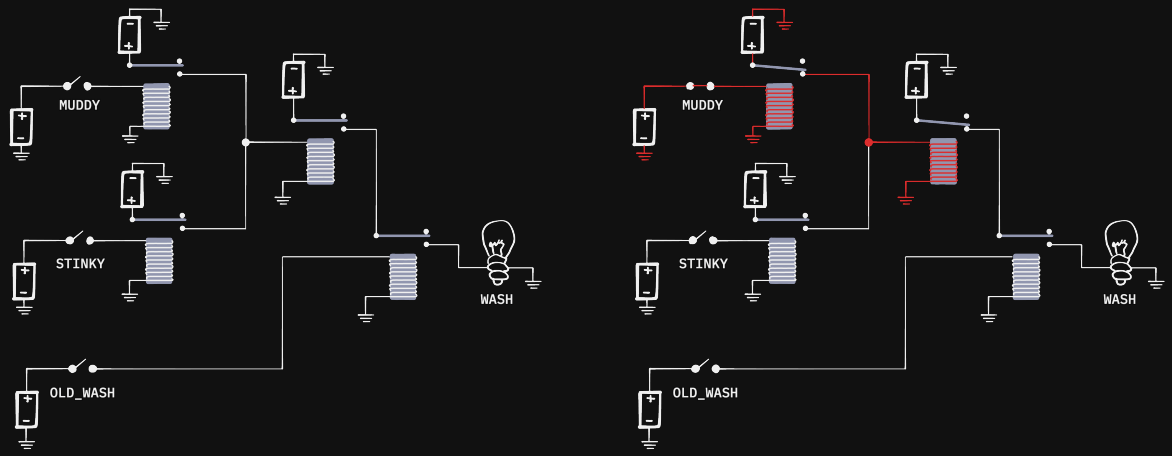


Diagram 4.4. An electronic OR gate.

That is an OR gate using relays. Now here is the full dog washer circuit up to this point:



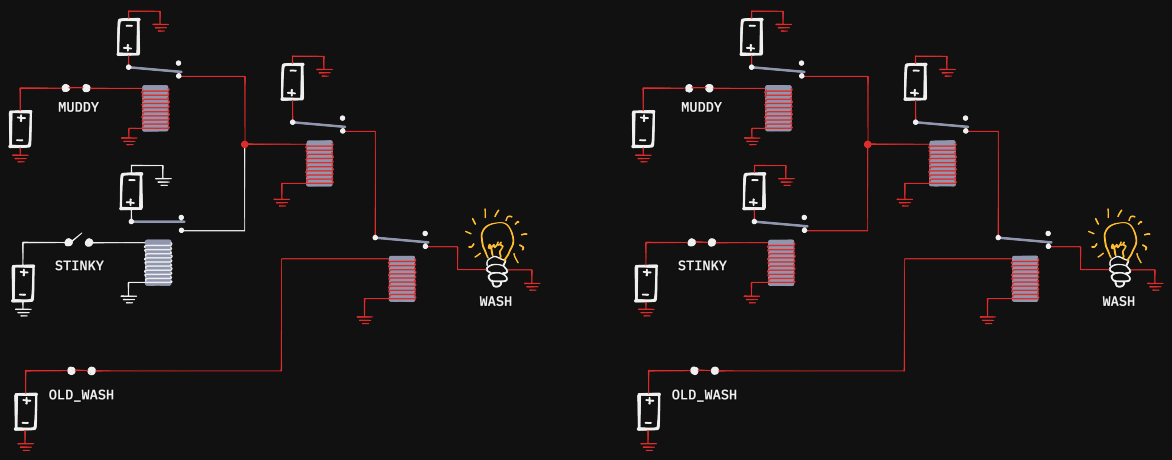


Diagram 4.5. The full dog washer circuit built with relays.

The animation does not show every possible combination of switches, only a handful. But in a nutshell, if `MUDDY` or `STINKY` is on, and `OLD_WASH` is also on, the bulb turns on.

Notice the OR that checks `MUDDY OR STINKY`, it's bulb is gone. Its output is now a wire, feeding straight into the next relay's coil. When `MUDDY` or `STINKY` pulls its arm, that wire gets connected to `+`, current flows through the coil, and the next relay pulls. When neither pulls, the wire is connected to nothing at all, so no current flows and the relay lets go.

That works, but from now on, our gates' outputs will mostly feed other gates' inputs, so the output needs to be a wire, not a bulb, and we want that wire to always be in one of exactly two states: connected to `+`, or connected to `-`. Never disconnected (unless we are using specific gate that we will cover later).

A `1` output is a wire being driven high. A `0` output is not "nothing"; it is a wire being driven low. Remember this information; it will come in handy later on.

Technically, with relays, we could get away with leaving a `0` output disconnected, like we have done here: an unpowered coil just stays off. We drive it low anyway. Real gates, built from transistors, need a `0` to actually be driven low for the logic to work correctly, and since we are using relays to replace them, we should still follow that rule.

All of our relay gates can be simply adapted to do this. As you can see in the after example below, even when the relay is not pulling the arm, even when the output is 0, it is still touching the negative end of that battery, so it is being driven to `-`.

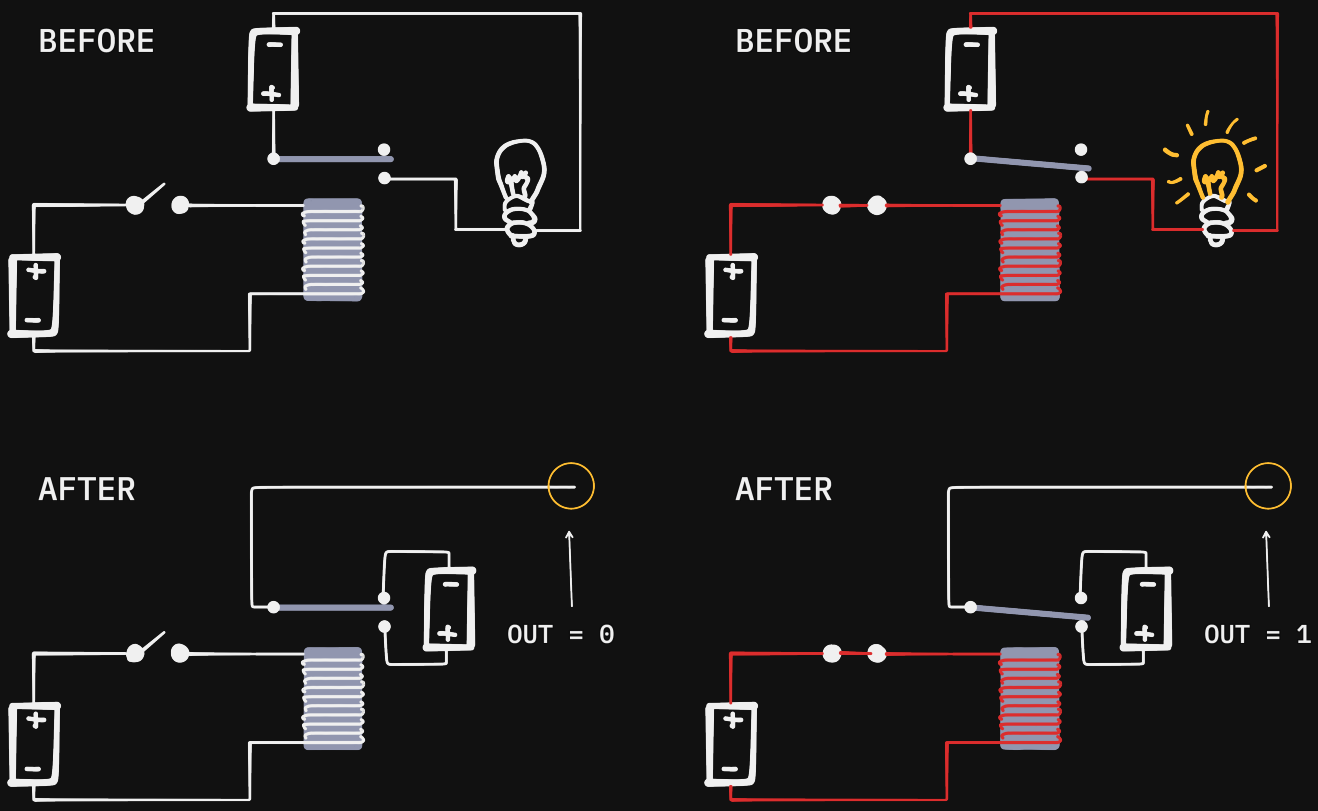


Diagram 4.6. Driving an output wire.

In this diagram, red wire means current is actively flowing. That's why `OUT = 1` is still white. Later when we start drawing logic gates, and more high-level diagrams, red wire will just mean high, or 1.

Okay, back to the dog washer. Let's introduce one last input, or "sensor": `RAIN_SOON`, whether it is predicted to rain soon. The rules of the circuit change once again:

```
((MUDDY OR STINKY) AND OLD_WASH) AND NOT RAIN_SOON
```

The parentheses indicate the groupings of logic, similar to math. So in plain English: if the dog is muddy or stinky and it's been more than 5 days since the dog's last wash and it's not going to rain soon, then wash the dog.

Let's focus on this NOT for a second. NOT just flips its input: if the input is on, the output is off, and if the input is off, the output is on.

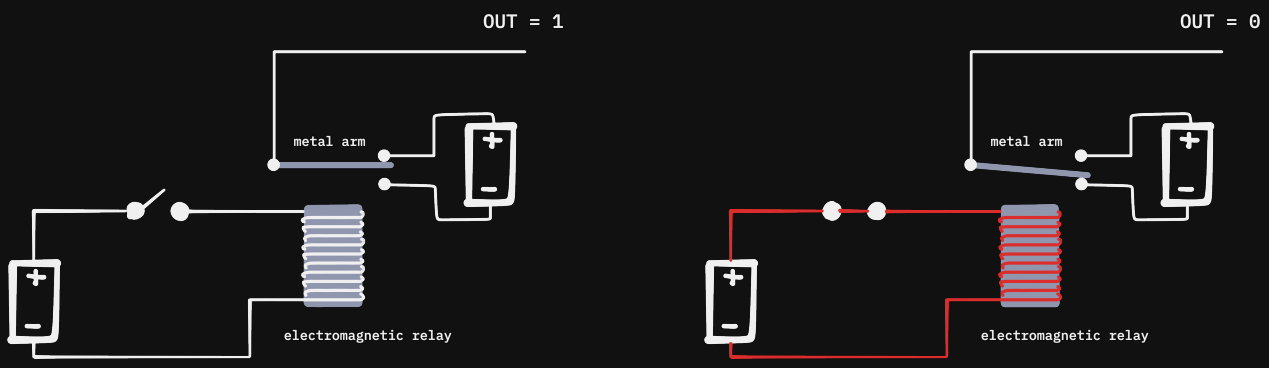


Diagram 4.7. A NOT gate.

Now before we look at the completed circuit, let's learn some basic logic gate symbols. An AND gate is drawn like this:

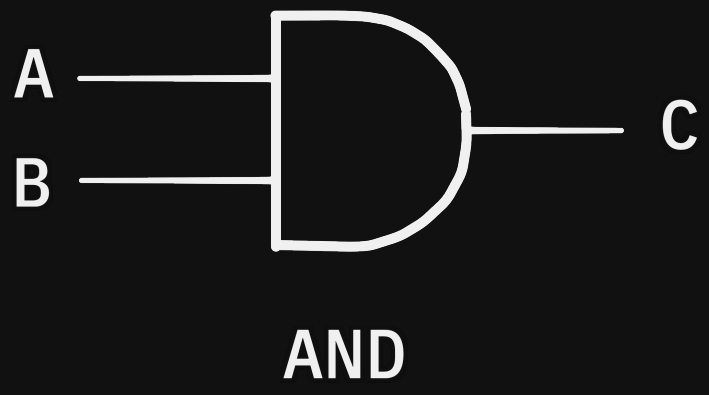


Diagram 4.8. An AND gate.

This symbol represents the AND circuit we made previously, except instead of turning a bulb on and off, it drives an output wire. An OR gate is drawn like this:

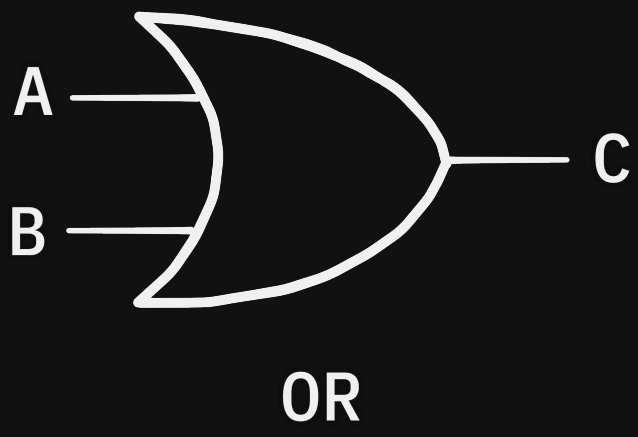


Diagram 4.9. An OR gate.

This symbol represents the OR circuit we made previously. Whenever I use these symbols moving forward, they can almost directly translate to the circuits with the relays I showed you previously, just instead of turning on a light bulb, they drive a wire. Here are three more useful gate symbols:

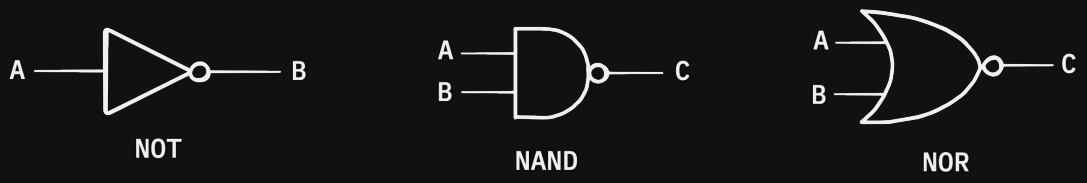
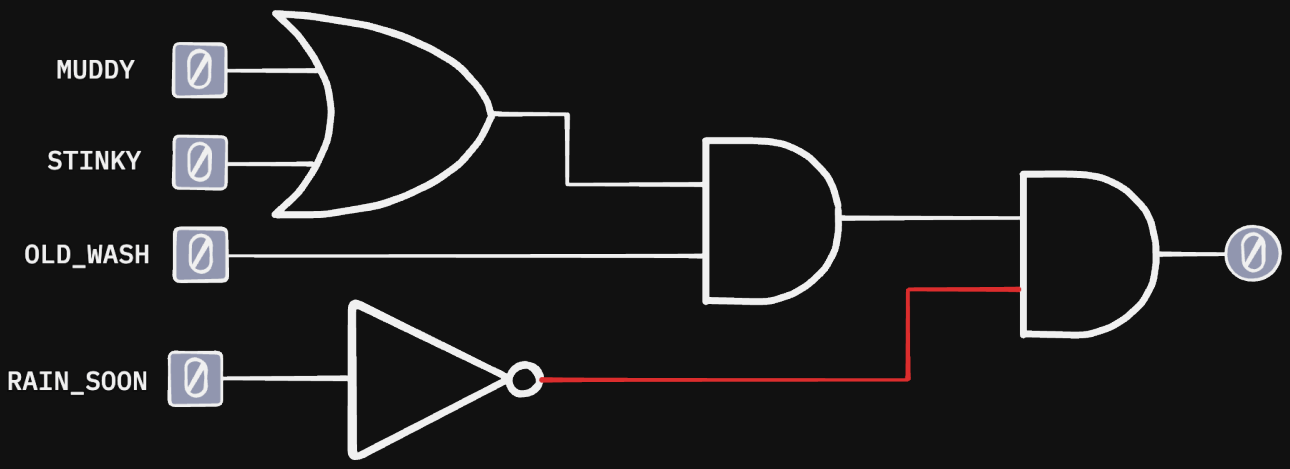


Diagram 4.10. NOT, NAND, NOR gates.

NAND is AND with the output flipped. NOR is OR with the output flipped. That little circle at the end of a gate means "flip the output." With our knowledge about logic gates, let's create the "should-I-wash-my-dog 5000" machine!



The full animation at bytecolony.computer/writing/the-cpu/#f=dog-washer-v2:1.

Diagram 4.11. The final dog washer circuit.

Again, this animation doesn't cover all possible states.

Keep in mind these electromagnetic relays we used in the examples are quite big and slow. Relays aren't the only solution. They are simply an early solution, and an easy one to understand, and many real computers like the Harvard Mark I actually used these types of relays.

In modern computers, similar behavior is achieved by using transistors. A transistor is kind of like a relay, but with no mechanical moving parts. A voltage on one pin decides whether current can pass between the other two. This makes transistors much faster and smaller.

This is how circuits make yes/no decisions. The circuit doesn't "understand" what `MUDDY` means. It just makes simple logical decisions.

A wire is just a wire. We gave these wires meaning. We decided that one wire means `STINKY`, another wire means `MUDDY`, and another means `RAIN_SOON`.

I don't know about you, but addition seems like a pretty logical next step after these logic gates. But not so fast.

To make a CPU, we need to give wires a different kind of meaning: numbers. Before we can build a circuit that adds, we need a way to represent numbers using only on and off. That is what the next section is about.

1.4. Counting With Wires

First, let's define some terms. A wire driven low is `0`, and a wire driven high is `1`. Let's call one wire one bit. A bit can either be `0` or `1`. These are labels that represent the state of a wire.

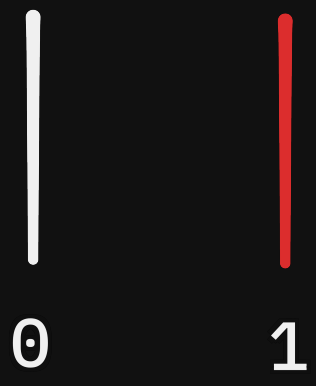
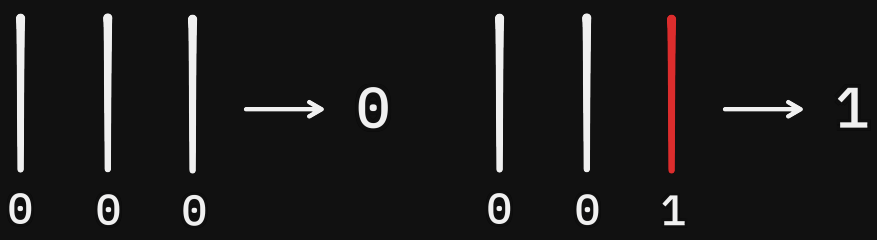


Diagram 5.1. One wire can represent two states: `0` or `1`.

If we want to represent numbers using wires, we are going to need more than one wire, because one wire can only represent up to two numbers, since it only has two possible states: `0` or `1`.

But two wires have `2^2`, or four states, and three wires have `2^3`, or eight states. That would allow us to represent more numbers. Here are all the possible states we have with 3 wires:



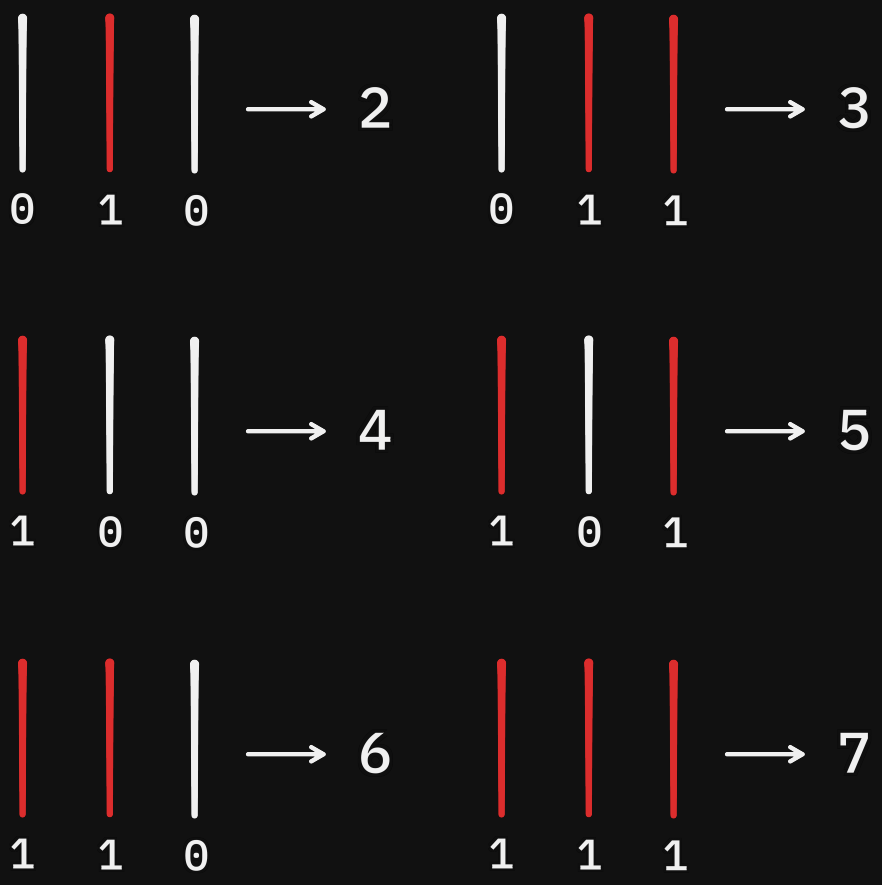


Diagram 5.2. States with 3 wires.

We can represent 8 numbers just like this. Keep going to 8 wires and there are `2^8`, or 256, possible patterns. A group of 8 bits is called a byte. If we use those patterns to represent non-negative numbers, one byte can represent 0 through 255.

But why does `010` mean 2? Why does `101` mean 5? Is it just randomly assigned? Not exactly. To understand this, let's take a quick detour to decimal, a.k.a. base ten.

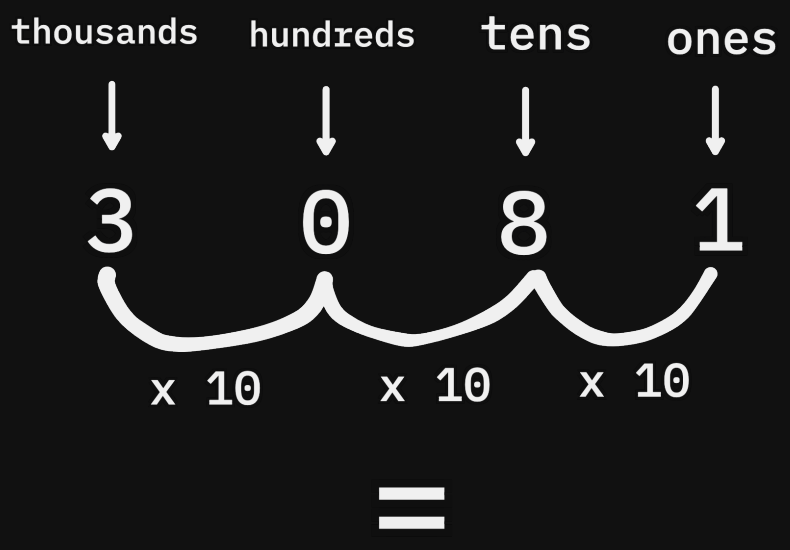


Diagram 5.3. The decimal system.

In our decimal counting system, each place-value is a power of 10. That is because we have ten digits: 0-9. This exact same place-value logic can apply to the binary system too. We have two digits, 0 and 1, so each place is a power of 2.

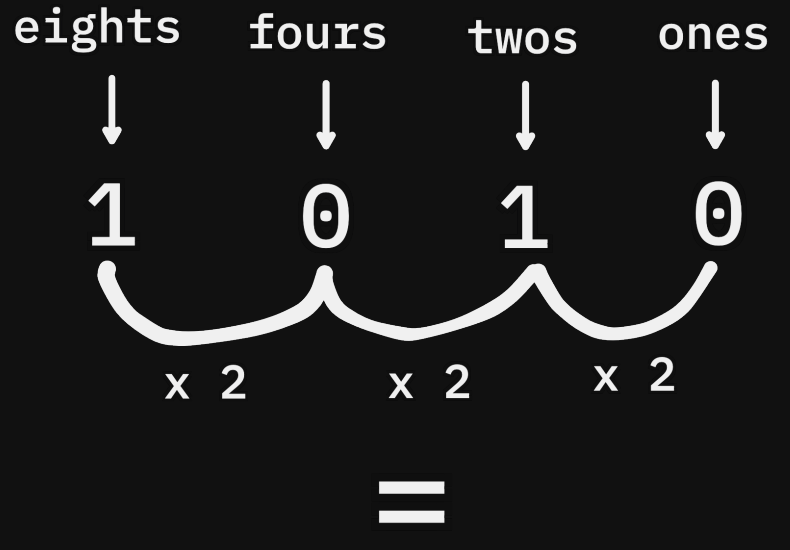


Diagram 5.4. The binary system.

Binary is, at the end of the day, the same place-value idea as decimal, just with two digits instead of ten. A few examples:

- `101` means 5
- `1101` means 13
- `101010` means 42
- `1100011` means 99

You don't need to do these problems in your head, but I hope the idea of how binary works makes sense. Let's walk through `1101` together.

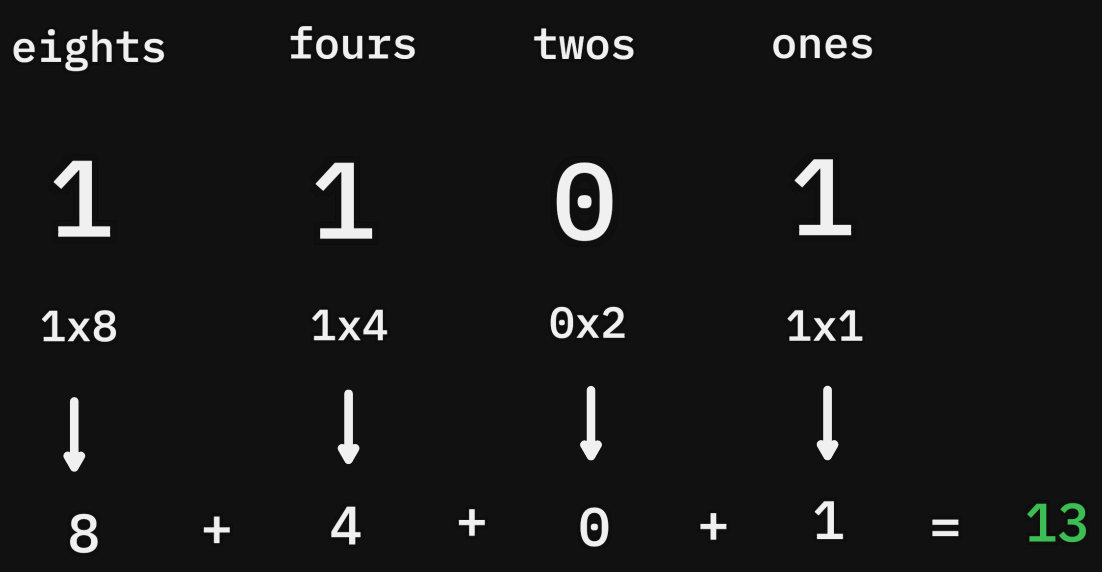


Diagram 5.5. An example in binary.

So now that we can represent numbers with wires, how can we add numbers together? That is what the next section is all about.

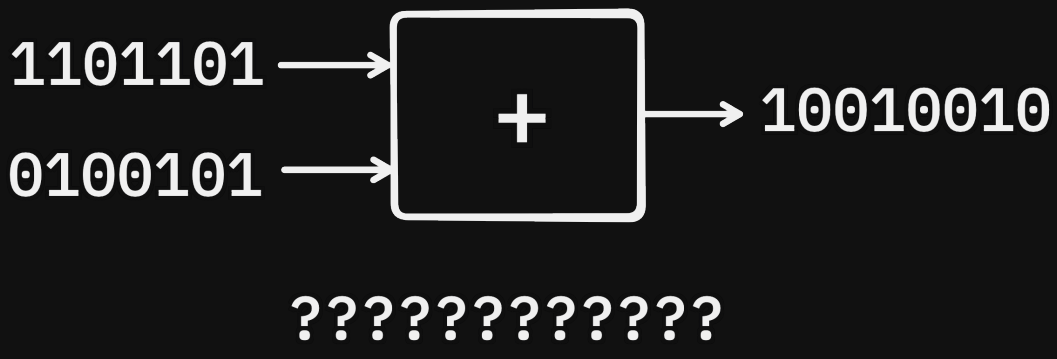


Diagram 5.6. Math?

1.5. Math?

Let's start with a brief reminder of how we algorithmically add two decimal numbers.

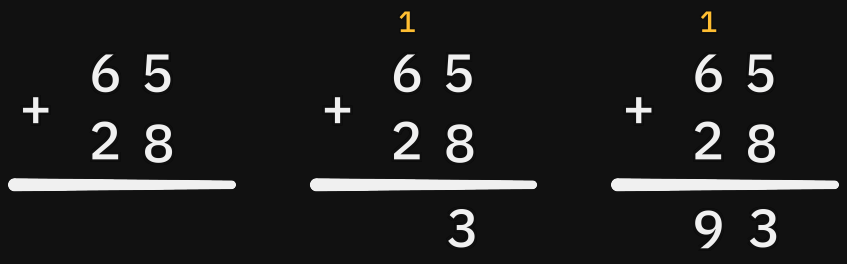


Diagram 6.1. Standard decimal addition.

We start at the rightmost column, do `5 + 8`, get 13, and carry the 1. So we write 3 as the sum, and 1 as the carry. We then move left and repeat over and over remembering to add any carry-in values. Binary addition works the same way.

Diagram 6.2. Binary addition.

`1 + 1` gives `10`, which is binary for 2.

So the sum bit for that column is `0`, and the carry is `1`. `1 + 1 + 1` gives `11`, which is binary for 3. So the sum bit is `1`, and the carry is `1`, and so on.

How do we build a circuit using logic gates that performs this standard addition algorithm? Well, let's start with the rightmost column. If we think about it, all the possible states are:

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

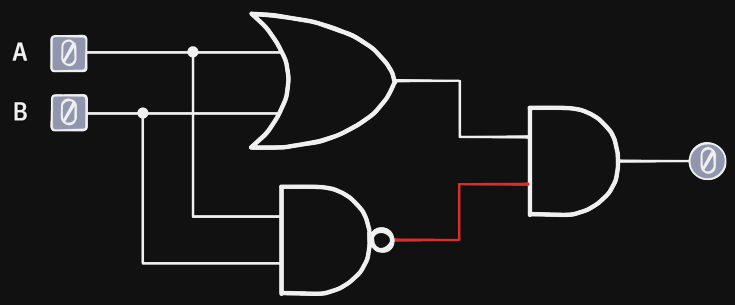
So just `0 + 0`, `1 + 0`, `1 + 1`, or `0 + 1`. If we can make a tiny circuit that takes two inputs, and produces two outputs that match these combinations, we have added the first column.

This is called a half adder. A half adder adds two bits, but it does not handle a carry-in value. That is the job of a full adder.

Let's first build this half adder. Let's start by computing the sum, not the carry-out. This is what we want our circuit to do:

A	B	Sum
0	0	0
0	1	1
1	0	1
1	1	0

The sum is `1` only when exactly one input is `1`. This is called XOR, short for exclusive OR. If we combine an OR gate and a NAND gate, and AND them together, we get XOR:



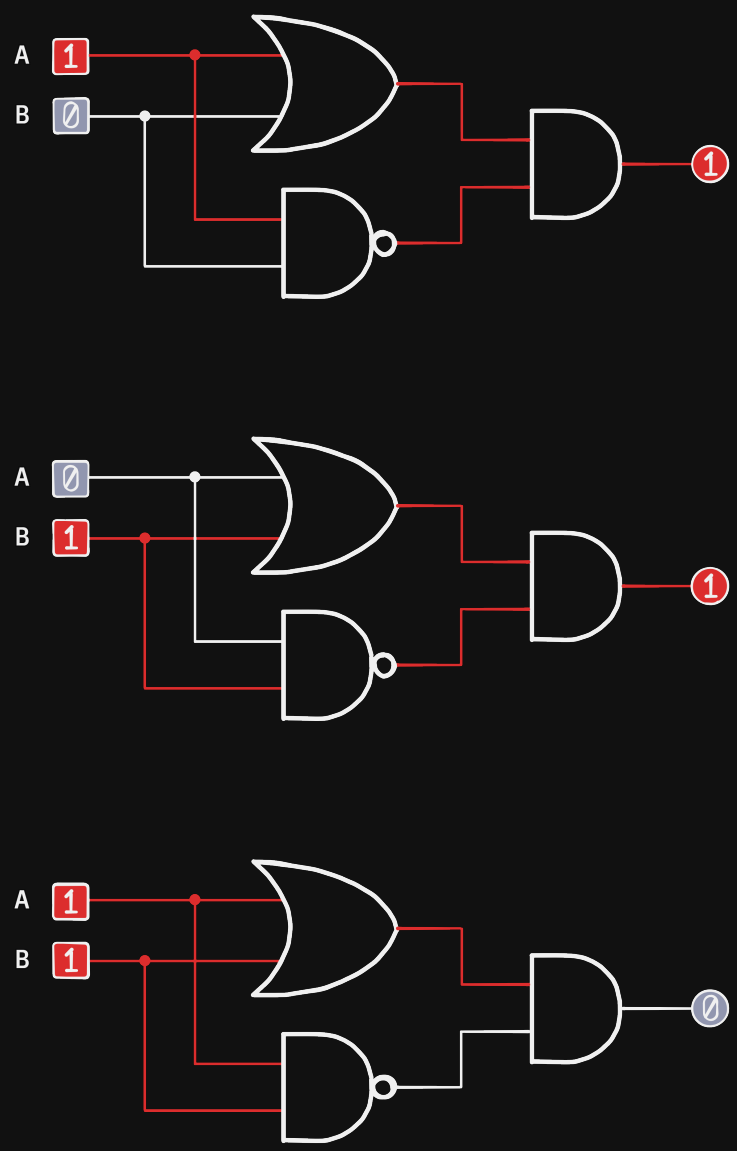


Diagram 6.3. Half adder sum / XOR.

OR checks that at least one input is on, and NAND makes sure the inputs aren't both on. Here is how an XOR gate looks:

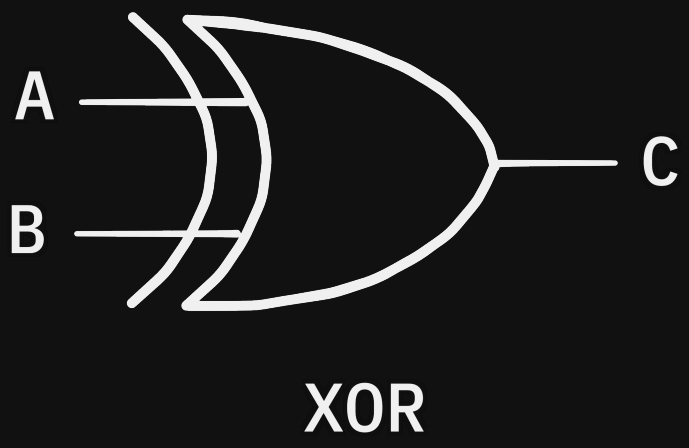


Diagram 6.4. An XOR gate.

Now let's do the carry value. The carry is simple! We only want to carry if we are doing `1 + 1`, so we use an AND gate to check if both inputs are on. Here is our half adder:

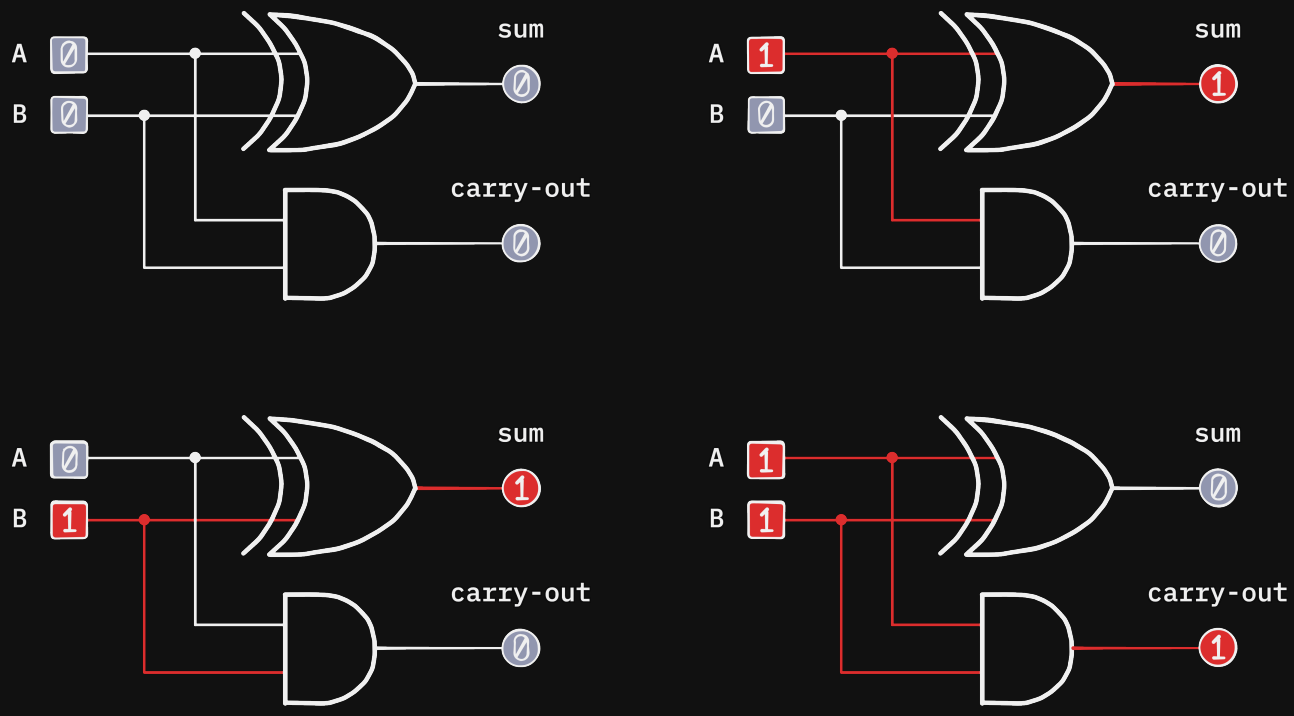


Diagram 6.5. A half adder.

As you can see, it works! `0 + 0 = 0`, `1 + 0 = 1`, `0 + 1 = 1`, and `1 + 1 = 10`.

Now let's package up our half adder into a little box. From now on, I will call these packaged-up circuits chips:

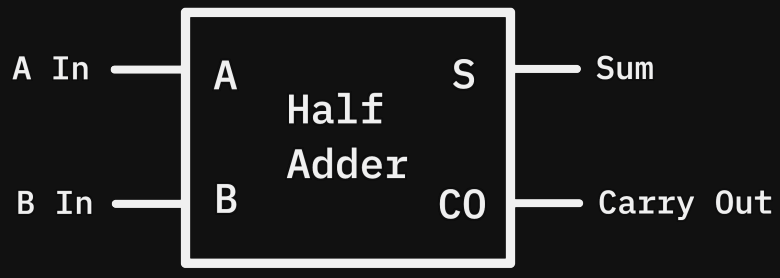


Diagram 6.6. A half adder chip.

Now that we have a half adder, we can add the rightmost column. That works because the rightmost column has no carry-in from a previous column. It only needs to add two bits. So if we have a number like this:

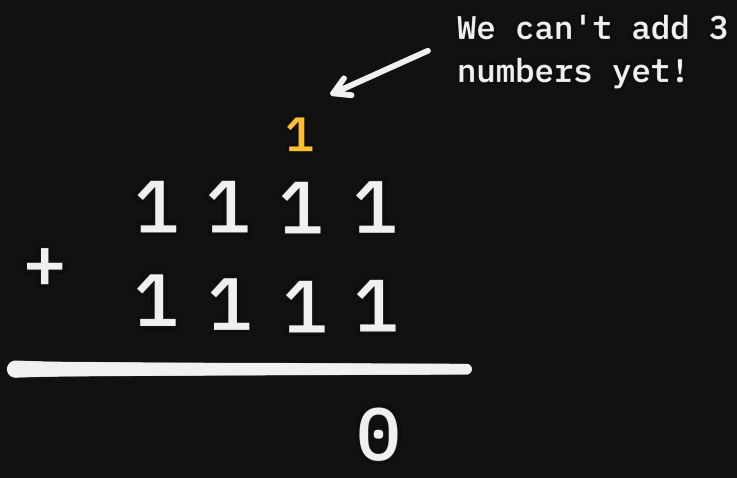
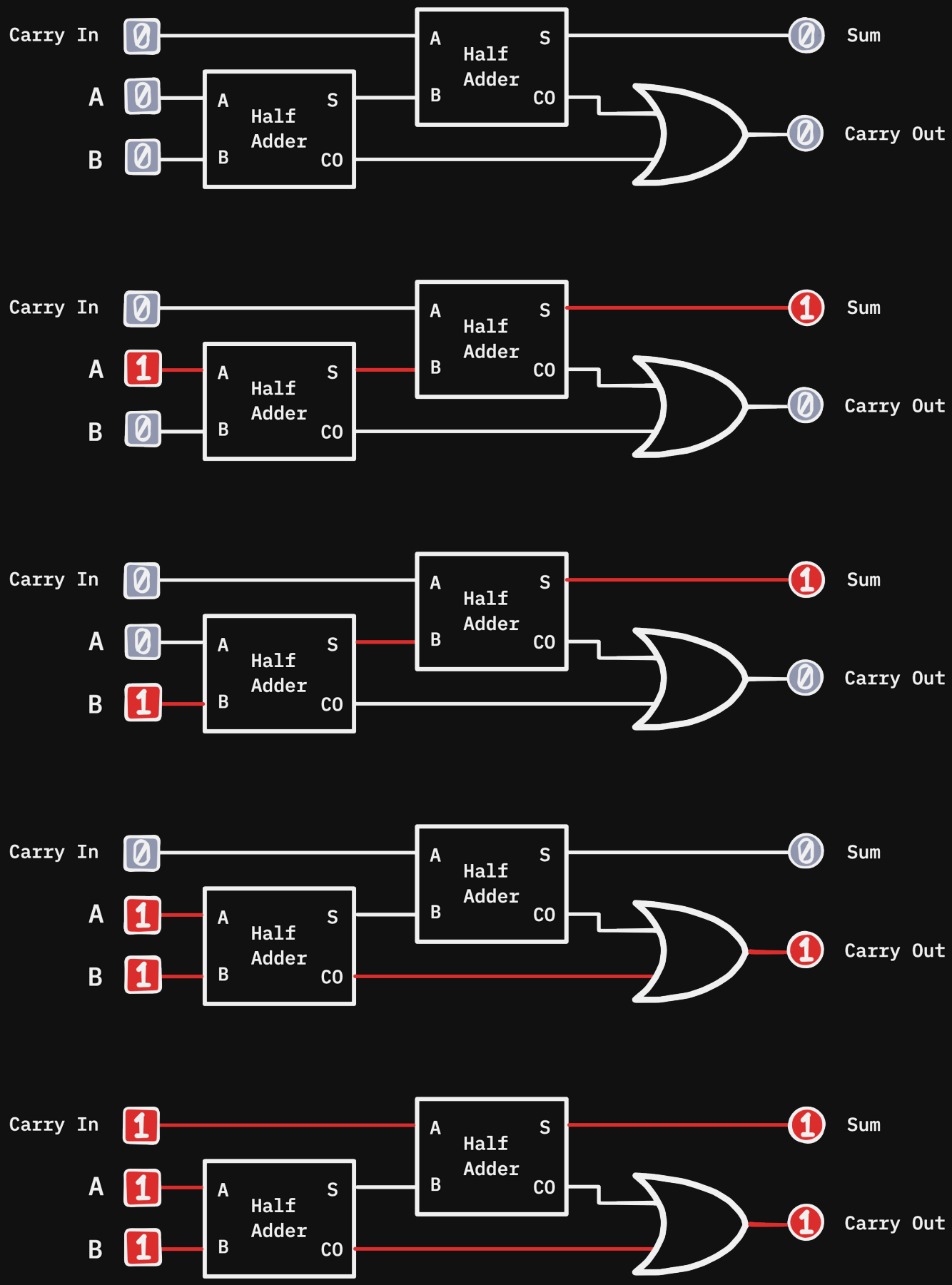


Diagram 6.7. The next column has to add two bits plus a carry-in.

The half adder can handle the first column: `1 + 1`. That gives us a sum bit of `0` and a carry-out of `1`.

But now the next column has three things to add: `1 + 1 + 1`. The two original bits, plus the carry from the previous column.

A half adder cannot do that. It only accepts two inputs. To continue adding up the other columns, we need a circuit that can take in three inputs: `A`, `B`, and `carry-in`. To add three bits, we use two half adders and an OR gate:


Carry In
A
B
Half Adder
A
B
S
CO
Sum
Carry Out

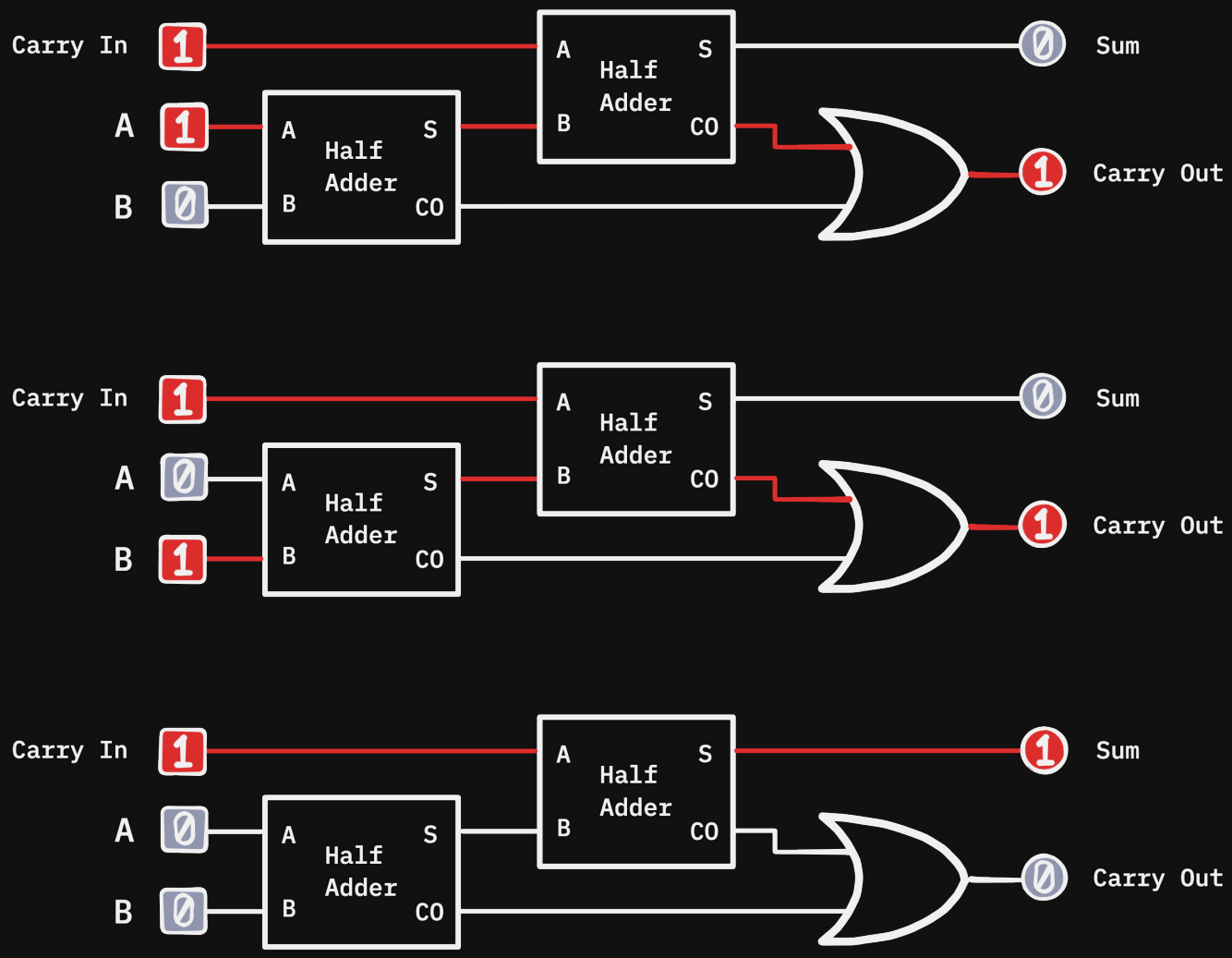


Diagram 6.8. A full adder.

This might look confusing at first. What if both half adders output a carry-out at the same time? That actually never happens. If the first half adder outputs a carry, its sum bit is 0, and that 0 is one of the second half adder's inputs, so the second one can't carry. So both can never output carries at the same time. Take a moment to think about this if you are confused. So we can confidently OR the two carry outputs together. If either one is `1`, the full adder's carry-out is `1`.

Let's again package this up into a chip:

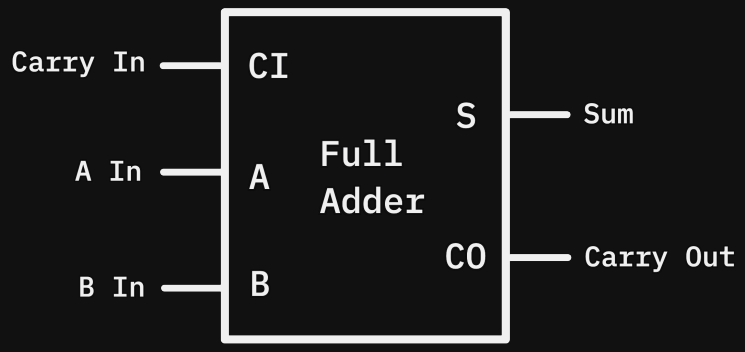


Diagram 6.9. A full adder chip.

We have made a full adder! Now we can chain full adders together to add two 8-bit numbers. Since 8 bits make one byte, this is an adder that can add two one-byte numbers: anything from 0 to 255.

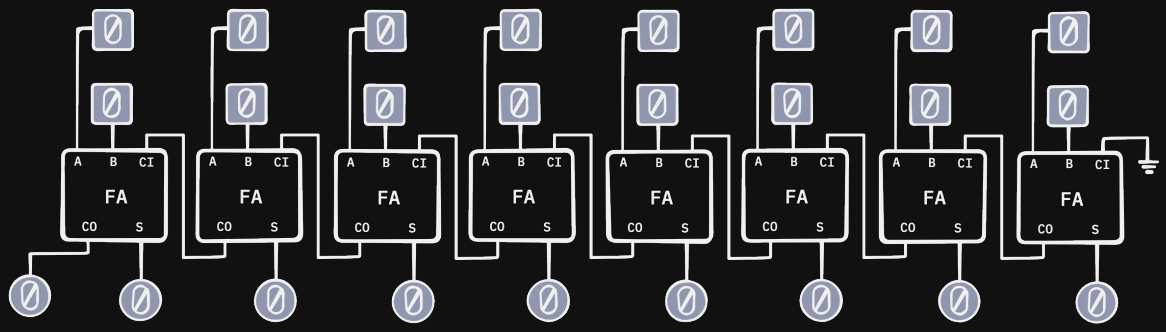


Diagram 6.10. An 8-bit adder.

Each full adder handles one column. The carry-out from one column becomes the carry-in for the next column. That is it! That is all addition is!

Keep in mind, carry-in for the first adder is set to ground, a.k.a. 0. In the chip version below I bring that first carry-in out as a `CI` pin instead, so the chip can be chained too. For now we just feed it 0. Also, notice how we have 9 outputs, not 8. That is because two 8-bit values can add up to a number too big to fit in eight bits. It's like how adding two 2-digit numbers could result in a three-digit number for us. Like `50 + 50 = 100`.

Let's package this up into a chip once again:

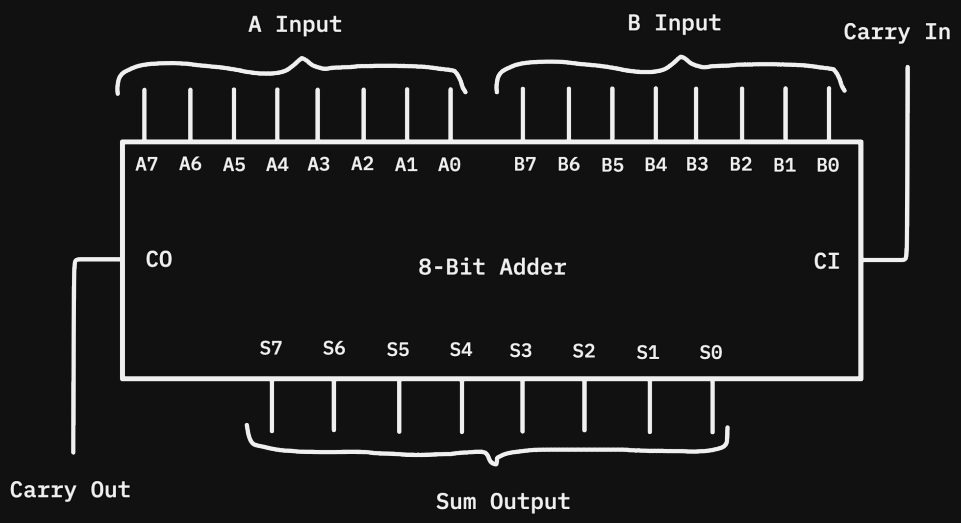
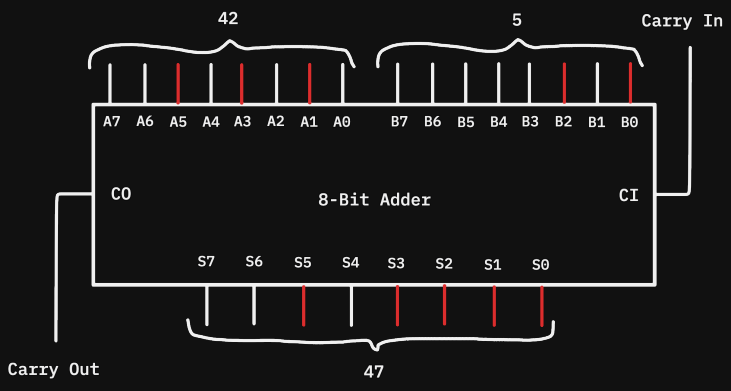
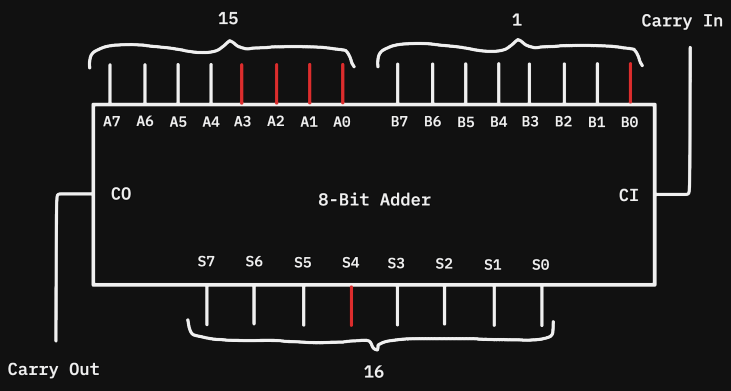


Diagram 6.11. An 8-bit adder chip.

Now we have the carry-out and carry-in as separate inputs and outputs and the whole adder nicely organized into this chip. Let's have a look at some example problems:





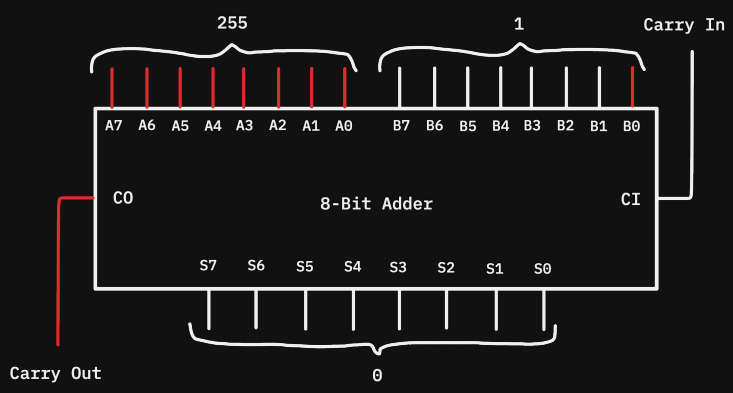


Diagram 6.12. Some examples on the adder.

As you can see in the third example, adding 1 to 255 turns every sum bit to `0` and turns the carry-out on.

This doesn't mean the adder got the wrong answer. In fact, `255 + 1` is `1 00000000` in binary: eight `0` output bits, plus one extra carry-out bit on the left. If we only look at the one-byte output, the result looks like `00000000`, or 0. If we also look at the carry-out, we can see that the real answer was 256.

That is called an overflow: the result was too large to fit inside one byte, so the extra information spilled out into the carry-out bit.

With a bit of extra circuitry around the adder, we can also make little status wires, called flags. For example, if the answer is `00000000`, a `ZERO` flag can turn on. If addition spills past one byte, a `CARRY` flag can turn on. So `11111111 + 00000001` gives `00000000` with carry-out `1`.

I don't want to go deep into flags yet. Just remember that the adder can output little yes/no facts about the sum. That matters later for instructions like "jump if zero." But let's not get ahead of ourselves.

We have just built addition! But we also need something else: storage. For example, let's say we want to build a circuit that counts by ones, like 1, 2, 3, 4, ...

The obvious idea is to feed the output of the adder back into one of its inputs. Start with `00000000`, add `00000001`, get `00000001`. Feed that back in, add `00000001` again, get `00000010`. Then `00000011`, then `00000100`, and so on.

That seems correct at first glance. But there is a big problem. An adder just looks at its current inputs and computes an output.

So if we wire the output straight back into the input, there is no stable value anymore. The adder is basically being asked to make a number equal to itself plus one:

```
input = input + 1
```

That can never settle. As soon as the output changes, the input changes too, which means the output has to change again, which means the input changes again. With relays, you might physically see this mess play out. With transistors, it would happen almost instantly.

There is no boundary between the old value and the new value. There is no clean "step 1, step 2, step 3."

So this is not enough. We need a circuit that can hold a value, then update it only when we tell it to. That is the next problem: memory.

Part 2: Machinery

2.1. Storing a Bit

In the last section, wiring the output of a circuit back into its input led to chaos. But with a smarter configuration, we can use this idea to store a bit.

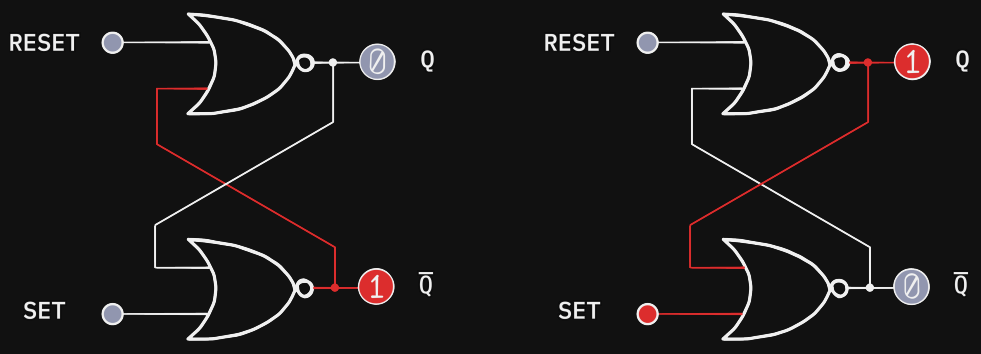
The simplest circuit that stores a single bit is called an SR latch. SR stands for set-reset. The value `Q` is the output we really care about. If it is `1`, that means the latch is storing a `1`; if it is `0`, the latch is storing a `0`.

The diagram below also shows a second output written as a Q with a bar over it. That is just how engineers write `NOT Q`, pronounced "not Q". It always holds the opposite of `Q`. I will write it as `NOT Q` in the text.

The two inputs are `SET` and `RESET`, drawn as little buttons in the diagram: gray means not pressed, red means pressed. Pressing `SET` forces `Q` to `1` and pressing `RESET` forces `Q` to `0`. For this circuit to be used properly, `SET` and `RESET` should never be on at the same time.

The cool part is, if both `SET` and `RESET` are `0`, then `Q` is whatever we last did to it! The output loops back into the circuit, so the current state keeps reinforcing itself. This is the basic concept behind memory.

This diagram should help this make sense:



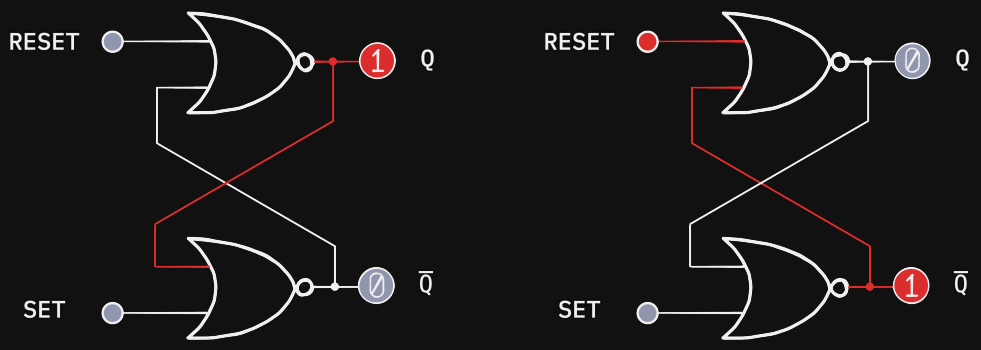


Diagram 7.1. An SR latch.

One quick reminder before we trace through this: a NOR gate only turns on when both of its inputs are off. Now let's trace through it.

If `SET` is on, the bottom NOR gate has to output `0`, because one of its inputs is on. That makes `NOT Q` equal to `0`.

Now the top NOR gate sees two `0` inputs: `RESET` is `0`, and `NOT Q` is `0`. So the top NOR gate outputs `1`, making `Q` equal to `1`.

Then even if we turn `SET` back off, the latch stays in that state. `Q` is still `1`, which keeps forcing `NOT Q` to `0`, and `NOT Q` being `0` allows `Q` to stay `1`.

`RESET` works the other way. If `RESET` is on, it forces `Q` to `0`, which allows `NOT Q` to become `1`. Then even after `RESET` turns off, `NOT Q` keeps forcing `Q` to stay `0`.

The circuit has state. Its output depends not only on the current input, but on what happened before.

Now that we have the core mechanism, let's refine the interface. Right now `SET` and `RESET` are super clunky. While they demonstrate the mechanism, what we would really like to have is two inputs.

- `Data` (`D`)
- `Enable` (`E`)

When the enable wire turns on, `Data` gets stored in `Q`. Or in other words, when we turn the enable wire on, `Q` mirrors `D`. Then when we turn `E` off, `Q` stays stable with whatever `D` was last.

This type of latch is called a D latch, D meaning data. It can be made using the SR latch and a few extra logic gates.

It basically checks: if `D` is true and `E` is true, `SET` is true, and if `D` is false and `E` is true, `RESET` is true. That's it, so let's not worry about the exact implementation.

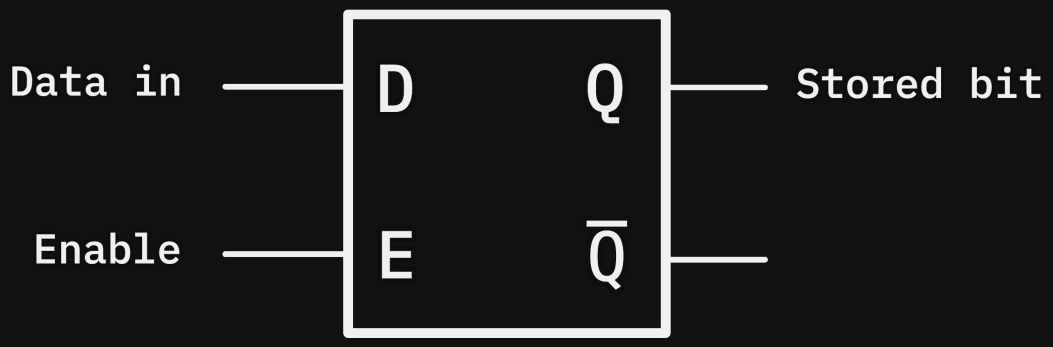
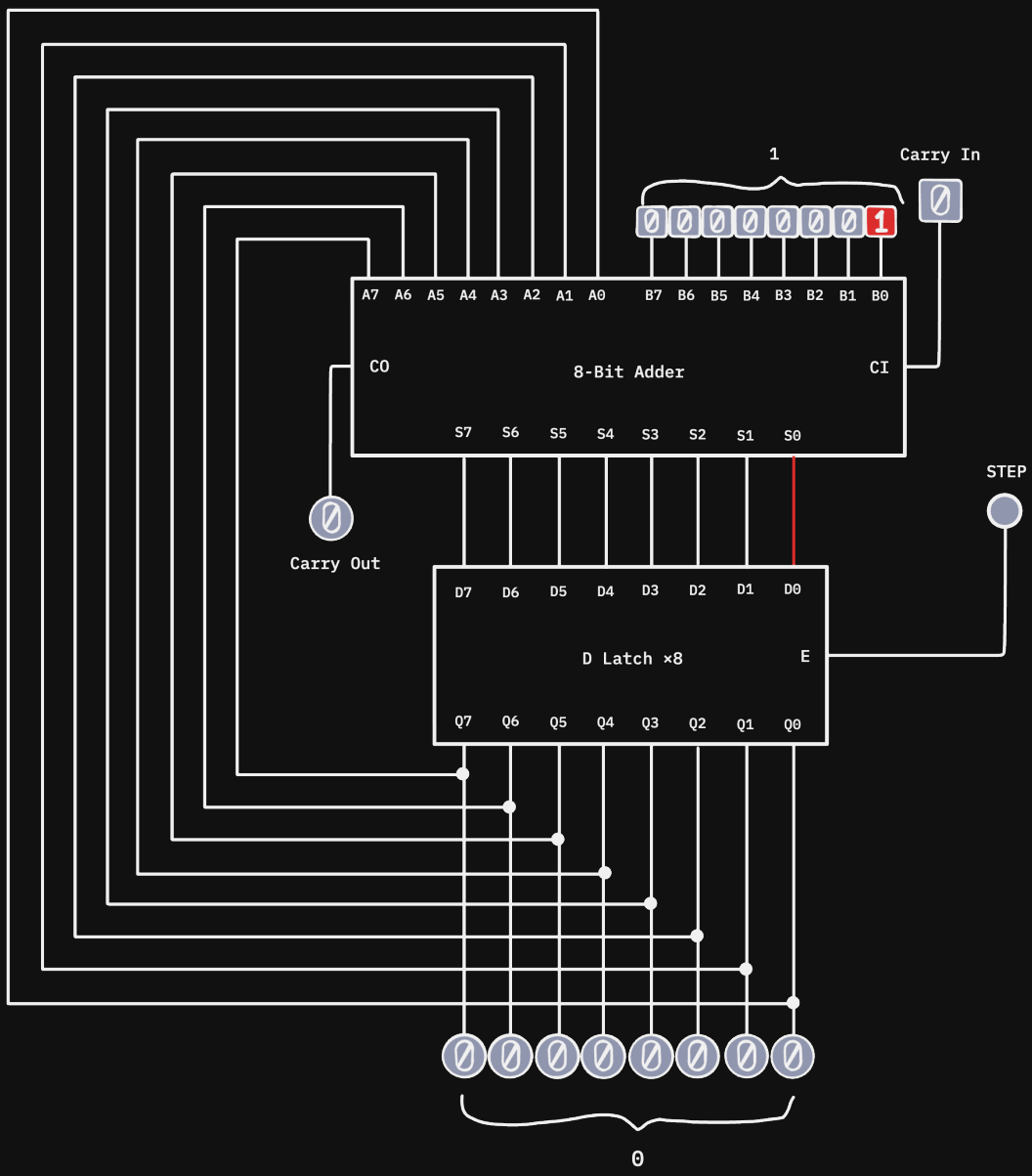


Diagram 7.2. D latch.

But we have a problem. Let's say we now try to use 8 of these D latches to hold the result from our adder, which would then feed back into the input for our accumulator (that counting circuit from last section, I'll call it an accumulator from now on). It still wouldn't work.

Here is the problem: say we have the enable wire hooked up to a button. When that button is pressed down, the enable wire is on, so `Q=D` for that time. But if `Q` feeds back into the adder, and the result of the adder `D` changes quickly enough, `Q` can change again, jumping unpredictably based on how long we hold that button for.

If we want the accumulator to work correctly, we need the enable wire to turn on for an instant and then turn back off. That is just hard to do.



The full animation at bytecolony.computer/writing/the-cpu/#f=d-latch-accumulator:1.

Diagram 7.3. D latch accumulator.

As you can see in this diagram, even pressing the button quickly jumps the result up by 5. With real transistors, even if you try to physically tap the button, it could count up by millions, overflowing these 8 bits thousands of times. How long you hold the button decides the answer. It doesn't count in ones.

But what if we had a storage circuit that only copied `D` into `Q` at the exact instant `E` turns on?

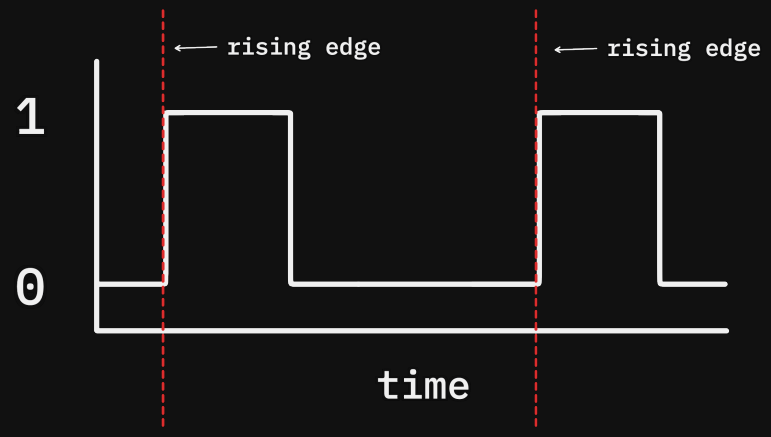


Diagram 7.4. The rising edge of a signal.

This graph shows the state of a wire. When the line is at the top, it is on. When it is at the bottom, it is off.

When a switch is flicked or a button is pressed, a transition happens. That transition is called an "edge", and when the wire turns from off to on, it is a rising edge.

Now what if we only set `Q` to `D` on that transition, at the rising edge? The edge is an instant of time, not a duration.

The circuit that does this is called a D-type edge-triggered flip-flop. This might sound like a mouthful, but D-type just means it takes in a data input, edge-triggered means it triggers on the edge of a signal, and flip-flop means it is a storage circuit similar to a latch, but usually edge-triggered.

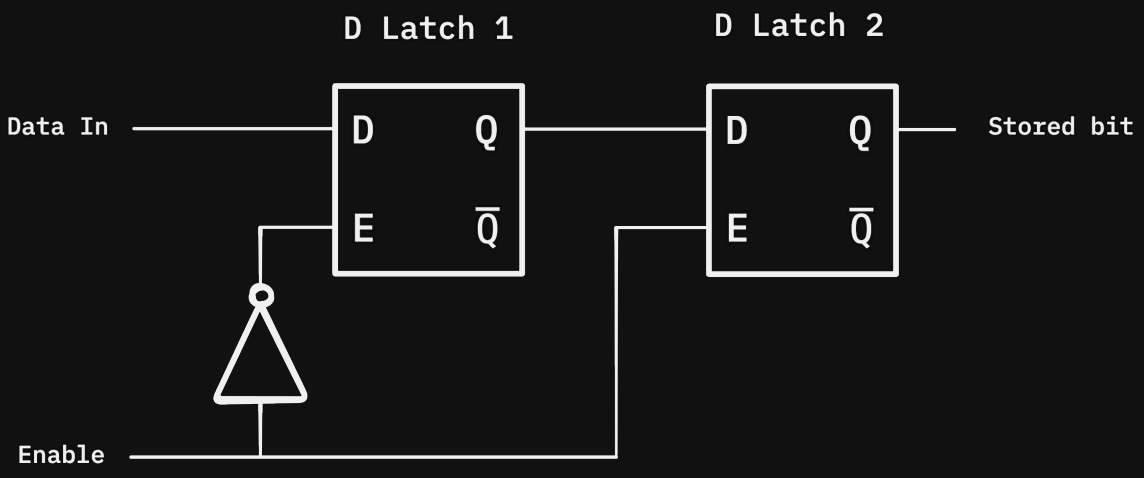


Diagram 7.5. A flip-flop.

How it works is, when the enable wire is off, the first latch mirrors `D`. That is because the NOT gate flips the enable signal, so the first latch sees it as on.

Then when enable turns on, the second latch stores the output of the first one. And because enable is now on, the first latch is locked, so it can't change!

So if `D` changes while enable is off, we are all good because the second latch is locked. But if `D` changes while enable is on, we are fine because the first latch is locked. So `Q` can only change at the exact moment `E` turns on, that is our rising edge!

Here is one storage cell, which is just the flip-flop we showed above:

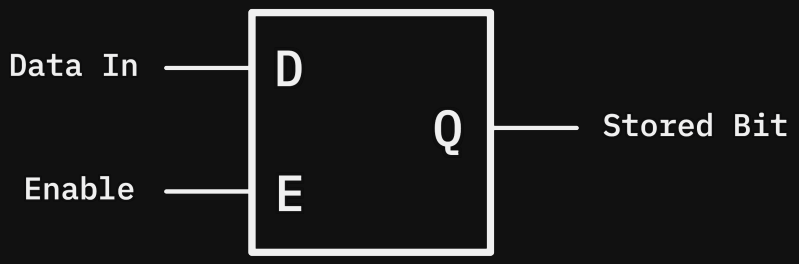


Diagram 7.6. A one-bit storage cell.

If we connect 8 of them side by side, we get one byte of storage:

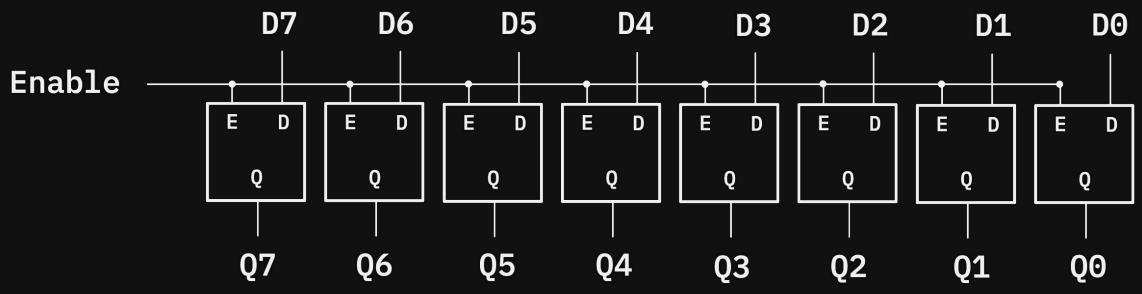


Diagram 7.7. Eight storage cells.

And we can put all that into a chip called an 8-bit register:

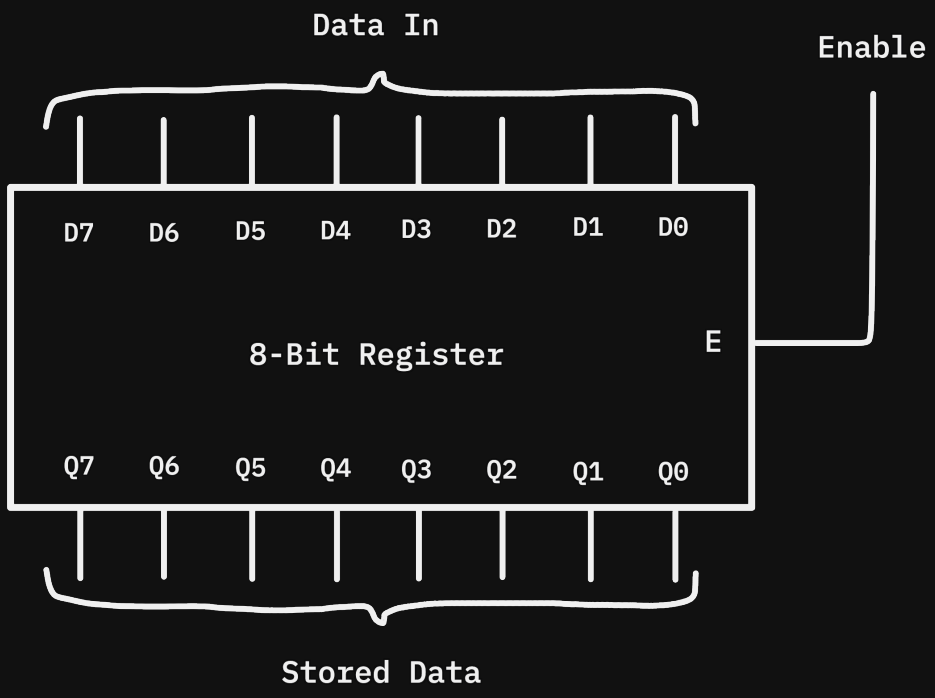
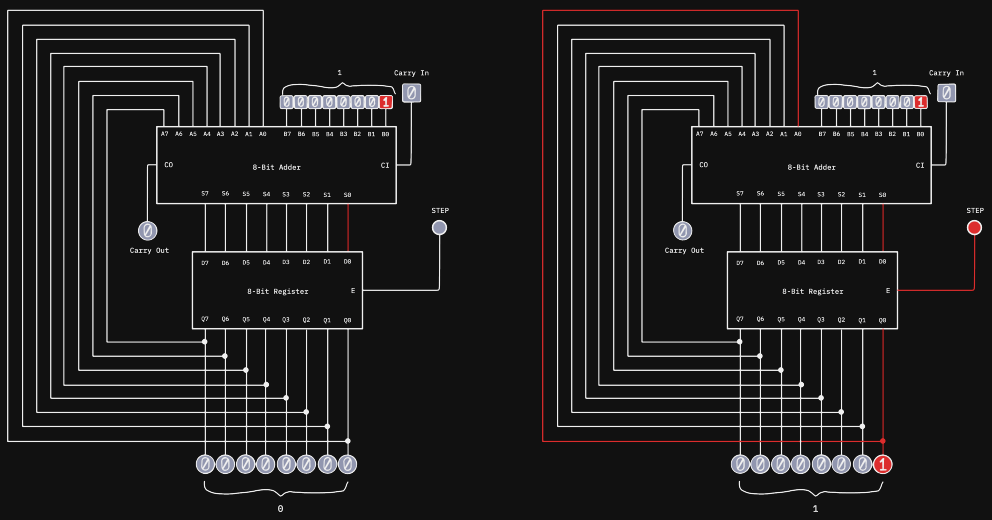


Diagram 7.8. An 8-bit register.

Now with this register, let's build a basic accumulator/adder circuit.



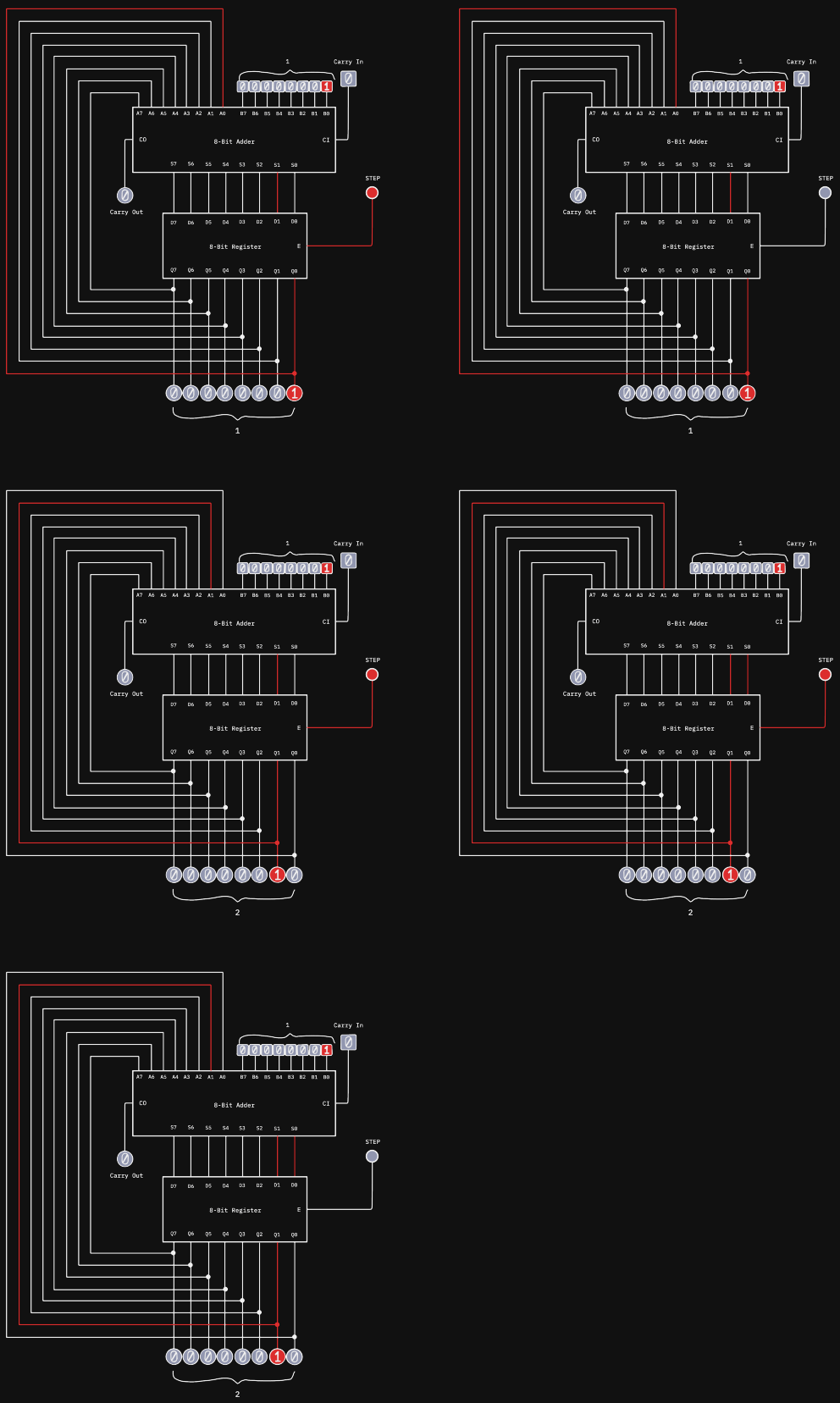


Diagram 7.9. Our full accumulator.

As you can see, the circuit kindly waits for us, and is incrementing by ones! How this works is, when the **STEP** button is pressed, the output from the adder gets saved into the register on the rising edge of that press. This then changes the input to the adder, which changes

its output, but the register holds its value because it only captures on the edge of the press. Holding `STEP` down does nothing special. So each press increments the register's value by, in this case, 1.

Now, a real computer would need to do these kinds of things millions or billions of times per second, and we don't have some human clicking a step button. What we have is a circuit that automatically goes on, off, on, off billions of times per second. This is called a clock. Each rising edge of the clock acts like one press of `STEP`.

Here is the basic concept of a clock:

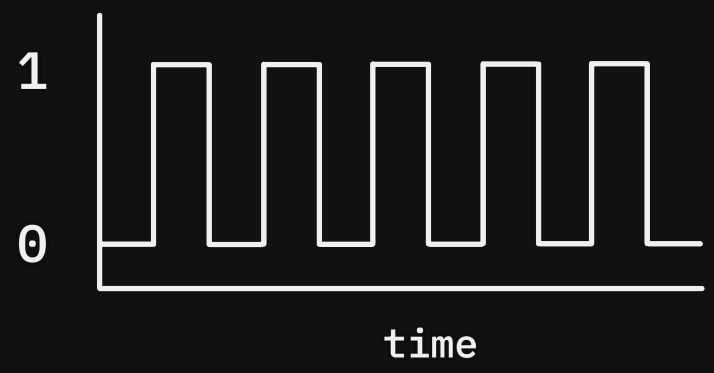


Diagram 7.10. A clock signal.

This repeating on-off behavior can be achieved in different ways. A rough toy example is feeding the output of a NOT gate back into its input, so the signal keeps trying to flip back and forth between on and off.

Real clocks are built in more sophisticated and reliable ways, often using crystals or other oscillator circuits. But we do not need to build the clock itself here. For now, we can treat it as a little chip that repeatedly produces the same on-off signal.

Just imagine the new accumulator with a clock signal instead of a `STEP` button. I am too lazy to draw it for you.

We now have some storage. A register that can hold a byte, and update exactly when we want. But registers on their own are not enough. We need to be able to move numbers between registers, the adders, and the main memory, which we will build later.

2.2. Public Transport?

One simple solution to move bytes around would be to give every component its own bundle of 8 wires to every other component, but that would become a mess very quickly.

A simpler solution is to have one single 8-bit data highway, where components can put data on and take data off. This collection of 8 wires is called a bus. I hope the title of the section now makes sense for any of you confused readers, sorry not sorry.

One more thing: moving forward, when I want to draw a collection of 8 wires, instead of drawing each wire, I will just draw a thick arrow that represents 8 wires.

To show the state of the wires, I can write a number in the arrow; the number 0 for example means the wires are all off, and the number 2 would mean the wires are `00000010` which is 2 in binary.

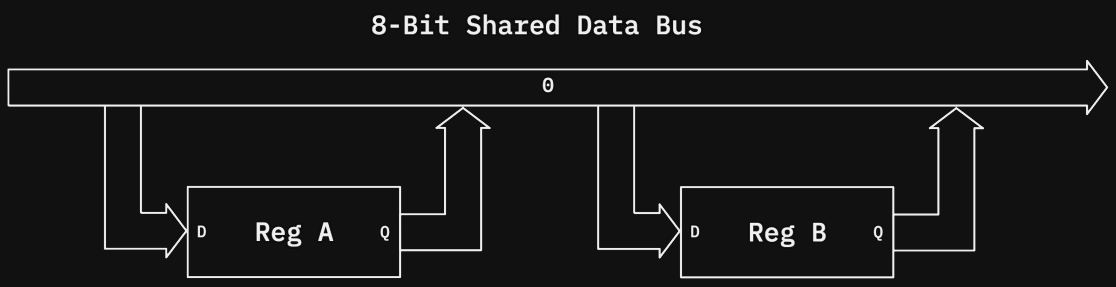


Diagram 8.1. Two registers sharing a bus.

But we have an issue: this diagram is technically not possible yet. Say register `A` is outputting `00000000` and register `B` is outputting `00000001`, both onto the same 8 wires. Look at the last wire. `A` is driving it low, so that wire is connected to `-`. `B` is driving the same wire high, so it is also connected to `+`. What happens?

Yep. A short-circuit.

We need a way to connect these registers to the bus, but also let them get out of the way when they are not supposed to actively drive a wire to `-` or `+`, like I discussed previously.

Just setting the output wires to `00000000` is not enough. On a shared bus `00000000` is not nothing. It is actively driving the bus to `-`. Of course, in our relay world technically we don't need to drive to `-` like I mentioned before, but in the real world, where transistors are used, we certainly do and that causes problems.

One clean way to solve this problem is by using something called a tri-state buffer. It has two inputs, `E` and `D`, which stand for enable and data.

We have seen `E` before, in the context of "enable writing", but now we are using `E` in the context of "enable outputting".

If `E` is on, the output will just be whatever `D` is, so either `0` or `1`. If `E` is off, no matter the value of `D`, the output will be `Z`.

`Z` means the buffer's output is disconnected from the bus. Engineers call this "high impedance", which is where the letter `Z` comes from. It is not driving the bus to `+` or `-`, so another component can safely drive the bus without a clash or short-circuit.

Or in other words, this buffer is not touching the wire. `0` is very different: the component is actively driving the wire to `-`. So we have three states:

```
1: driven high
0: driven low
Z: disconnected
```

This relay diagram of how a tri-state buffer works should make this concept crystal clear.

Also I have drawn everything the output wire is currently touching in yellow. Yellow is just there so you can follow the path with your eyes; it doesn't mean anything.

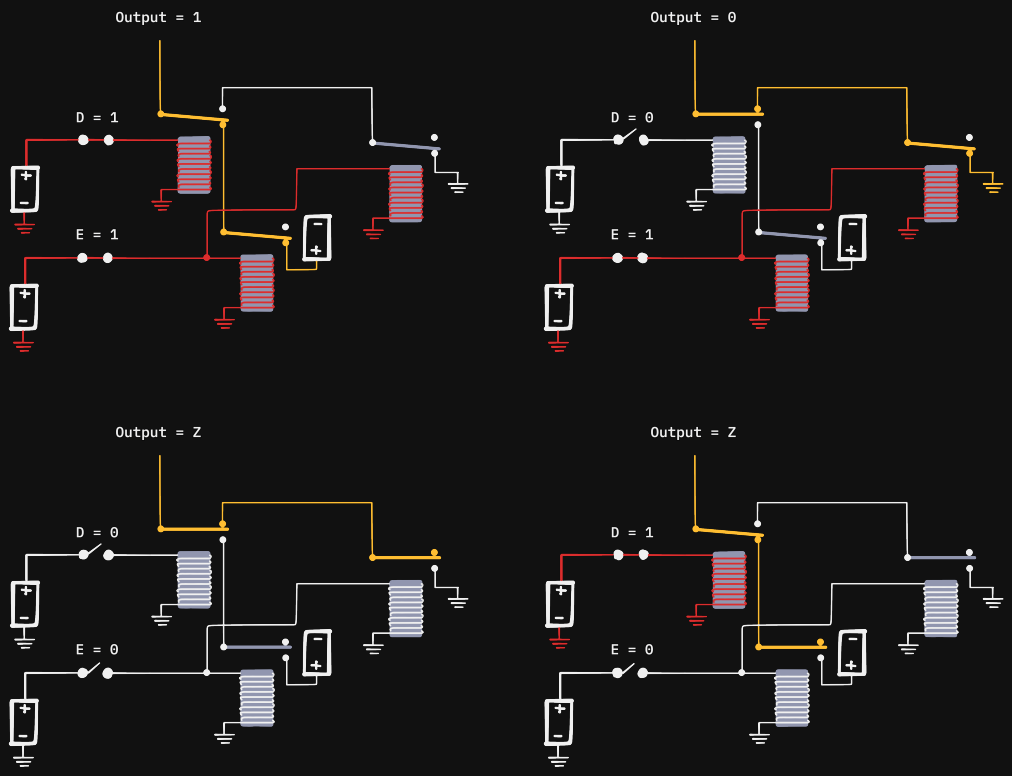


Diagram 8.2. A tri-state buffer built with relays.

This looks complicated, so let me break it down. First, ignore the two relays on the right and look only at the `D` relay at the top. Its arm is attached to the output wire, and it works just like the output driver from before. When `D` is `1`, the arm is pulled down onto the wire that leads toward the battery, `+`. When `D` is `0`, the arm goes up onto the wire that leads toward ground. Remember, ground is just the `-` side.

The important idea is that neither of those wires is directly connected to `+` or `-`. Each one has a relay between it. Both of those relays are controlled by `E`.

When `E` is `1`, both gaps close. The output is now connected to whichever side `D` picked, so it is driven to `1` or `0`.

When `E` is `0`, both relays touch a point connected to nothing. Both wires lead to a dead end. The output wire is touching nothing. That is `Z`. Here is the full table:

E	D	Output
1	1	1
1	0	0
0	0	Z
0	1	Z

This is the logic gate diagram for a tri-state buffer:

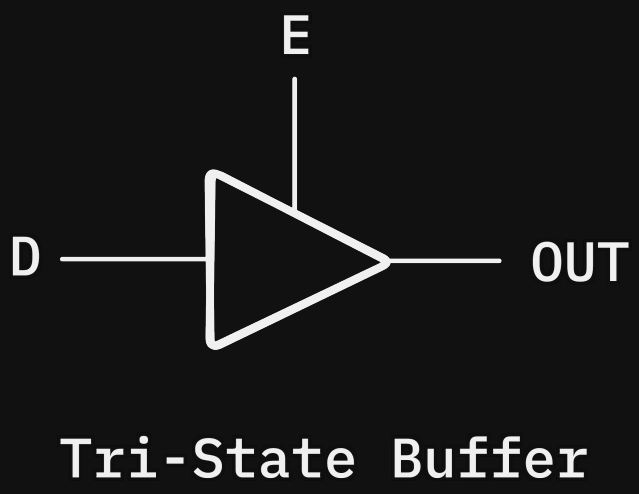


Diagram 8.3. A tri-state buffer logic gate.

Let's address this enable conundrum. We have two uses for the word enable, with completely different meanings and contexts. One means enabling writing, and the other means enabling output. From now on, we will use two separate terms to avoid confusion: `WRITE` and `OUT`. This `OUT` has nothing to do with the output of the tri-state buffer.

We can make a new type of register, one with `WRITE`, `OUT`, a `D` input, and a `Q` output that shows the stored bits 0-7. By `D`, I simply mean the input data that we can store when `WRITE` is enabled.

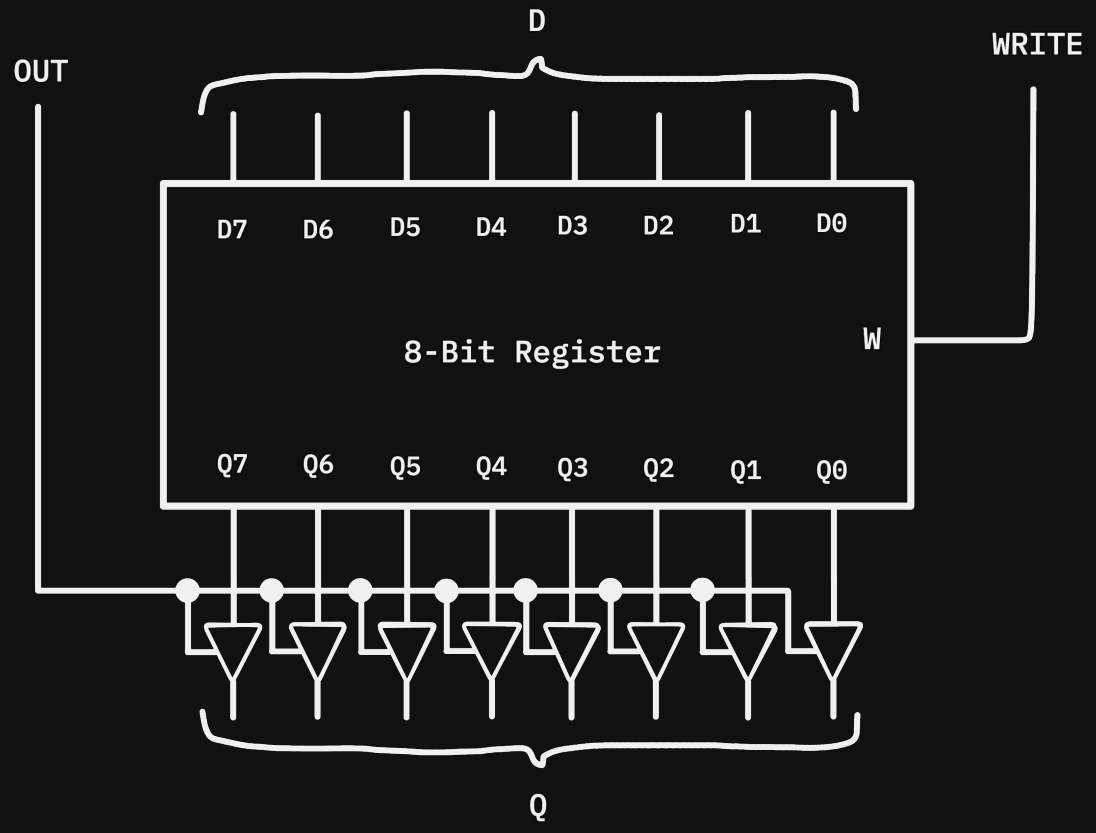
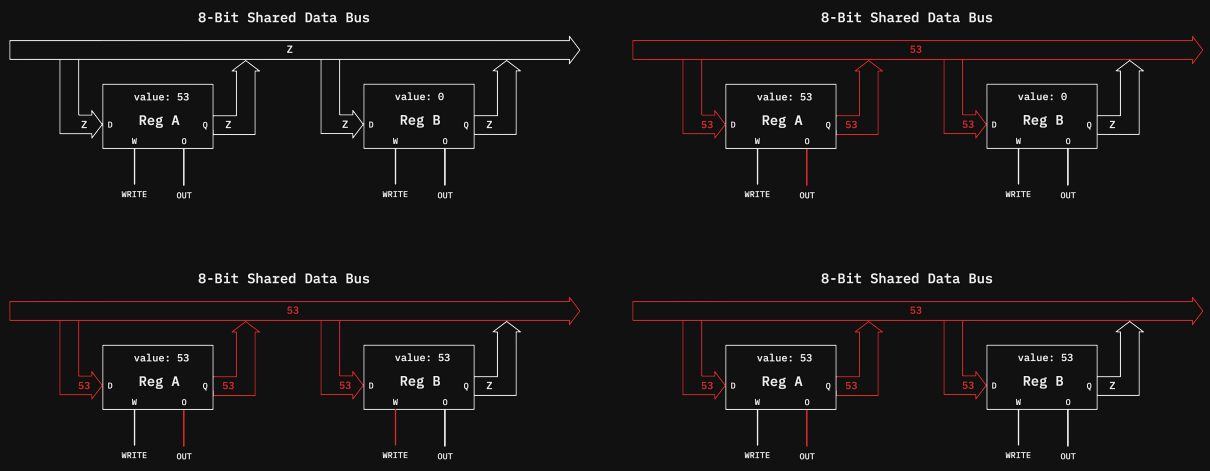


Diagram 8.4. Our register with an OUT input.

We are just connecting `OUT` to all of the enables in the tri-state buffers. So if `OUT` is `0`, `Q` will be all `Z`, and if `OUT` is `1`, `Q` will be whatever is stored in the register.

With that, we can use these new registers with a common bus to move data. Here is an example where the content of register `A` gets copied into register `B`.



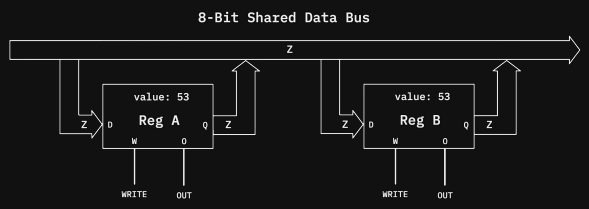


Diagram 8.5. Copying register A into register B through the shared bus.

I have some text inside the register that shows what it is storing. We of course have `W` and `O` which are `WRITE` and `OUT` as well as `D` and `Q` which are the inputs and outputs.

Of course, on the second frame, when `OUT` of register `A` is enabled, the `D` wires of both registers are also going to be 53 because they are directly connected to the bus.

Also generally in this diagram, register `B`'s output is sometimes shown as `Z` even when the bus is 53. That is because register `B`'s `OUT` is off, so register `B` is not driving the bus. It may be connected to a bus currently at 53, but the 53 is coming from register `A`. `Z` describes what register `B` is doing, which is nothing.

By the end of this sequence, we have copied the value 53 to register `B`! We can have many more registers sharing a common bus, as long as only one is driving the bus at a time.

Now we can store a byte, compute a sum, and move bytes around! The next problem is organization and scale. How do we organize many stored bytes so the machine can choose one slot, read it, and write back to it? A handful of registers aren't enough.

2.3. Organizing Data

We want to build a system that organizes data into a simple structure.

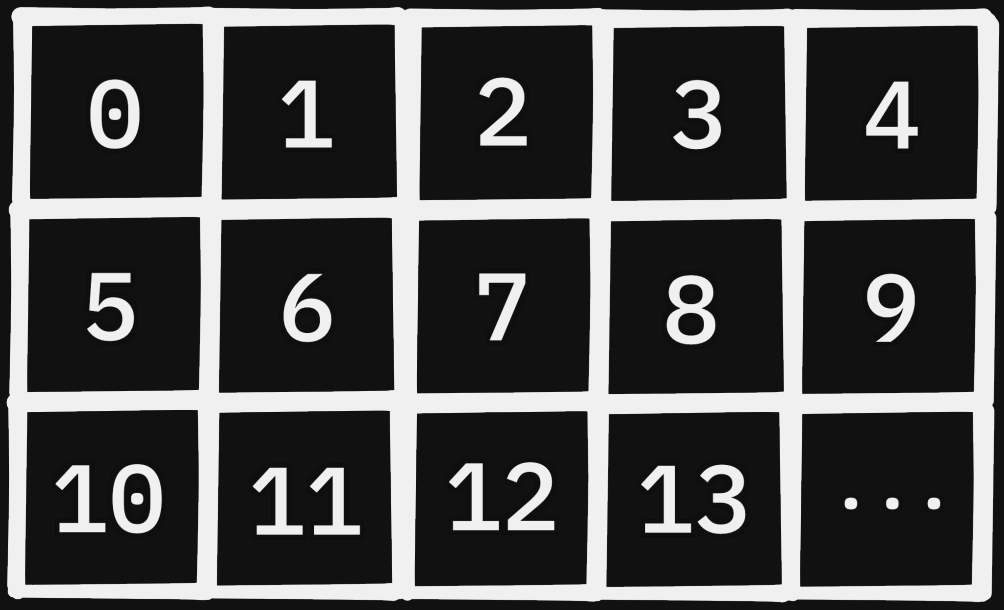


Diagram 9.1. Our data structure.

Many slots, each with its own address. This system is known technically as `RAM`: Random Access Memory. It is called `RAM` because when the CPU wants to access a slot, it just knows the number and can access any slot at will. It is not like flipping through a book looking for the right page. It is more like grabbing a book from a bookshelf, where you already know exactly where the book sits.

Now let's think about exactly what we would want this `RAM` chip to do.

- `address`: the slot we wish to access
- `WRITE`: whether we want to write a value to this address
- `OUT`: whether we want to output the value onto the bus
- `data in`: the value we would like to write
- `data out`: the data output line

To be clear, `WRITE` and `OUT` are control signals, so 1 input wire each. For this demo `RAM`, `address` is only 4 input wires. `data in` and `data out` carry bytes and are both connected directly to the common bus. This only works if no other part is driving the bus when `OUT` is enabled.

So we are going to build a minuscule 16-byte `RAM`: 16 addresses, with each address storing one byte. This design can be scaled up easily. Our address will be 4 bits long, because `2^4` is 16.

We could do this as a tall stack of 16 registers, but a grid is nicer. So we will split the 4-bit address in half. The bottom two bits pick the row, and the top two bits pick the column:

```
top 2 bits = column
bottom 2 bits = row
```

Two bits can choose 4 values, so this gives us a 4×4 grid of memory slots. That is 16 total bytes! Once the address selects a slot, two things can happen:

- If `WRITE` turns on, the selected slot stores `data in`.
- If `OUT` is on, the selected slot drives its stored byte onto `data out`.

Let's start with building a simple decoder. This decoder will take 2 bits of our address and, based on that number, turn on exactly one of its 4 outputs.

In the diagram, the top bit is the bigger bit (the 2's place) and the bottom bit is the smaller bit (the 1's place).

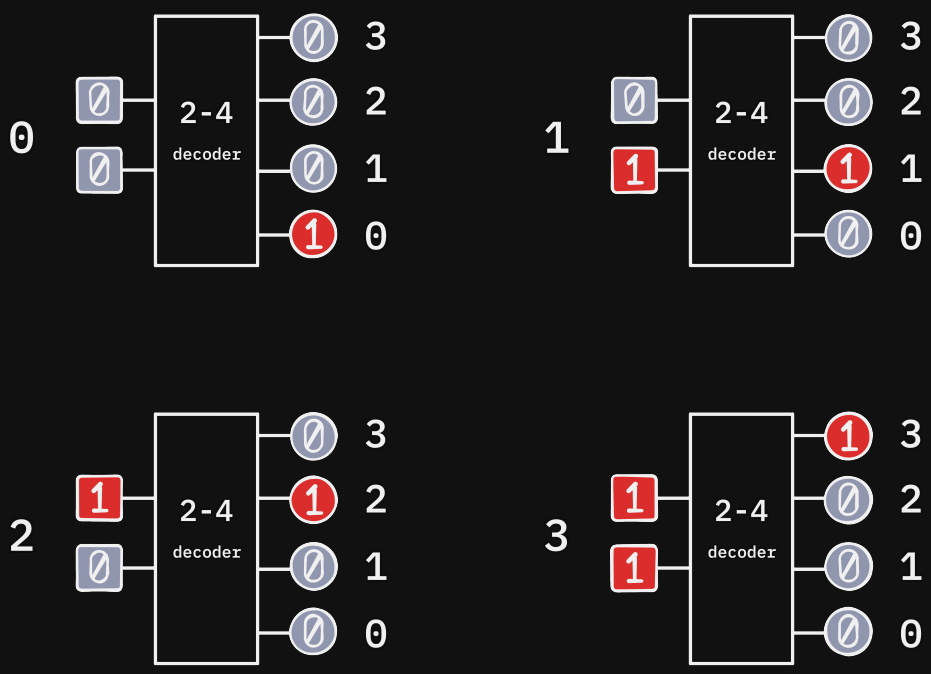


Diagram 9.2. How a decoder works.

As you can tell, no matter the inputs, exactly one output wire is on at a time. How a decoder works is extremely simple. It uses a bunch of logic gates to ask these simple questions.

- If `00` → turn on output 0
- If `01` → turn on output 1
- If `10` → turn on output 2
- If `11` → turn on output 3

Here is how it works if you care:

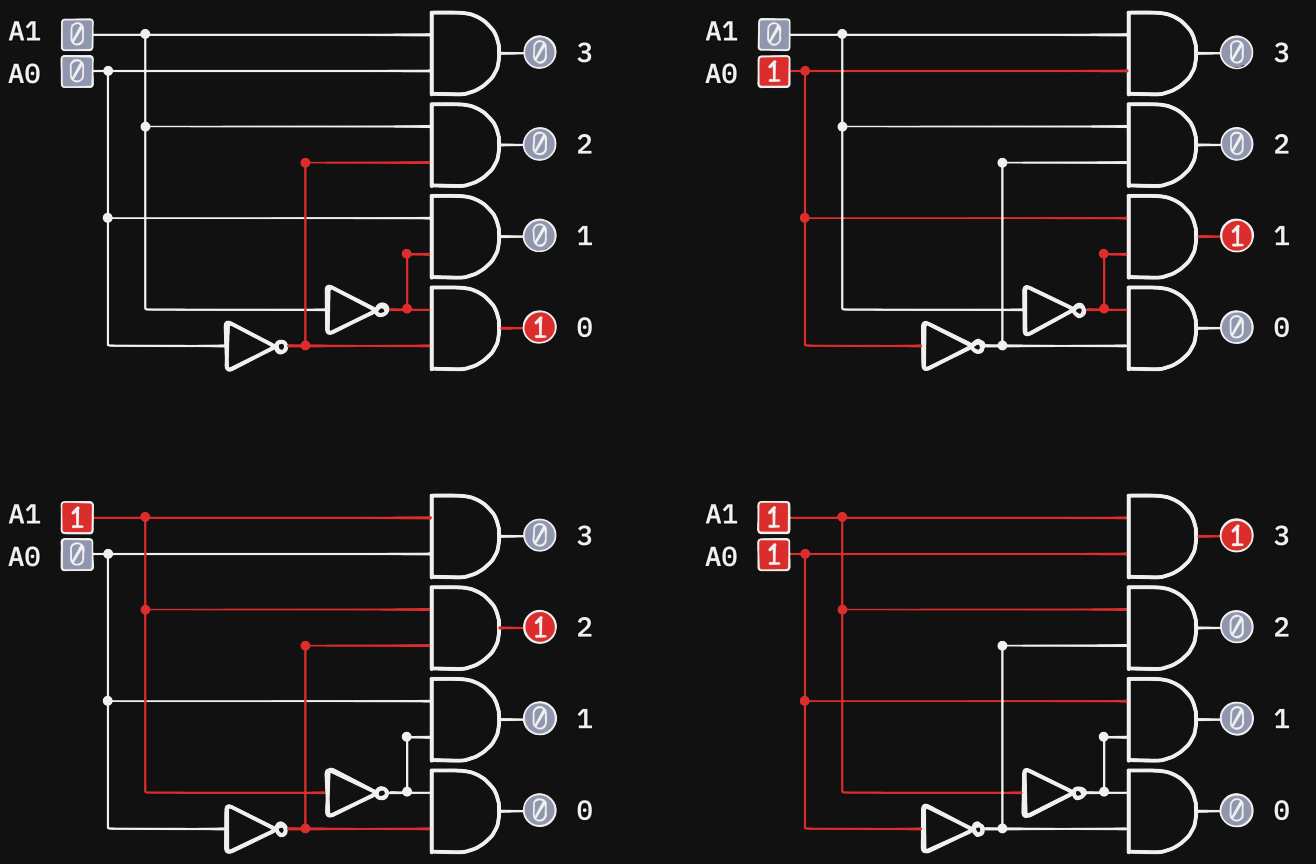
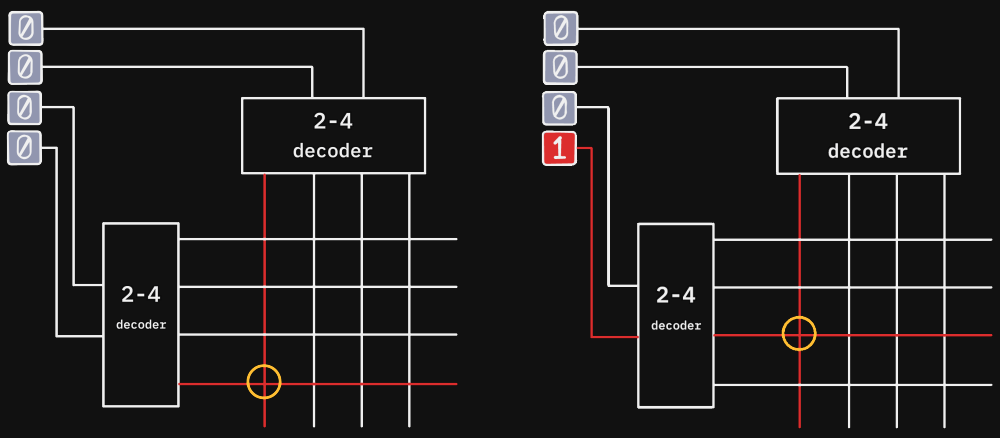


Diagram 9.3. 2-4 decoder internals.

We use one `2-4` decoder for the rows and another `2-4` decoder for the columns. Where the selected row and selected column cross, that is the byte we want to target. This diagram shows a few addresses as examples. Each address gets its own little intersection.



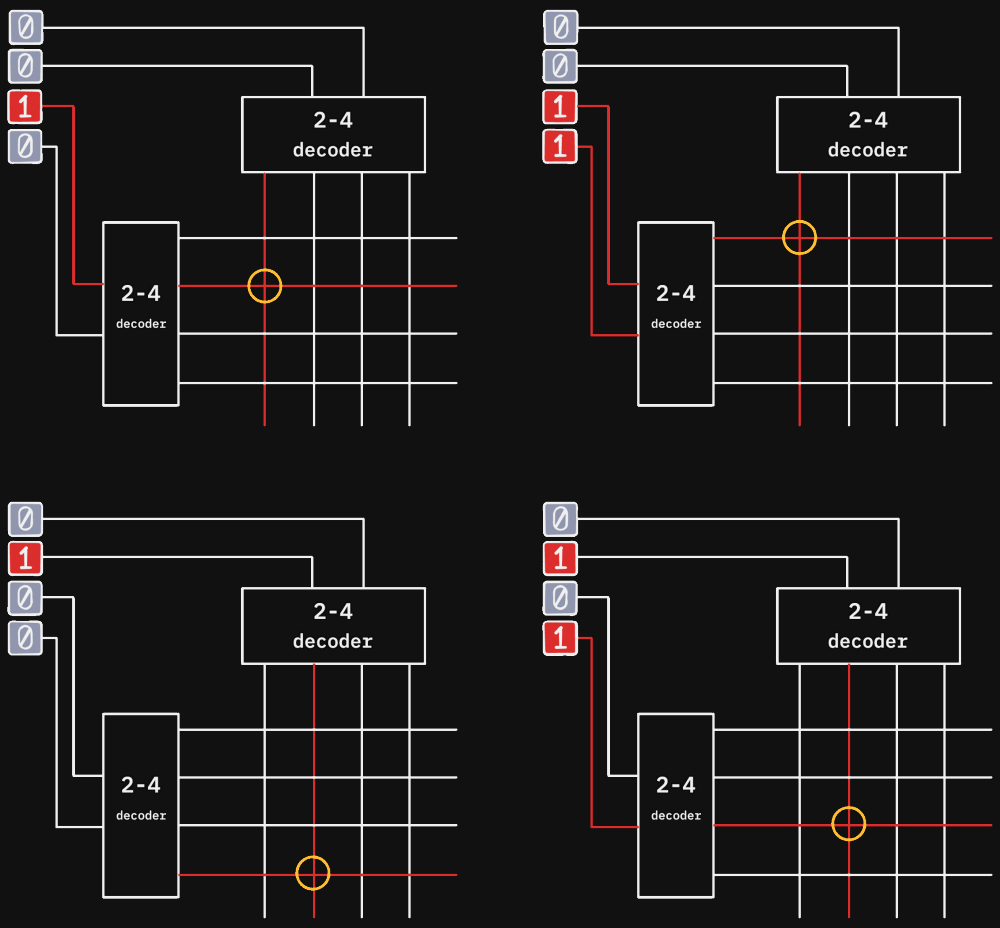


Diagram 9.4. Where the row and column meet.

Honestly? That's it. We can use two decoders, sixteen registers, some wires and buses all mashed together with some extra logic gates and BOOM! We have some `RAM`.

In this diagram, blue lines are 8-bit data buses. `OUT` is yellow, and `WRITE` is orange. They are still ordinary wires; the colors are only there to make the diagram easier to follow.

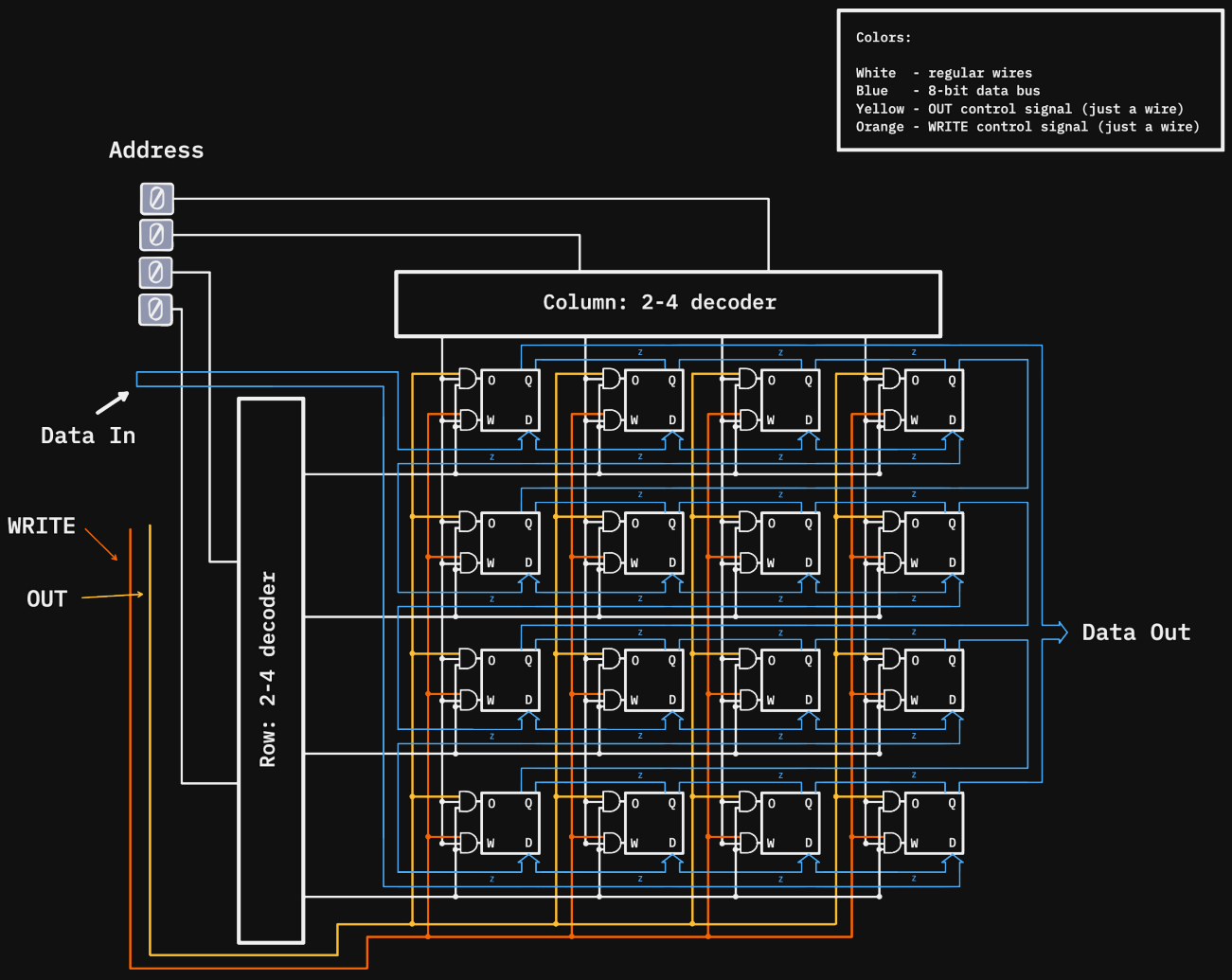


Diagram 9.5. A zoomed out `RAM` diagram.

This is kind of a lot to unpack, so let me explain the high level parts before we zoom in and take a closer look.

We have a blue **`data in`** bus that is fed into the bottom of all of the registers. We also have another blue **`data out`** bus that comes out of the top of all the registers and combines into one output. Every slot is connected to both buses, but only the selected slot is allowed to use them. If the slot's row and column are selected, it can either read from **`data in`** when **`WRITE`** turns on, or drive **`data out`** when **`OUT`** is on.

Let's look at one cell more closely:

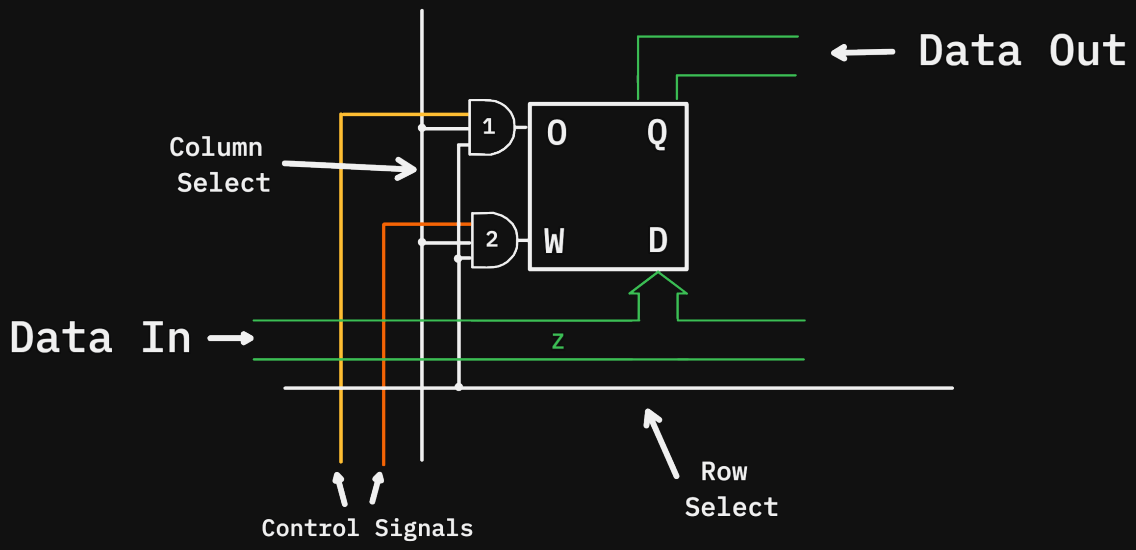


Diagram 9.6. A zoomed in `RAM` cell diagram.

What AND gate 1 checks is, if `Row Select` and `Column Select`, and `OUT` is on, then that means we have selected that register to output its value, thus we turn on `OUT` and the register will output something on the `data out` bus.

AND gate 2 checks, if `Row Select` and `Column Select`, and `WRITE` is on, then that means we have selected that register to write to, thus we turn on `WRITE` for that register, and it will write the data on the `data in` bus.

Both AND gates take in 3 inputs. If you are wondering how that works, think of two AND gates chained together.

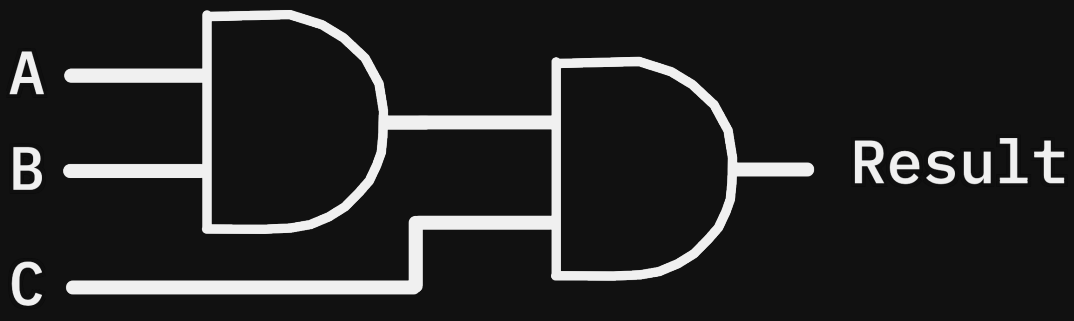


Diagram 9.7. A three input AND gate.

So now that we have built `RAM`, let's pretend that instead of 16 registers, we have a `RAM` array with 256 registers. The same logic can be copied, just with two `4-16` decoders instead of two `2-4` decoders and 8 address inputs rather than 4.

Diagram 9.8. Our `RAM` chip.

Technically, there are still two buses inside the `RAM`: a data-in path and a data-out path. That is basically what we saw with the register in the bus section.

But drawing two separate data buses every time is cumbersome. From the outside, we can abstract this as one shared data bus with a double-headed arrow called `I/O`, which stands for input/output. It is practically just like having two buses, one for input, one for output.

When `W` turns on, `RAM` copies the value from the data bus into the selected address. If `O` is on, `RAM` drives the selected address's value onto the data bus.

So from now on, instead of drawing registers connected to a common bus like this, where we have a separate `D` and `Q`, we can just draw them like this:

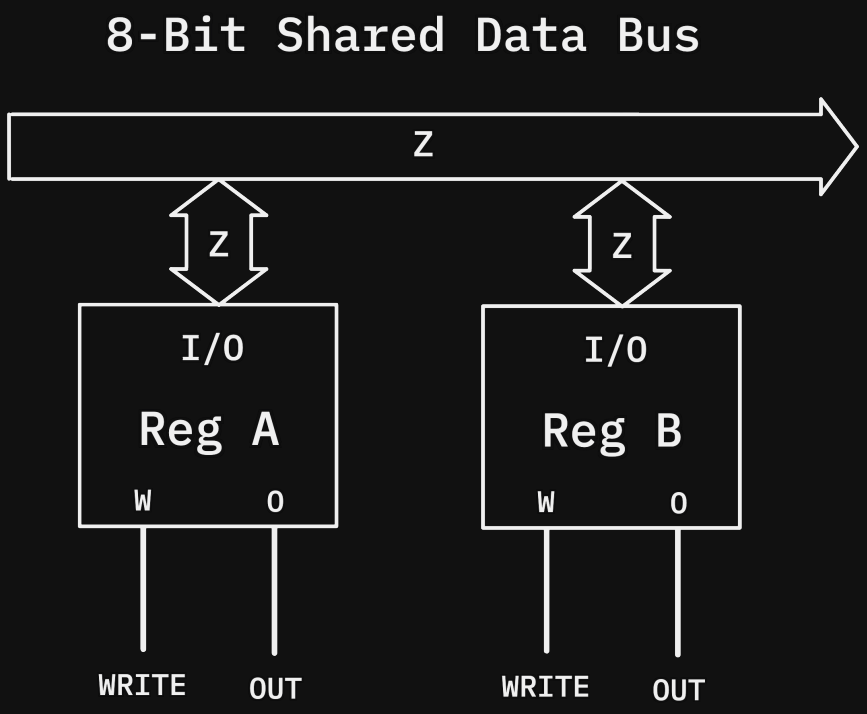


Diagram 9.9. I/O Registers.

They both are the same technically, just this is easier to draw, so moving forward, instead of drawing two buses for `D` and `Q` I'll just draw one double-headed `I/O` bus.

Now back to the **RAM** chip. If you pay close attention to the diagram, you will notice that the address input is not directly connected to the common data bus.

That is intentional. The data bus is for moving values around the CPU. During a **RAM** operation, it needs to carry the value being written to **RAM** or the value being read from **RAM**. So it cannot also keep holding the address at the same time.

We need something to hold the address while the data bus is being used for the actual I/O. This is called the Memory Address Register, or **MAR**. The **MAR** is just a regular 8-bit register with no **OUT** control signal as it is always outputting directly into **RAM**.

We first put an address on the data bus and turn on **MAR_WRITE**. The **MAR** stores that address. Then the **MAR** keeps sending that address to **RAM**, leaving the data bus free to carry the value being read or written.

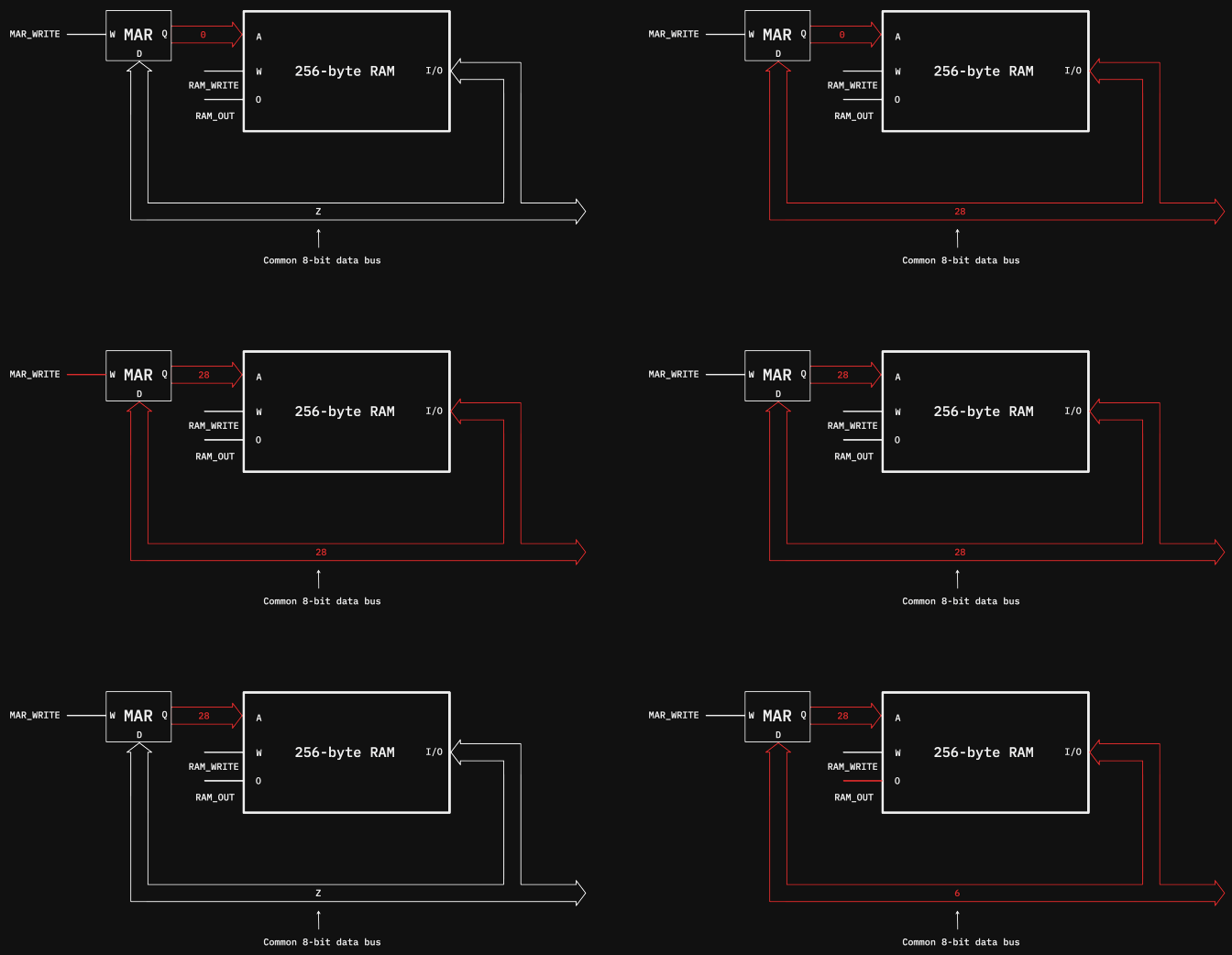


Diagram 9.10. How the `MAR` works.

So first, we put 28 onto the common bus. Enable `MAR_WRITE` and store that into the `MAR`. We then remove 28 from the common bus and enable `RAM_OUT`, and we get 6 as the value stored in slot 28. Cool.

2.4. The ALU

If we have a handful of registers and `RAM`, we can now move bytes around using this common data bus. But what we really need is a component that can "process" numbers, a component that can do arithmetic and logic. Thus we have "The Arithmetic and Logic Unit," or `ALU` for short. Imagine a chip where we could input two numbers, an operation, and output a result, along with some other information.

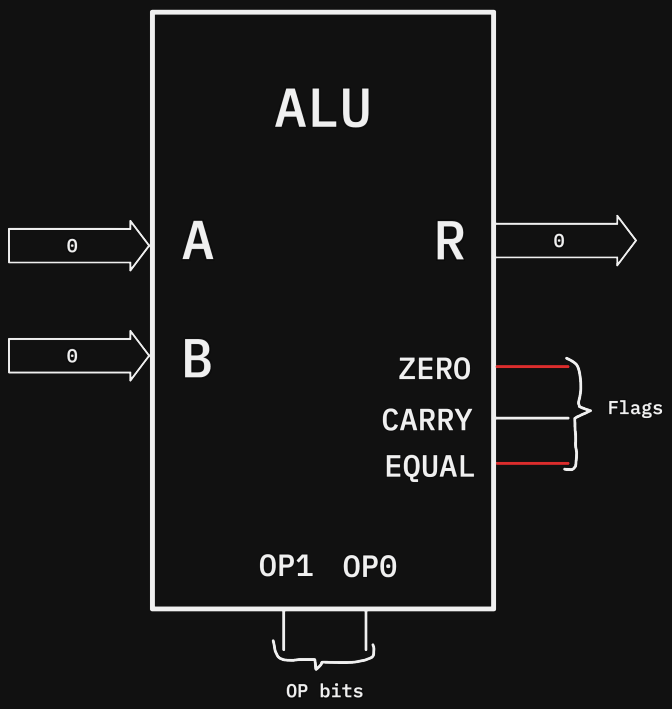


Diagram 10.1. The `ALU` chip.

For this CPU, I am keeping the **ALU** simple. It will have two operation-select bits, which gives us four possible operations:

OP	Operation	Meaning
00	ADD	output A + B
01	AND	output A AND B
10	OR	output A OR B
11	XOR	output A XOR B

The **ALU** will also output a few flags, which are just extra yes/no facts about the result or the inputs:

Flag	Turns on when
ZERO	the result is 00000000
CARRY	addition spills past 8 bits
EQUAL	A and B are the same

So, in the previous diagram, we did 0+0 which is 0, so the **ZERO** flag is on, and the **EQUAL** flag too because both inputs are equal.

This next diagram uses a new component. It is a mix of two registers we have already seen. Remember the D latch, the one with the **D** and **E** inputs, before we made it edge-triggered? While its enable was on, **Q** simply equaled **D**. No edge-triggering or anything like that. This new register is a D latch with the enable permanently on: it never really holds anything. Its output just follows its input.

But like our newer registers, its output goes through tri-state buffers with an **OUT** control signal, so we still decide when it outputs. In total: a data input that is always being stored, an **OUT** control signal, and an output.

Also, you'll see me feeding two buses into a single logic gate. "How does that work?", you might think. Well, there are really just eight gates, one per bit. One gate takes **A0** and **B0**, the next takes **A1** and **B1**, and so on. Eight output wires, which is just another bus.

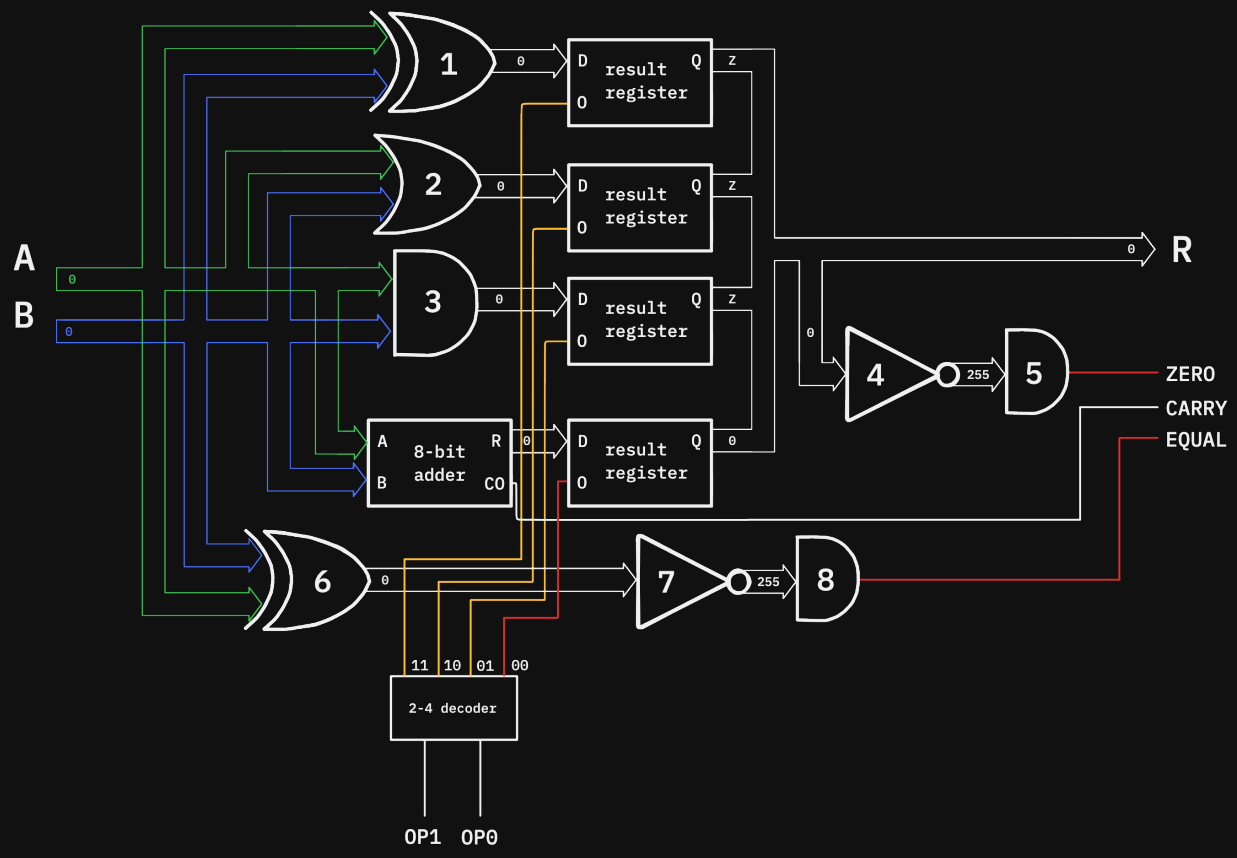


Diagram 10.2. `ALU` internals.

At its crux, the **ALU** works by routing **A** and **B** into all 4 operations at once, in this case XOR, OR, AND, and ADD. We store each of the results in a corresponding result register.

To decide which one to output, the **OP** bits go into a **2-4** decoder that enables exactly 1 of the result registers onto the **R** bus.

Now for the flags. First, the **ZERO** circuitry. It consists of:

NOT → AND

The NOT gate on the right, labeled 4, is just like before: there are actually eight NOT gates, each flipping one wire of the bus. So it takes in a bus, and outputs a bus.

But the AND gate next to it, labeled 5, takes in a bus and outputs just one wire. Like the three-input AND from the **RAM** section, this AND gate takes in 8 inputs and produces 1 output. It just checks if all of its inputs are on.

So if we flip each bit, and then check if all of them are on, we get the **ZERO** flag. This makes sense because all the bits going into the AND gate can only be on if they were originally `00000000`, a.k.a. zero!

Next, let's cover the **EQUAL** circuitry. It consists of:

XOR → NOT → AND

The XOR gate, labeled 6, is basically like last time: we XOR each pair of bits from `A` and `B`, and create an eight-bit bus. Remember, XOR outputs `0` when its two inputs are the same. So if `A` and `B` are equal, we get `00000000` as the output.

Then we flip the bits with gate 7, getting `11111111`, and if we AND them all together with gate 8, we can check if they are all true. If even one pair of bits differs, that wire ends up `0` after the flip, and the AND outputs `0`. Simple.

The `CARRY` flag is simple: we just connect the adder's carry-out, `CO`, straight out. Of course, it only means anything when we are actually adding. Boom, that's it, the `ALU` is complete. We can now piece together this whole CPU.

2.5. The Big Picture

Here is a big picture diagram of the whole CPU.

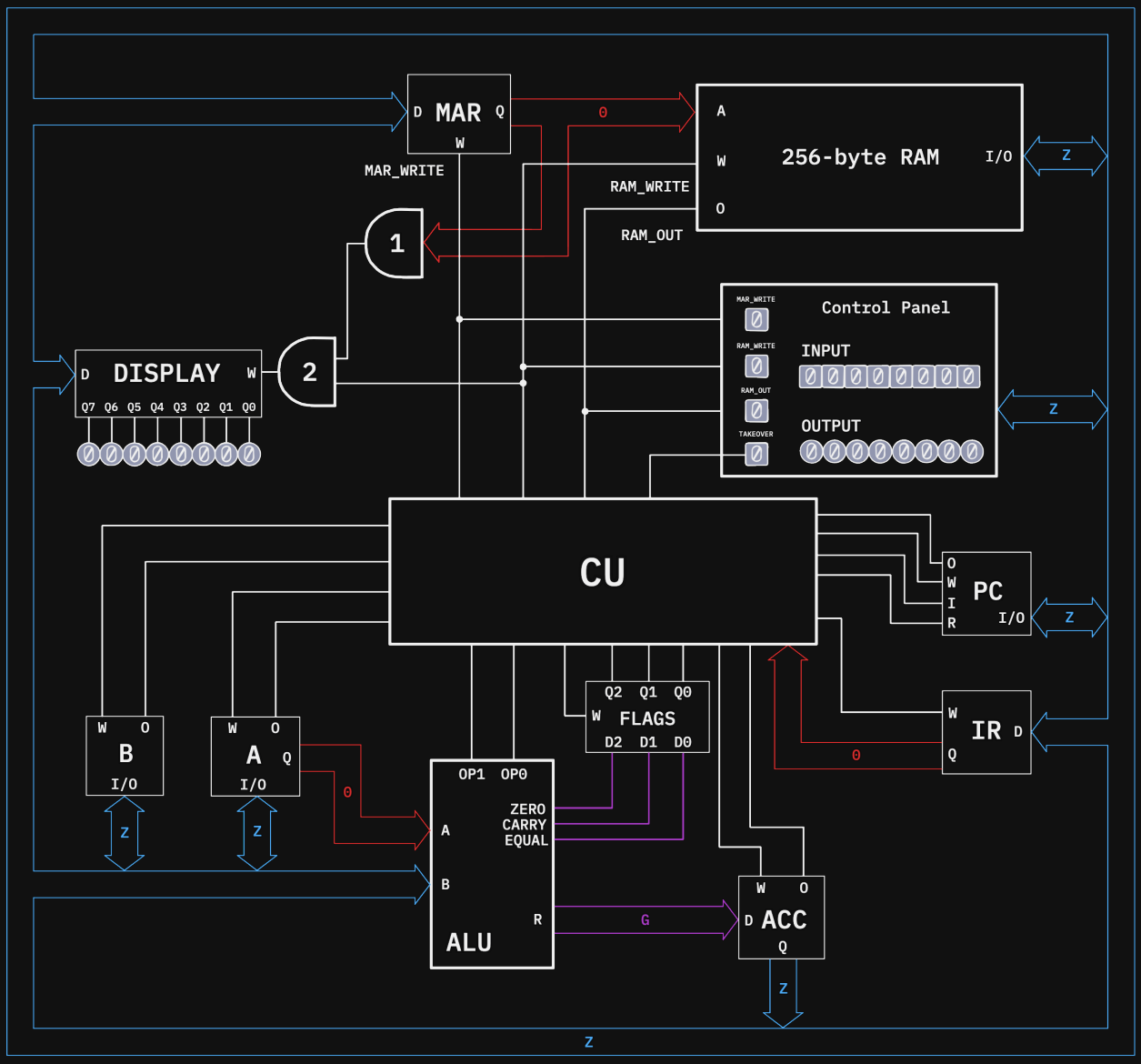


Diagram 11.1. The whole CPU.

Hey? Doesn't this diagram look familiar? Well, yes, we have come full circle from the intro, except this time the CPU isn't something completely foreign. Of course we still have a lot to learn and build, but wow. Just wow.

Some quick notes on the diagram before we dig in. The 8-bit data buses are blue when not being driven. If a control wire is red, it means it's on. If a bus is red, it means it's being driven like normal. But we have something new here that we haven't seen before.

Purple wires and buses mean garbage. This is not the same as `Z`. A `Z` wire is one that nothing is driving. A `G` wire is being driven, with an actual value on it. That value is just nonsense, hence the name garbage. So where does that garbage value come from in this diagram?

Remember, the **`ALU`** never stops computing. It always takes its inputs and has a result instantly. One input comes from register **`A`** (we will talk more about how register **`A`** works later), but the other input comes from the bus, and when nothing is driving the bus, the

input is sitting at `Z`. The thing is, in this CPU an input at `Z` would just behave like 0. Our gates are built from relays, and a relay coil with nothing driving it is simply off, exactly as if you fed it 0. So really, with `OP` set to ADD like it is here, the `ALU` would be computing `A + 0` if the second input was `Z`.

But we mark it as garbage anyway, because in a real CPU made from transistors instead of relays, a `Z` value does not settle to a clean 0; it drifts and can get really funky. That is why we treat those values as garbage: we just ignore them. If the bus is being driven then we can of course use those values.

Now back to the diagram. Almost every single chip here, we have already built. The `RAM` and `MAR` combination, we have seen how that works previously. `A`, `B`, `ACC`, `IR`, `FLAGS`, and `DISPLAY` are all just slight variations of registers. `PC` is an accumulator-style circuit, and we know how the `ALU` in the bottom left works.

Let's examine each element of the CPU closely and see what details changed and why. First, register `B`. It is a normal tri-state 8-bit register with `W` and `O` control wires. We will talk about its use in more detail later.

We then have register `A`, which is just like register `B`, but it also has a `Q` output in addition to `I/O`. This `Q` output is simply the value stored inside the `A` register, bypassing the tri-state buffers and feeding directly into the first input of the `ALU`. So even though the output (`O`) control signal is off, that `Q` is still driving the `ALU`'s first input, in this case to 0.

Next we have the `ALU`. Its second input comes from the common bus, so the other number it works with is just whatever is on the bus at the time. Other than that it is as normal, but the flags are all going into a register called `FLAGS`. This register is a 3-bit register: it only stores 3 bits. These three bits are just the values for each flag. This 3-bit register has no `O` control signal, thus no tri-state buffers, so it is always outputting.

Then we can see the result of the `ALU` operation is being fed into another regular 8-bit register called `ACC`. It is slightly different to register `B` because its input doesn't come from the common bus, but from the `ALU`, so it has a separate `D` and `Q` instead of just one `I/O`. It still has the regular `W` and `O` control signals though.

Register `IR` is a simple register without an `O` control signal. It takes input from the bus, and outputs it into the `CU`. We will talk about what exactly the `CU` is and the jobs of these different parts a little later on.

Next we have `PC`, which is a combination of an adder and a register, something like what we have seen previously. The register part of `PC` has its regular control signals `W` and `O`, and the `STEP` button from before has been replaced with an `I` control signal, which stands for increment. Inside, a few gates pick which value reaches the register's input: the bus when `W` is on, the adder's output when `I` is on. We also have the `R` signal which resets the register back to 0 with some more logic gates and wires. Nothing too fancy.

Our `MAR` and `RAM` are the same as before, but if all of the bits stored in `MAR` are 1, meaning the address is 255, then that first AND gate in the diagram, labeled 1, turns on. We then AND that with `RAM_WRITE`: if we are writing to address 255, we also write into the `DISPLAY` register.

What ends up happening is that DISPLAY stores whatever is stored in RAM address 255, and displays that value on 8 bulbs for us to see. You can think of this as our simple output for any programs we might write.

Then we have the Control Panel. Its job is to load a program into RAM in the first place. It can read and write to any RAM address it wants.

How it works is, first you flip the TAKEOVER switch, which inside the CU basically freezes the computer and resets PC. Then, while TAKEOVER is on and the RAM_OUT button is off, the value on the panel's input switches is put onto the common bus. If RAM_OUT were on, then RAM would try to drive the bus at the same time as the Control Panel. That's why we need to make sure it's off before we can safely drive the bus.

From there you can hit MAR_WRITE, which stores that as the address you want to work with in RAM. To write, you then flip the switches to the value you want and press RAM_WRITE. To read, you instead press RAM_OUT, which makes the Control Panel stop driving the bus, because RAM will then drive the bus, and the output bulbs will turn on to that value.

Also, the panel's buttons only actually drive their control wires if TAKEOVER is on, using tri-state buffers. This is to ensure that while the CPU is running like normal, the CU and the panel don't drive the wires at the same time. This works the other way too: when TAKEOVER is on, the CU makes sure not to drive those RAM control wires, again using tri-state buffers.

Now, about this mysterious box labeled CU. What is it exactly? It is the control unit. Think back to most of our previous demos. Turn O on, then W, flip this control signal and then flip this control signal. The CU does all of that flipping for us. It turns the control wires on and off in the right order, at the right times, to make the computer actually work.

But how exactly does the computer work in the first place? First we load our program into RAM: a list of instructions, each one just a number. The control unit then fetches the instruction from RAM at the address stored in PC, so if PC is 0, it fetches the instruction at address 0 and stores it in IR. How? It copies PC into the MAR and flips on RAM_OUT, exactly like we have done before. Next it figures out what that instruction means, and executes a bunch of steps to actually do that instruction, then it increments or changes PC and repeats that whole cycle.

This is kind of a simplified version of what our CPU will do, but we will dive into it in the following sections. You can think of this cycle as fetch, decode, execute. Fetching gets us the instruction, decoding figures out what it means, and executing actually flips those control wires on and off to accomplish that instruction!

But before we can figure out how the CU works, we need to figure out how instructions work. What does an instruction even look like?

Part 3: Execution

3.1. Instructions Are Numbers

Instructions tell the CPU what to do, and a list of instructions is a program. There are game programs, calculator programs, email programs. Everything on your computer is a program. Just a long, long list of instructions telling the CPU to do simple things like add these two numbers, move this value here, etc. All programs are just a list of instructions stored in `RAM`, each instruction coded as a number.

So here is what number each instruction in our CPU translates to:

#	Binary	Mnemonic	Bytes	Action
0	`0000`	`NOP`	1	do nothing
1	`0001`	`LOAD A, n`	2	next byte → A
2	`0010`	`LOAD A, [addr]`	2	RAM slot → A
3	`0011`	`STORE A, [addr]`	2	A → RAM slot
4	`0100`	`MOV B, A`	1	copy A into B
5	`0101`	`MOV A, B`	1	copy B into A
6	`0110`	`JMP addr`	2	addr → PC
7	`0111`	`JE addr`	2	if `EQUAL` flag: addr → PC
8	`1000`	`ADD`	1	A = A + B
9	`1001`	`AND`	1	A = A AND B
10	`1010`	`OR`	1	A = A OR B
11	`1011`	`XOR`	1	A = A XOR B
12	`1100`	`JC addr`	2	if `CARRY` flag: addr → PC
13	`1101`	`JZ addr`	2	if `ZERO` flag: addr → PC
14	`1110`	`HALT`	1	stop the clock

Table 12.1. The full instruction set.

The number for each instruction is called its opcode. Let's write a tiny program. We store `00000001` in address 0 of `RAM`. Then store `00000101` in address 1 of `RAM`. Then store `00000011` in address 2, then `11111111` in address 3, and finally `00001110` in address 4. Our `RAM` looks like this:

```
0: 00000001
1: 00000101
2: 00000011
3: 11111111
4: 00001110
```

If `PC` starts at 0, which it does, then what do you think this program would do? The first byte is `00000001`, which is 1, and the table says 1 is `LOAD A, n`: load the next byte into register `A`. The next byte is `00000101`, binary for 5, so those two bytes make up the instruction:

`LOAD A, 5`

The first byte tells us what the instruction is; the second tells us what value to load. The next instruction is `3`, `STORE A, [addr]`, which basically puts the contents of `A` into the address of `RAM` that we specify. Again, the first byte tells us the instruction and the second byte tells us the address, in this case `11111111`, which is address 255.

Hey? Isn't address 255 different from the rest? Yep, if we write to address 255 we also write to the `DISPLAY` register where we can see that value on some light bulbs. Have a look at this diagram for a recap.

The last byte is `14`, which the table says is `HALT`. It stops the clock, the whole machine freezes, and the program is over.

But, why do we even need to stop the CPU? Well, if we don't, `PC` would just keep counting up and the CPU would keep fetching. It doesn't know our program ended at address 4. It would march straight into the rest of `RAM` and run whatever is sitting there as if it were instructions, leftovers from the last program, or whatever junk the memory happened to wake up holding. We need a way to stop.

This program puts 5 on the output and stops. Pretty simple. As you can tell, some instructions take two bytes, and some only need one. A 2-byte instruction is one where the first byte alone is not enough. `LOAD A, n` needs to know what value to load, so the next byte in `RAM` is that value.

One more thing: the order things are written in. For `MOV`, the destination comes first. `MOV B, A` copies `A` into `B`, not the other way around. `STORE` is the odd one out: `LOAD A` means a value going into `A`, and `STORE A` means `A` going out into memory.

Also, a number with brackets like `[5]` means whatever is stored at address 5, but just `5` means literally the number 5, which would be stored in the next byte of `RAM`. It's just an easier way for us humans to write it.

Now, about the jump instructions: `JMP`, `JE`, `JC`, and `JZ`. All of these instructions can set `PC` to the address stored in the second byte.

`JE`, `JC`, and `JZ` each check one flag. The flags from the last `ALU` instruction sit in the `FLAGS` register until the next `ALU` instruction overwrites them. So to make a decision, you run an `ALU` instruction, then jump based on what it found. If the flag is on, the jump sets `PC` to that address. If not, the program just carries on to the next instruction. This is the core element used in creating programs that can decide stuff and loop.

For example:

```
0: LOAD A, 1
2: STORE A, [255]
4: MOV B, A
5: ADD
6: JMP 2
```

One note: the number before each instruction is just the address in **RAM** where it would be stored. That is why you can see it jump in twos for 2-byte instructions.

This jumping back to the start of the loop is done with a jump instruction, so whenever **PC** reaches 6, it jumps back to 2 and repeats all over again. Each loop, this program displays **A**, copies **A** into **B**, and then adds. Both **ALU** inputs are the same number, so **A** doubles every loop, and the output shows 1, 2, 4, 8, 16, all the way up to 128. On the 8 output bulbs, that looks like a single lit bulb going across the display. Pretty neat.

But then 128 + 128 overflows: **A** wraps around to 0. And from that point on, the program is doubling 0 forever, so the display goes dark and stays dark. The CPU gets stuck in a weird spot.

To solve this overflow problem, and in general to make decisions, we can use the flags to jump to different parts of our program depending on what the last **ALU** instruction found.

```
0: LOAD A, 1
2: STORE A, [255]
4: MOV B, A
5: ADD
6: JC 0
8: JMP 2
```

Now, when the addition overflows, the **CARRY** flag turns on, and **JC** catches it: we jump back to the start of the program, address 0, which loads 1, and the program starts all over again. Forever. On every normal loop, **JC** checks the flag, finds it off, and the program just carries on to the **JMP**.

Because this computer is built from relays, and thus pretty slow, we can probably watch the light walk across the display with our own eyes. If we want to slow it down some, we can pad the program with some **NOP**s that waste clock cycles.

Real computers are insanely fast and usually have specialized timer hardware and such, but sometimes run "do-nothing loops" that just waste clock cycles for a set time. Here is an example of a simple do-nothing delay loop:

```
0: LOAD A, 1
2: MOV B, A
3: LOAD A, 0
5: ADD
6: JC 10
8: JMP 5
10: (rest of the program)
```

All this loop does is add 1 to **A** over and over, 256 times, until the addition overflows and the **CARRY** flag lets it escape. It computes nothing useful, it just eats up time.

But we don't really need these for our slow computer. Just know real computers use a combination of timer hardware, little counter circuits that tick along with the clock, and

loops like these to wait for the right amount of time. Some CPUs even have a sleep instruction that shuts them down completely until something wakes them up.

These jump instructions can help us branch on certain conditions, letting our program decide stuff, and also run in loops.

But how would the CPU actually execute an instruction? What control wires would it flip, and in what order? Let's take the `MOV B, A` instruction as an example.

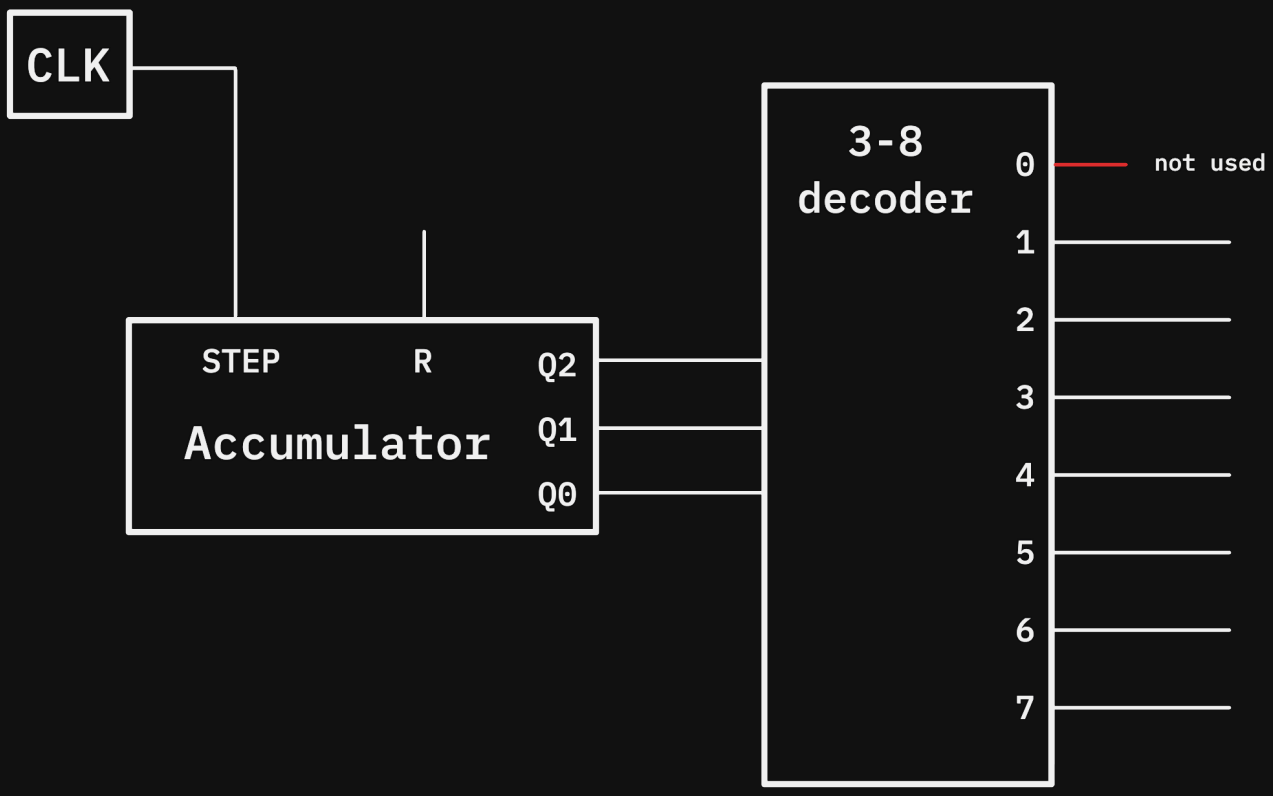
First the `CU` would flip the `O` signal on the `A` register, driving the bus to whatever is stored in `A`. Then the `CU` would flip the `W` signal on register `B`, writing the value on the bus into the register. Then it would turn off the `O` signal on register `A`, and boom, we are done. The `CU` had to do a few tiny things in exactly the right order to execute that instruction. Each of those tiny things is called a micro-step, and we will talk about them in more detail later on.

How does the `CU` know which wires to flip and when? That is what the next section is all about.

3.2. The Not-So-Clever Control Unit

The first thing the control unit needs is a step counter. Something that counts, step 0, then step 1, then step 2, and so on, back to step 0. That is the core flow of the control unit.

We can combine a tiny 3-bit accumulator and a `3-8` decoder. This will give us 8 steps, and because the accumulator overflows and naturally wraps back around to 0, the steps will loop correctly.



The full animation at bytecolony.computer/writing/the-cpu/#f=cu-counter:1.

Diagram 13.1. The step counter.

The accumulator part of the step counter is connected to the clock. The clock is the heartbeat of the computer, and it lives right here in the control unit.

Output 0 of the decoder is wired to nothing, so step 0 is an idle tick: the counter goes 1, 2, ... 7, 0, and every instruction takes 8 clock cycles whether it uses them or not.

One thing: remember the `TAKEOVER` switch? Well, when it is on, it basically resets this step counter's accumulator to 0, so after a reset the first thing that happens is step 1. This also means `TAKEOVER` is a reset switch. Flick it on and off, and both `PC` and the step counter get reset to 0.

Each of the decoder's output wires turning on is 1 micro-step. Take `MOV B, A` from before: enable the output of `A` onto the bus, then enable `W` on `B` to store that value.

If we tried to just turn `O` and `W` on at the same time, the register might capture the value on the bus before the other register has started driving the bus. The bus hasn't had time to settle. So this won't work. What we want is to turn `O` on, wait for the bus to settle, and then turn `W` on, right in the middle of `O` being on.

We could try spreading this across two steps: turn `O` on for step 1, then keep `O` on for step 2 and turn on `W` too. The problem is that when the counter switches from one step to the next, the decoder briefly flickers through the wrong output, so `O` can drop at the exact moment `W` fires. And every move would now cost two steps instead of one. It's not a good way to do it.

The best approach to this is just taking our clock signal and flipping it:

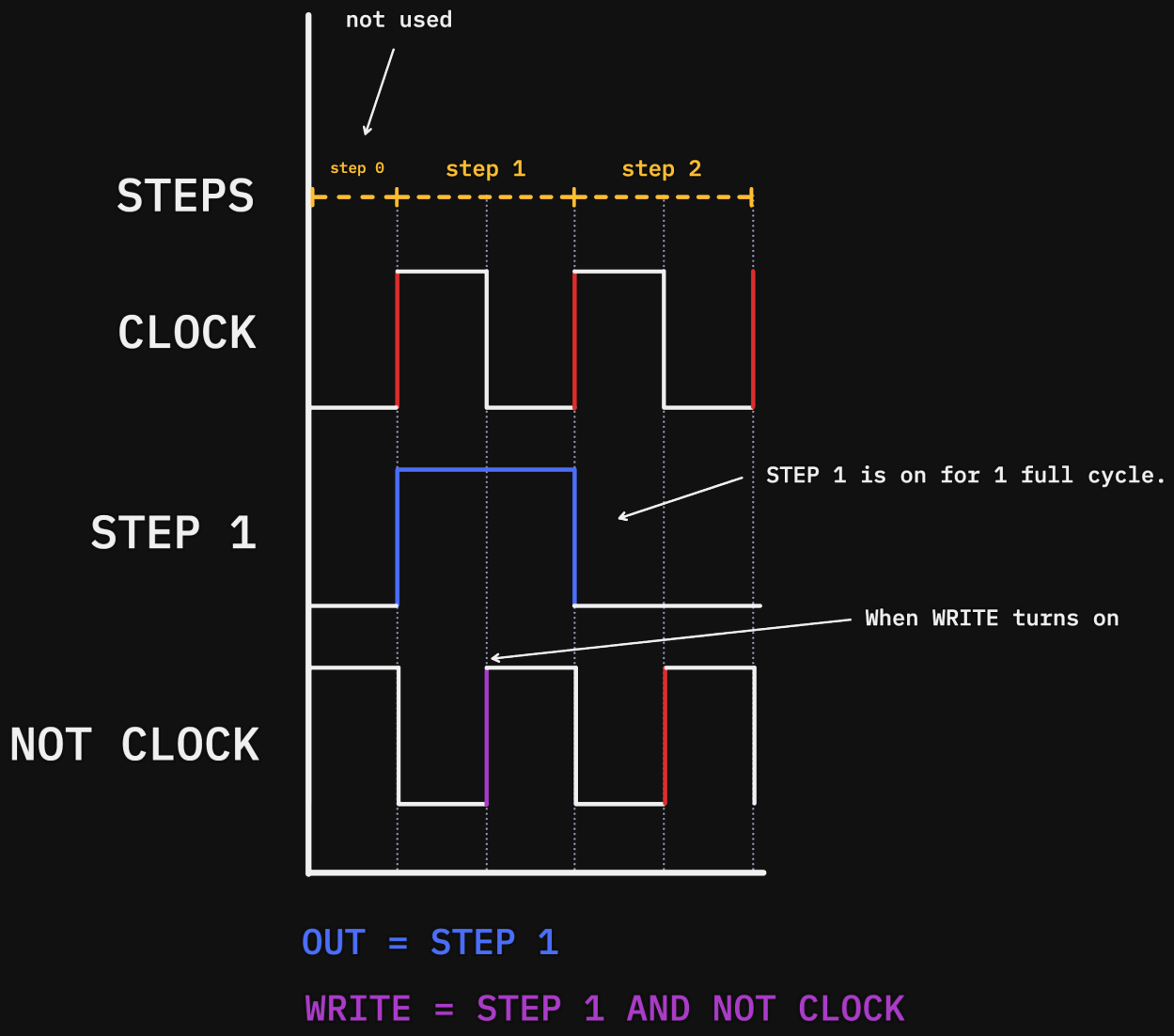


Diagram 13.2. Micro-step timing diagram.

Purple marks when `W` turns on, and blue is when `O` is on. Each step changes on a clock edge, and each step is a signal, 1 full clock cycle long, until the next rising edge.

`O` is directly connected to that step line. If we enable `O` on step 1, the bus is driven for that full cycle.

`W` is ANDed with `NOT CLOCK`, like `step 1 AND NOT CLOCK`. The thing is, step and `NOT CLOCK` don't come on at the same time. The step line is already on, and then halfway through the cycle `NOT CLOCK` goes up, and that is when the AND gate's output goes up. That is the edge `W` needs, right in the middle, after `O` has had some time to settle.

Now that we have covered that, let's figure out what should happen on each step. The first thing the `CU` will have to do for every instruction no matter what is to actually fetch that

instruction from `RAM` and store it in `IR`. `IR` stands for Instruction Register and is used to store the instruction so the `CU` can decode it and figure out what it means.

The register `PC` stands for Program Counter and all it does is store the address of the next byte that we need to fetch. Then after we have fetched that instruction it usually gets incremented by 1, but with jump instructions, the value in `PC` gets changed, so the program starts executing from somewhere else.

To fetch the next instruction, first we would need to put the value in `PC` into the `MAR`. Then activate `RAM_OUT`, putting that instruction on the common bus, and store that value inside the `IR` register. Then we increment `PC` so it points at the next byte in `RAM`. Sometimes this next byte is an instruction and sometimes it's the second part of a 2-byte instruction.

So to accomplish this, we only need 2 simple micro-steps:

1. `PC_OUT`, `MAR_WRITE`
2. `RAM_OUT`, `IR_WRITE`, `PC_INC`

We know how we can output and write to a register in one micro-step, but three things? How can we increment `PC` too? Incrementing `PC` has nothing to do with the bus and doesn't disrupt anything else going on, so we just can! And notice, these 2 micro-steps are exactly the same for every single instruction.

One thing I am not drawing: every `W` signal is ANDed with `NOT CLOCK`, like we just discussed. The same goes for `PC`'s `I` signal: incrementing stores a value too, it just comes from `PC`'s own little adder instead of the bus, so it waits for `NOT CLOCK` like every other store. Drawing that same little AND gate nine times would clutter everything up, so instead, every pin colored pink has it secretly. Pink pin means stores on `NOT CLOCK`.

One more thing: I have used tri-state buffers before the `RAM` and `MAR` control signals, and I have kept the enable on for now. You can mostly ignore this for now, but it will come back later.

NOTE: All of the pink control signals are ANDed with NOT CLOCK

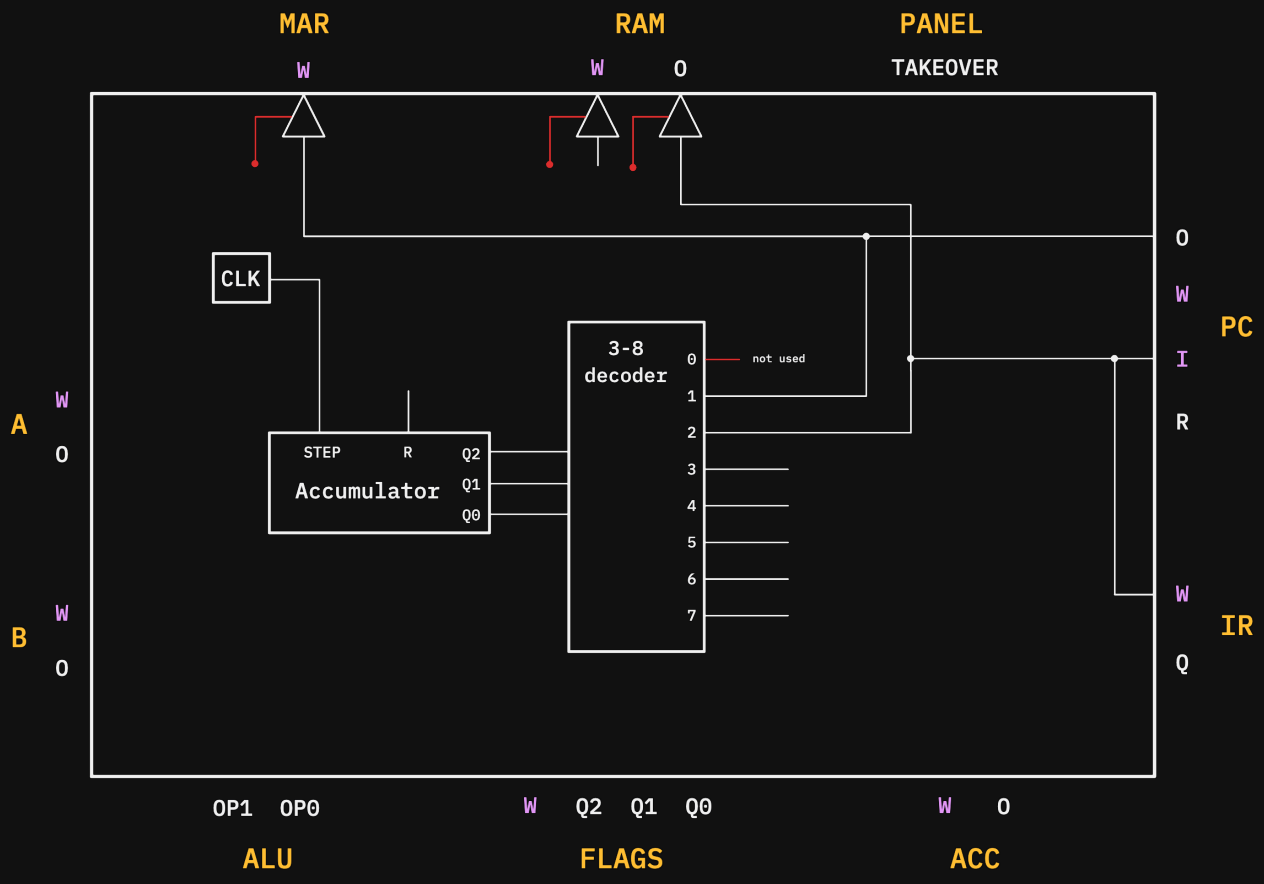


Diagram 13.3. The first two micro-steps.

That is the fetch. Everything from step 3 on depends on what we just fetched.

3.3. Decode & Execute

Steps 1 and 2 are the fetch, and they are the same for every instruction. The rest of the micro-steps depend on the instruction we want to execute, so we use another decoder, this time a **`4-16`** decoder. Each instruction is actually 8 bits long, one byte, but the opcode only needs 4 bits, so only the bottom 4 bits of `IR` are wired into the decoder and the top 4 are ignored. We only have 15 instructions (0-14) so we can ignore output 15 of the decoder, and also output 0, since `0` is a `NOP` which if we remember does nothing.

Let's first focus on the 2-byte instructions, the instructions that need to fetch a second byte. Instructions like `LOAD A, n`, `LOAD A, [addr]`, `STORE A, [addr]`, `JMP addr`, `JE addr`, `JC addr`, `JZ addr`.

Because all of these instructions need to fetch another byte, the third micro-step is always the same:

3. `PC_OUT`, `MAR_WRITE`

We can check, "is this instruction one of the 2-byte instructions?", using a giant OR gate, and if it is, on step 3, we can just execute the same micro-step. Steps 4 and 5 of these instructions all vary slightly, but nothing a few logic gates can't solve.

#	Instruction	Step 4	Step 5
1	`LOAD A, n`	`RAM_OUT`, `A_WRITE`, `PC_INC`	Nothing
2	`LOAD A, [addr]`	`RAM_OUT`, `MAR_WRITE`, `PC_INC`	`RAM_OUT`, `A_WRITE`
3	`STORE A, [addr]`	`RAM_OUT`, `MAR_WRITE`, `PC_INC`	`A_OUT`, `RAM_WRITE`
6	`JMP addr`	`RAM_OUT`, `PC_WRITE`	Nothing
7	`JE addr`	`RAM_OUT`, `PC_WRITE` if `EQUAL` else `PC_INC`	Nothing
12	`JC addr`	`RAM_OUT`, `PC_WRITE` if `CARRY` else `PC_INC`	Nothing
13	`JZ addr`	`RAM_OUT`, `PC_WRITE` if `ZERO` else `PC_INC`	Nothing

And that "if else" in the jump rows? It is just a NOT gate and two more AND gates. If the flag is on, then we do `PC_WRITE`. If the flag is off, we do `PC_INC`. In simple terms: if we want to actually do that jump, we store the new address into `PC`. If we don't, we just increment `PC` and move on with our lives.

Feel free to try and step through this diagram, or don't, it's a giant mess:

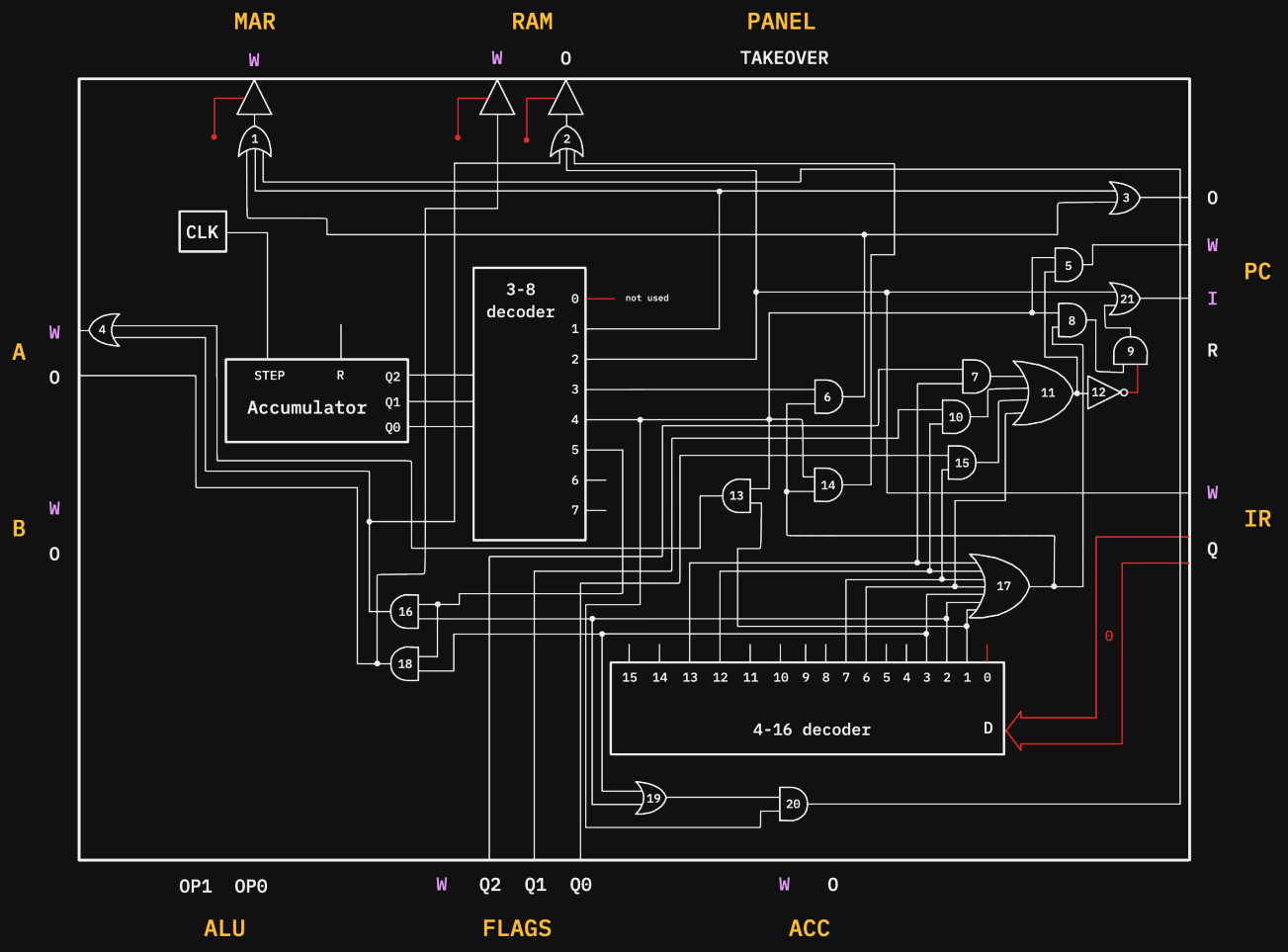


Diagram 14.1. All of the 2-byte instructions.

I know this is quite overwhelming, so let me break it down for you. We have the steps 1 and 2 wired just like before to fetch our instruction. Step 3 goes into gate 6, which is an AND gate. It basically checks, if it is step 3, and this instruction is a 2-byte instruction. Gate 17, the OR gate basically just checks if we have a 2-byte instruction or not.

So back to step 3, if we have a 2-byte instruction and it is step 3, then it enables `PC_OUT` and `MAR_WRITE`, and the `MAR_WRITE` store actually fires on `NOT CLOCK`. Step three is the same for all 2-byte instructions since we always need to fetch that second byte, but what we do with that second byte changes.

On step 4, every 2-byte instruction needs to use that second value, so we always call `RAM_OUT` if it is a 2-byte instruction. That is what gate 14 does. Then each instruction does something slightly different with that number.

Instruction 1 just stores it straight into register `A`. Gate 13 checks if it is instruction 1 and step 4, and if so enables `A_WRITE`, gated of course, only actually enabling on `NOT CLOCK`.

Instructions 2 and 3 are different. For those, that second value is not the number we want, it is the *address* of the number we want. So instead of storing it, we feed it into the `MAR`. Gate 19 checks if it is instruction 2 or 3, then gate 20 checks if it is also step 4, and if so it

enables `MAR_WRITE`. Now the `MAR` is storing the address of the right slot, all ready for step 5 to do the read or write.

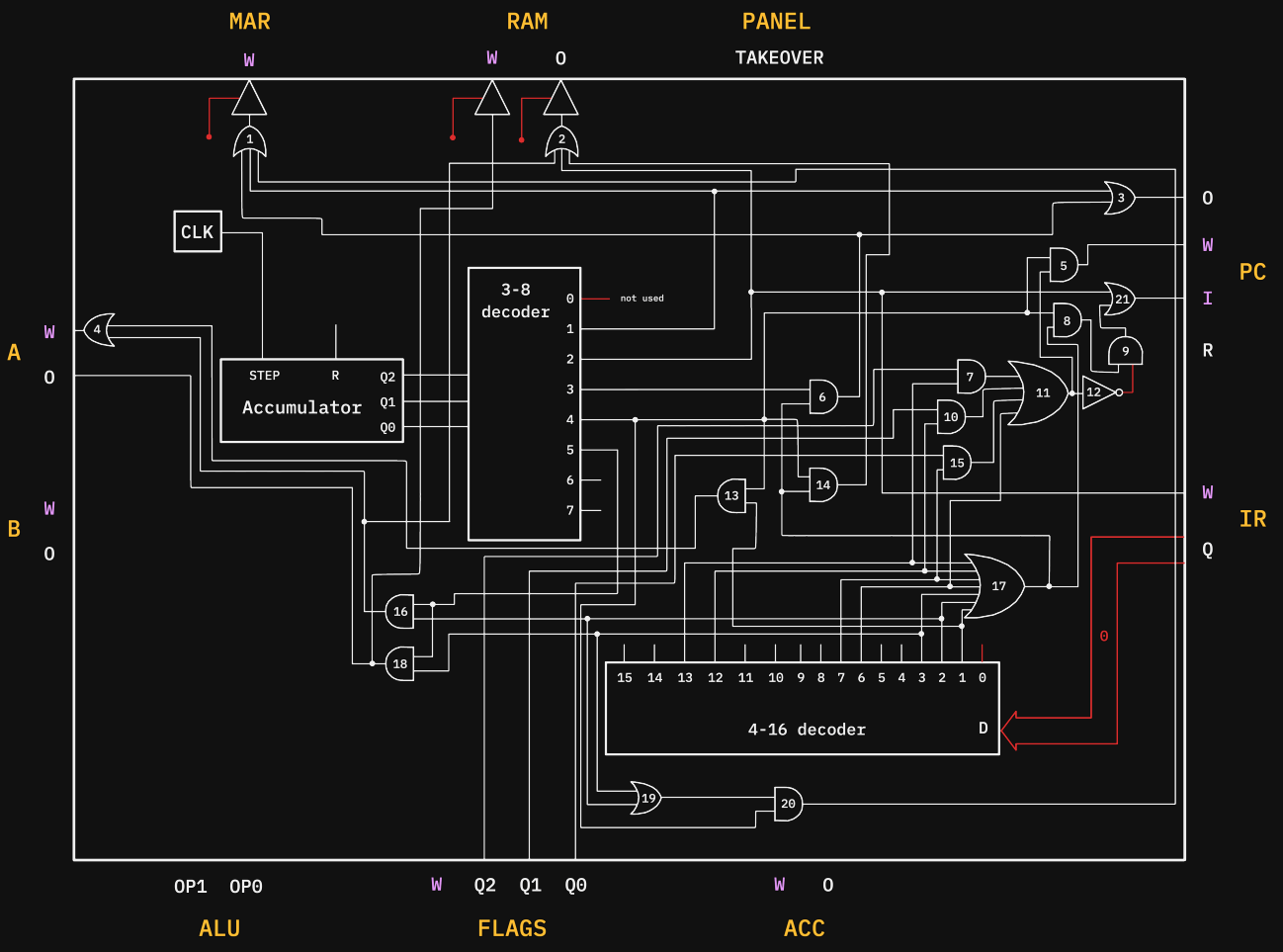
Now the jumps. For these, the second value of the instruction is a target address, so it needs to go into the `PC`. But first we have to figure out whether the jump actually happens. Gates 7, 10, and 15 do that: each one pairs a jump with its flag, so `JE` is checked against `EQUAL`, `JC` against `CARRY`, and `JZ` against `ZERO`. Those, plus a plain `JMP` which always jumps, all feed into gate 11, one big OR. If any of them fire, gate 11 turns on, meaning "this jump is happening".

From there it is simple. Gate 5 takes that signal and checks it is step 4, then enables `PC_WRITE`, loading the target address into `PC`, so then the program will start executing from that target address on the next fetch. Gate 12 then flips gate 11′s output, and gate 9 ANDs that with gate 8, which checks "step 4 and 2-byte" again. So when a 2-byte instruction reaches step 4 and the jump does not happen, it enables `PC_INC` instead, letting the program carry on to the next instruction. That is the whole "if else" we were talking about in that table. Quite straightforward.

At this point only two instructions even use step 5, `LOAD A, [addr]` and `STORE A, [addr]`. Gates 16 and 18 check if it is step 5 and either one of those instructions, then the appropriate control wires get turned on. Pretty simple.

Also, you might notice, control signals that get activated for different reasons have a "collector" OR gate instead of everything being wired together. This is just to prevent a short-circuit, one wire driving low and another driving high. A simple OR gate solves this. Gates 1, 2, 3, 4, and 21 are these collectors. Feel free to check them yourself.

Here is the process of executing the instruction `LOAD A, [addr]`. We first have to fetch the instruction, then fetch the `[addr]`, then store whatever lives at that address into register `A`.



The full animation at bytecolony.computer/writing/the-cpu/#f=cu-internals-two-demo:1.

Diagram 14.2. `LOAD A, [addr]` animation.

When the label for a control signal turns red, that means it's on. For the gated signals colored in pink, they will only actually turn on at `NOT CLOCK`. Next we do the `MOV` instructions.

#	Instruction	Step 3
4	`MOV B, A`	`A_OUT, B_WRITE`
5	`MOV A, B`	`B_OUT, A_WRITE`

Quite simple:

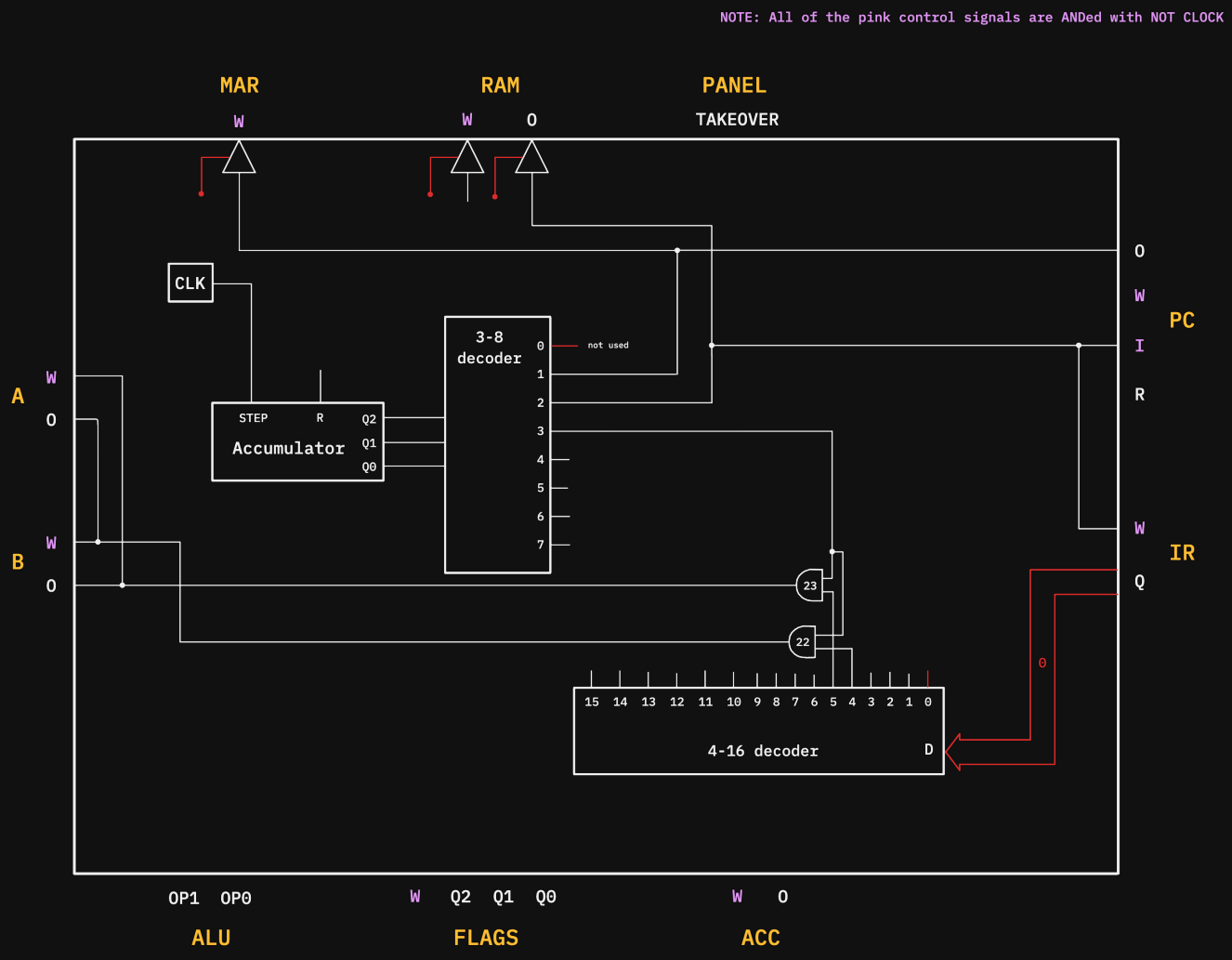


Diagram 14.3. The `MOV` instructions.

I have left out all of the 2-byte instructions from this diagram, just so it's easier to read. Next we need to decode all of our `ALU` instructions. Now this is where things get a bit weird, and I have some explaining to do.

In our CPU, if you remember, register `A` has a special `Q` that bypasses the tri-state buffers, and feeds directly into the `ALU`. So to use our `ALU`, the programmer would first put their first number into register `A` using some instructions of their choice, and put their second number into register `B`. Then any `ALU` instruction would automatically use registers `A` and `B` as the two numbers, and the `CU` copies the result back into register `A`.

#	Instruction	Step 3	Step 4
8-11	`ADD/AND/OR/XOR`	`B_OUT, ACC_WRITE, FLAGS_WRITE`	`ACC_OUT, A_WRITE`

That is why on step 3 we only have `B_OUT` and no `A_OUT` since `A` is directly connected to the `ALU` already.

But what is `ACC` you might be asking? `ACC` is kind of like a temporary register that the `ALU` uses to store its result. The `ALU` can't directly put the result onto the bus, since register `B` is driving the bus.

So on step 3, the `CU` lets `B` drive the bus, the `ALU` computes its result, and the `CU` stores the flags in `FLAGS` and the result in `ACC`. Then in step 4, when `B` stops driving the bus, we let `ACC` drive the bus and store that value into register `A`.

But now the big question: if all 4 of these instructions run the same micro-steps, what makes them different? Have a look at the last 2 bits of the number for each of these instructions in the instruction table. Then have a look at the `ALU`'s operation table.

Yep, the last 2 bits of each opcode directly match the `ALU`'s `OP` bits for that specific operation. Must be a coincidence…

Anyway, because of this, we can just directly wire those 2 last bits into the `ALU`'s `OP` bits. Since we only actually store the result of the `ALU` on these instructions, sure a `STORE A, [addr]` instruction might make the `ALU` XOR, but it doesn't matter because we don't store the result, it's garbage. So here we are:

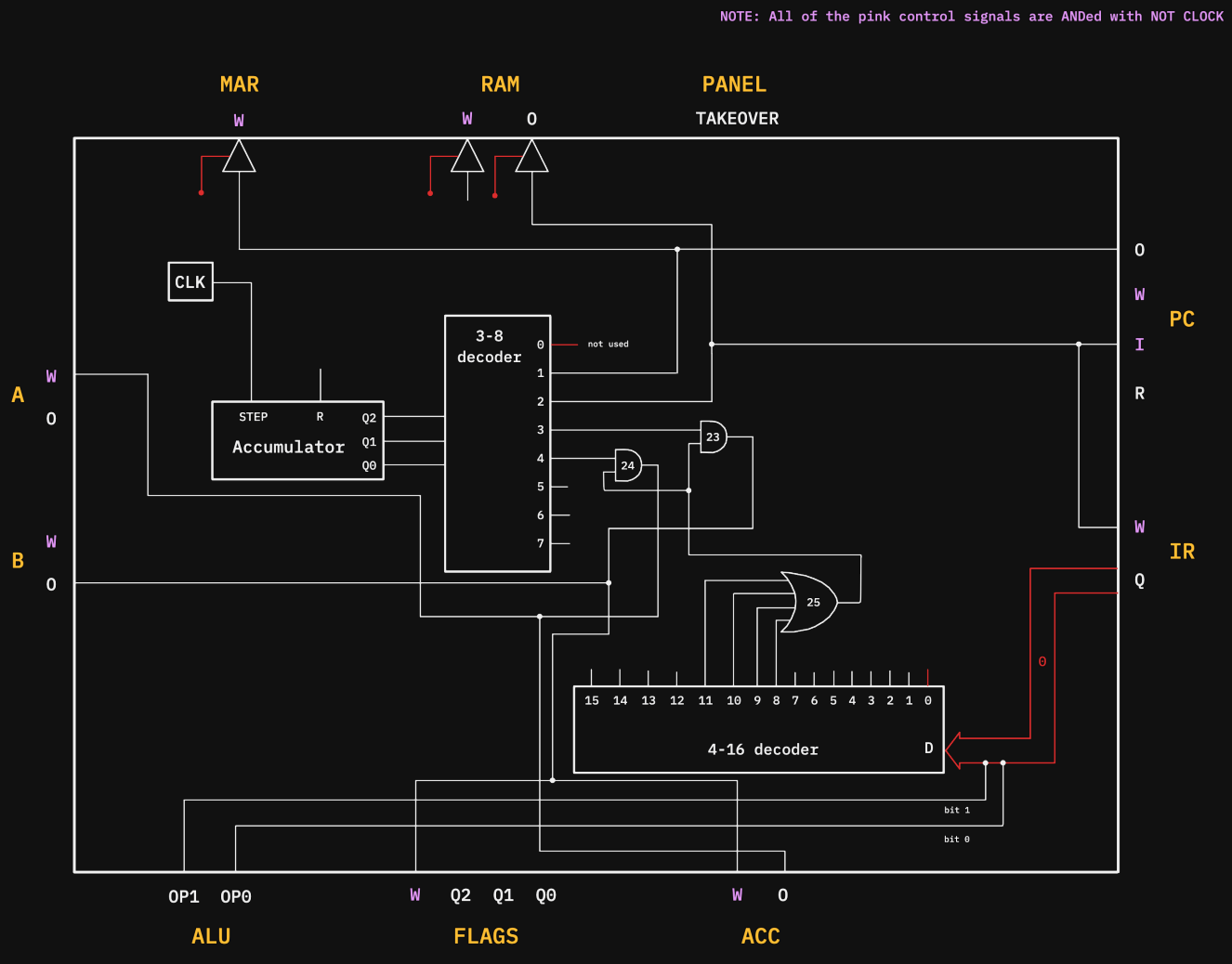


Diagram 14.4. The `ALU` instructions.

Again I am leaving out all the instructions apart from these 4, and again this wiring is a mess. All you need to know is that the bottom 2 bits of the opcode are tapped out and connected directly to the `ALU`'s `OP` bits. And the rest of the logic just checks if it is an `ALU`

instruction and if so, on the appropriate steps, it enables the correct control wires according to the table above.

One last thing: `HALT`. This instruction just stops the clock. The way we do this is by using an SR latch, whose `Q` is inverted and then ANDed with the clock signal itself. So, on step 3 of instruction number 14, we just set that latch, and boom! We froze the computer.

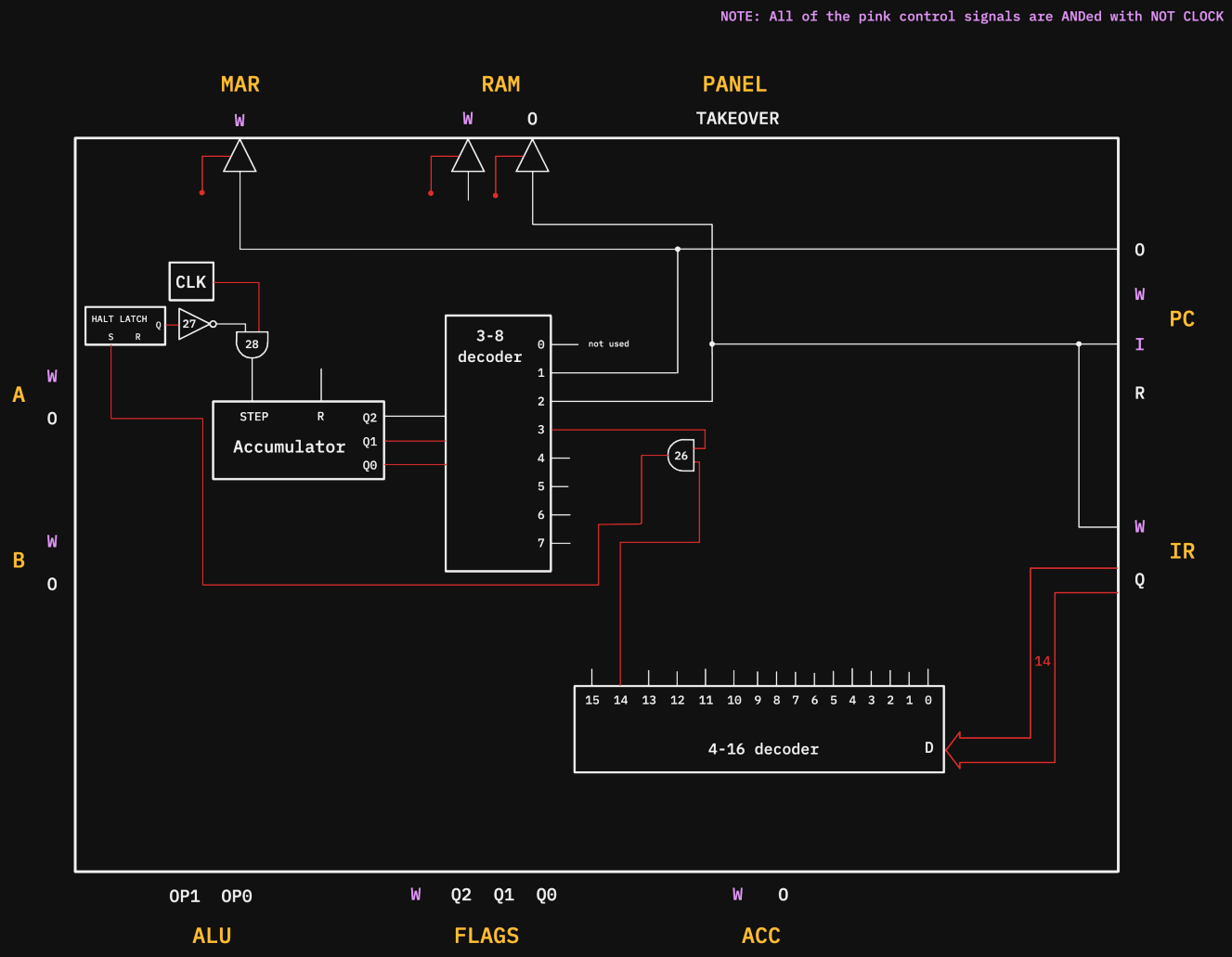


Diagram 14.5. The `HALT` instruction freezing the computer.

The clock keeps ticking, but the AND gate never lets it increment the counter, so the computer stays frozen. But how do we restart the computer? Well, remember the `TAKEOVER` switch I was telling you about? When we flick it on, it resets that latch. But its own inverted signal is ANDed in with the clock and the latch too, so a 3-input AND gate basically. So the computer stays frozen, but if you flick `TAKEOVER` off, then it will continue going.

`TAKEOVER` is also connected to `PC_RESET` which resets the program counter to 0, so the computer will start executing from the first instruction when `TAKEOVER` is turned off. It also resets the step counter, moving it back to micro-step 0 in preparation for when the computer turns back on.

`TAKEOVER` also disables the `CU`'s tri-state buffers and enables the panel's tri-state buffers for `MAR_WRITE`, `RAM_WRITE`, and `RAM_OUT`, so the panel can control those signals.

Here is how the HALT and TAKEOVER wiring would look in the CU:

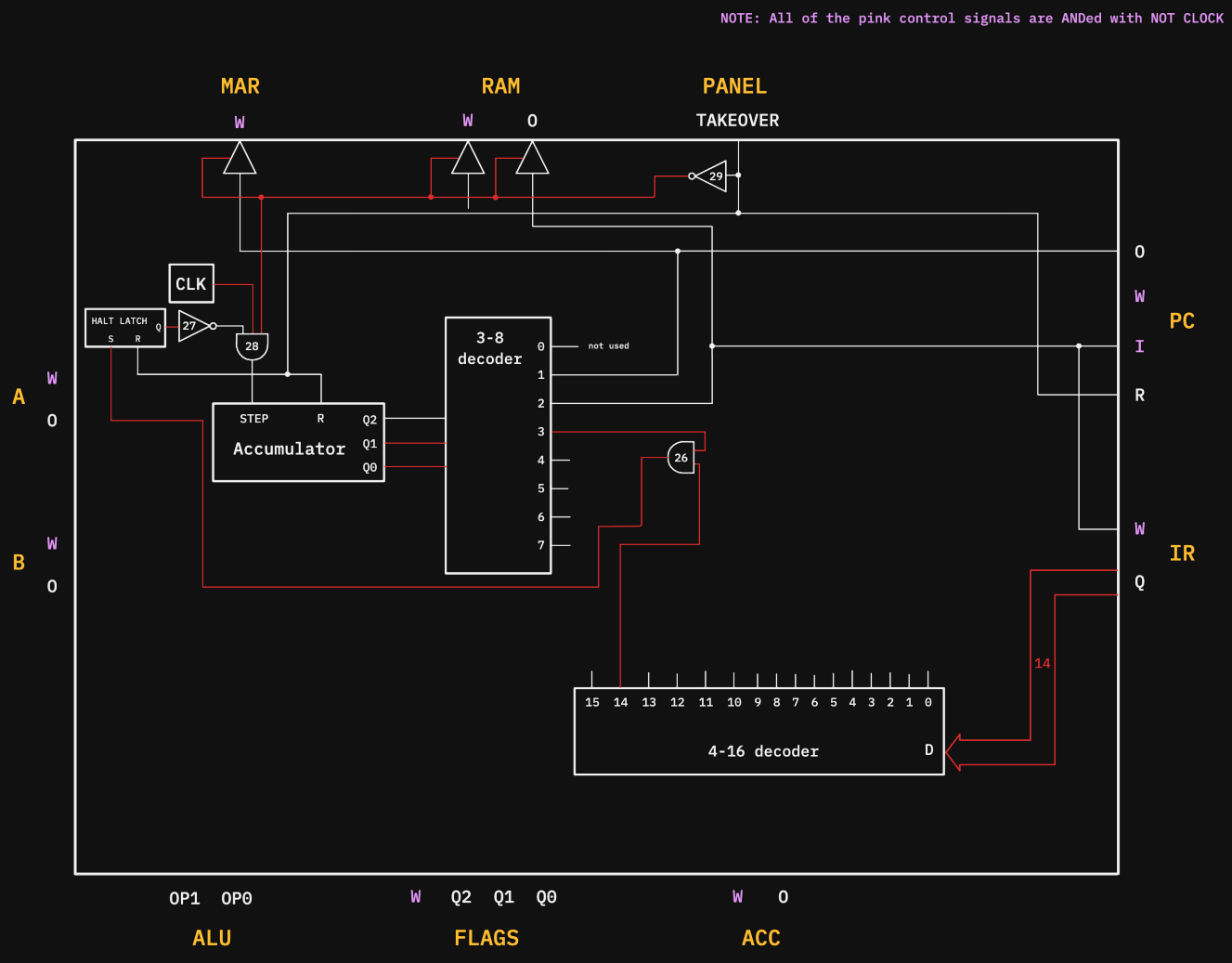


Diagram 14.6. The HALT and TAKEOVER wiring.

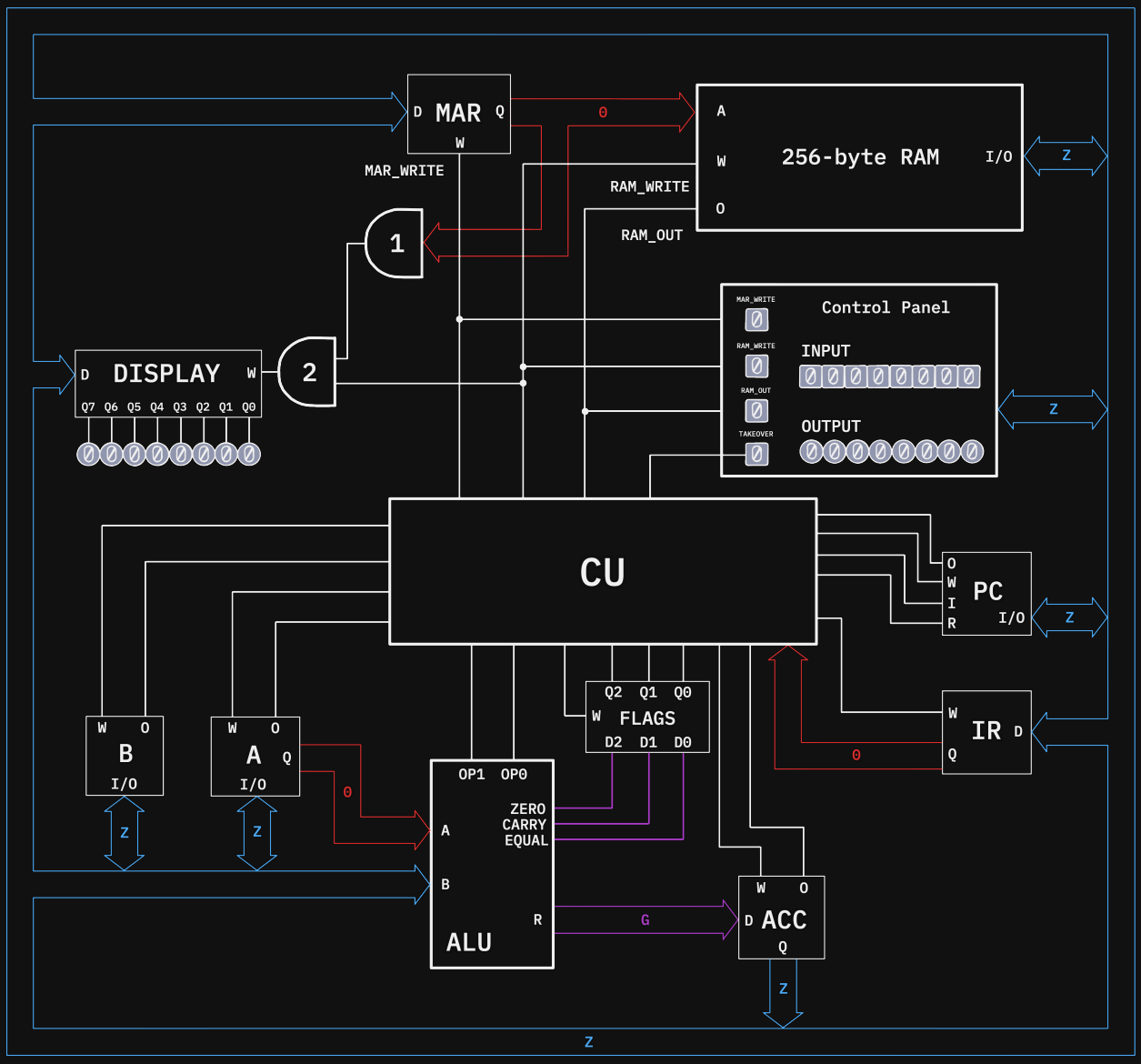
Wait, is that the whole CU? Umm, yes, that's it. Just combine all of those diagrams together, add in some extra OR gates and boom. That is the whole CU. You understand it, every wire of it. In the next short section, let's see how our computer can run a simple program.

3.4. Finally, The Fun Part

That's it, huh? A whole computer. To test it, let's watch it run. Let's start with the simplest program imaginable, just putting a number on the DISPLAY lamps:

```
0: LOAD A, 42
2: STORE A, [255]
4: HALT
```

Here is our CPU executing it:



The full animation at bytecolony.computer/writing/the-cpu/#f=program-demo:1.

Diagram 15.1. The CPU running a simple program.

For this animation, I am not animating the idle steps where nothing happens, and I am not animating 2 frames per micro-step. That's why you might see an `OUT` and a `WRITE` turn on at the same time, but you and me both know what is really happening.

And there it is, 42 sitting on the lamps. Now let's try something real, a multiplication program. Put one number in slot 100, the other in slot 101, and the product ends up on the `DISPLAY` lamps, as long as it fits in a byte.

```
0: LOAD A, 0
2: STORE A, [102]

4: LOAD A, 0
6: MOV B, A
```

```
7: LOAD A, [101]
9: ADD
10: JZ 30

12: LOAD A, [100]
14: MOV B, A
15: LOAD A, [102]
17: ADD
18: STORE A, [102]

20: LOAD A, 255
22: MOV B, A
23: LOAD A, [101]
25: ADD
26: STORE A, [101]

28: JMP 4

30: LOAD A, [102]
32: STORE A, [255]
34: HALT
```

The running total lives in slot 102. The program adds the value in slot 100 into slot 102, and then slot 101 counts down by one each time, until it hits 0. This is basically all the program is:

```
total = 0 (slot 102)
while counter is not 0: (counter is slot 101)
    total = total + number (number is slot 100)
    counter = counter - 1
show total on the lamps
halt
```

You might notice something weird:

```
20: LOAD A, 255
22: MOV B, A
23: LOAD A, [101]
25: ADD
26: STORE A, [101]
```

This little sequence adds 255 to slot 101. It loads the right values into the CPU's registers, adds 255, and stores that back into slot 101. But why?

On this computer, adding 255 is just like subtracting 1! This is because the byte throws away anything past 255, so adding 255 and dropping the overflow is the same as taking 1 away. For example, let's say we have the number 5, we add 255, we get 260, but a byte runs out after 255 and rolls back over to 0, so 260 wraps around to 4!

One more oddity:

```
4: LOAD A, 0
6: MOV B, A
7: LOAD A, [101]
9: ADD
10: JZ 30
```

Our goal with this chunk of code is to just check if slot 101 is zero. If so, we are done and can jump to some other part of the program. Because this CPU is slightly more primitive than real CPUs, the programmer (me) had to use some weird trickery to check if it is zero. In this case I just added 0 to the value to set the `ZERO` flag.

My point with showing you all of these tricks is that, even though this CPU might not be the most advanced, it can do real things!

Modern CPUs have proper solutions to things like subtraction, negative numbers, division, more flags, and a lot more that smart computer science people just thought of. You can go learn about them yourself. But here is the thing: those dedicated circuits exist mostly for speed. Our funky little CPU can already do all of it, just slower, with tricks like the ones above.

Modern CPUs are also just bigger in general. Instead of 8 wires in a bus, 64. Instead of 256 bytes of `RAM`, billions. Instead of 2 registers you can program with, dozens. Instead of a few ticks per second, billions of those too. But it is the same core loop, fetch, decode, execute, just bigger and faster.

Our program could also be seen as just a bunch of numbers:

```
1 0 3 102 1 0 4 2 101 8 13 30 2 100 4 2 102 8 3 102
1 255 4 2 101 8 3 101 6 4 2 102 3 255 14
```

And in `RAM`, it is actually just this:

```
0: 00000001 12: 00000010 24: 01100101
1: 00000000 13: 01100100 25: 00001000
2: 00000011 14: 00000100 26: 00000011
3: 01100110 15: 00000010 27: 01100101
4: 00000001 16: 01100110 28: 00000110
5: 00000000 17: 00001000 29: 00000100
6: 00000100 18: 00000011 30: 00000010
7: 00000010 19: 01100110 31: 01100110
8: 01100101 20: 00000001 32: 00000011
9: 00001000 21: 11111111 33: 11111111
10: 00001101 22: 00000100 34: 00001110
11: 00011110 23: 00000010
```

Some user would have to manually sit there at the `Control Panel` flipping these values into addresses of `RAM` one at a time. In the next closing section, we will touch a little bit on modern things like keyboards and monitors, all that other stuff that really turns a CPU into a computer.

3.5. The End

So what about keyboards, monitors? All the things that turn a CPU into a computer. Well, you have already seen the core idea. Imagine the `DISPLAY` register, just much bigger: thousands of memory slots instead of one, where each number sets the color of one tiny pixel on a screen. A keyboard is the same thing but in reverse, a slot you read instead of write, where the value is a code for the last key pressed.

Real computers are a lot fancier about it. They have dedicated I/O buses, whole controller chips that talk to the devices so the CPU doesn't have to, and devices that can even write into `RAM` directly on their own.

So to recap, a program is just a list of numbers sitting in `RAM`. The control unit fetches those numbers one at a time, decodes them, and executes each one as a few micro-steps, flipping control wires that move bytes between registers. Every one of those components is made up of simple logic gates, and every gate is made up of relays, or more commonly these days, transistors.

I hope you now understand why computers aren't "just" 1s and 0s, but a very tall pile of simple.